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Contents

Abstract This thesis analyses decision-making following penalties in rugby union, focusing on the choice between kicking at goal and kicking to the corner. I develop a structural decision framework that evaluates these options under two competing objectives: maximising win probability and maximising expected league points. The latter incorporates bonus-point incentives, which are central to league competition formats.

The model combines empirical estimation and simulation. A goal-kicking model is used to estimate conversion probabilities, while a stochastic framework models attacking sequences following kicks to the corner. Match outcomes are then simulated to evaluate the expected value of each decision under both objectives across game states defined by score margin, time, and try count.

The results show substantial divergence between the two objectives. While the win probability objective typically favours the safer option of kicking at goal, the expected league points objective frequently favours the corner, particularly when teams are close to securing a bonus point. These differences are most pronounced in late-game scenarios and can lead to economically meaningful losses when decisions are made using the wrong objective.

Overall, the findings highlight the importance of aligning decision-making with competition incentives. Strategies that maximise win probability are not always optimal in a league context, and incorporating bonus-point considerations can materially alter optimal play.

2 Introduction

In Round 18 of the 2024/25 Energia All-Ireland League Division 1B season, Dublin University led Nenagh Ormond 10-7 with twelve minutes remaining. They were awarded a penalty in a central position, 24 metres from the posts. The kick was makeable, around 95 percent. Three points would extend the lead to six. But Dublin University sat in eighth place on points difference alone, level on 29 league points with Highfield in ninth. Shannon, five points behind in tenth, were dismantling Queen's University Belfast in a concurrent fixture and would finish the day on 29 points themselves. Highfield were trailing Naas by two with time running out. If Highfield lost, Dublin University would secure safety regardless of their own result. If Highfield won, Dublin University needed five league points to guarantee eighth place.

Dublin University had scored one try. They were not chasing the try bonus. But the choice between posts and corner still mattered for survival. A successful kick would extend the lead but leave the team reliant on Highfield's result. Kicking to the corner offered a path to a second try, a larger margin, and reduced exposure to what was happening in Naas.

The framework developed in this thesis identifies this as a decision where win probability and expected league points favour the kick, but survival probability favours the corner. The difference is small in absolute terms. It is not small when the consequence is a relegation playoff.

Dublin University kicked at goal. They made it. They won 20-7. Highfield won 43-42 with the final play of their match. Dublin University finished ninth on points difference and entered the relegation playoff, surviving only after a last-minute kick in the playoff final.

This thesis argues that penalty decisions cannot be evaluated without reference to the objective being maximised. The choice that maximises win probability does not always maximise survival probability. The choice that maximises expected league points does not always maximise promotion probability. Teams judge success by promotion or survival, not points alone, and because these outcomes depend nonlinearly on final league position, optimal decisions can differ sharply from those implied by match-level logic.

Research Question

This thesis asks: what is the optimal penalty decision in rugby union when teams may be maximising different objective functions, and how do observed decisions compare to these benchmarks? The objectives considered include win probability, expected league points, promotion probability, and survival probability. The analysis covers 1,349 penalty decisions from 165 matches across two seasons of the Energia All-Ireland League Division 1B.

The central finding is that teams make suboptimal decisions in approximately 25 percent of cases, but not because they are systematically too conservative or too aggressive. Rather, they behave as though maximising the wrong objective, failing to account for the nonlinear incentives created by league structures. Decision accuracy varies by more than 24 percentage points between teams, and errors concentrate at specific score margins where these incentives bind most tightly.

Why This Matters

The penalty decision is a choice under uncertainty with state-dependent payoffs. Kicking at goal yields three points with some probability and zero otherwise. Kicking to the corner yields a distribution over outcomes: a try, a turnover, another penalty, or time expired. The value of each outcome depends on the score, the time remaining, the tries already scored, and the position of the team in the league table.

Expected points provides a useful but incomplete benchmark. If the expected points from kicking at goal exceed those from kicking to the corner, kick at goal. This logic underpins recent work applying expected points frameworks to rugby union. But expected points assumes that all points are equally valuable. They are not.

A team level on points with the team below them does not value league points linearly. Four points from a win might guarantee safety. One point from a losing bonus might require other results. Zero points might mean a playoff. The marginal value of points is discontinuous, and a decision that maximises expected points may minimise survival probability. The same logic applies at the top of the table: the marginal point that secures promotion is worth more than the marginal point that moves a team from third to second. Because final league position depends on the outcomes of all remaining matches, not just the current one, the value of any individual decision is shaped by the entire competition.

This thesis takes these nonlinear incentives seriously. It evaluates penalty decisions not only under expected points and win probability, but under the objectives that actually determine how seasons are judged.

Contributions

This thesis makes four contributions.

The first is a decision-theoretic framework for rugby union penalty strategy that, to my knowledge, is the first to evaluate choices under season-level objectives such as promotion probability and survival probability. Existing expected points models ask what an action is worth in terms of immediate scoring potential. This framework asks what an action is worth in terms of ultimate season outcomes, and derives optimal policy rules accordingly.

The second contribution is evidence that optimal decisions depend critically on the objective being maximised. Under expected league points, the framework recommends kicking to the corner in nearly 80 percent of situations. Under win probability, the split is closer to even. Under survival probability, the optimal action can reverse entirely depending on the league table. These disagreements arise in common match states and affect teams throughout the competition.

The third contribution is new evidence on decision quality. Teams achieve 74.8 percent accuracy against the framework's recommendations under expected league points. The median loss per suboptimal decision is 0.08 expected league points. Accuracy collapses at specific score margins, falling to 33 percent when leading by eight points. Relegation-threatened teams perform worst, achieving only 65 percent accuracy under the survival objective in late-season matches. These patterns echo the systematic inefficiencies documented in American football fourth-down decisions, though the mechanism differs: teams here do not exhibit consistent conservatism, but rather appear to optimise for match outcomes when season outcomes should dominate.

The fourth contribution is methodological. The framework integrates a continuous-time Weibull competing-hazards model for match scoring, a hierarchical goal-kick model that corrects for selection bias from unattempted penalties, and a state-transition model for corner attacks that tracks spatial position and time elapsed. These components feed into a league simulation that propagates individual decisions through to season-level consequences.

Preview of Findings

The framework produces clear recommendations in 95 percent of penalty situations. Teams are currently right about three quarters of the time, which sounds respectable until the errors are examined more closely.

The errors are not random. Teams systematically over-kick, choosing to take points in situations where attacking from the corner would yield higher expected value. The heuristic of always kicking to the corner agrees with the framework 84 percent of the time, outperforming observed captain decisions by nine percentage points. This does not mean teams should mindlessly attack. It means that the situations where kicking looks obviously correct are often exactly the situations where it is least optimal under the objectives that matter.

Decision accuracy varies dramatically by game state. Teams perform well at large deficits and large leads, where the optimal action is unambiguous. They struggle at intermediate margins where bonus point incentives create genuine tension. At a lead of eight points, where neither a penalty nor a try crosses a strategically meaningful threshold, accuracy collapses to 33 percent. At a lead of one point, accuracy is 60 percent. These are not obscure edge cases. They are common match situations where teams consistently get the decision wrong.

The gap between objectives is starkest for teams in contention. When expected league points and survival probability prescribe different actions, teams follow the expected league points recommendation 73 percent of the time. They are optimising for the wrong thing. The framework developed here makes the right objective explicit and provides the tools to act on it.

Roadmap

Section 2 reviews the literature on sports analytics, expected value frameworks, and decision-making under uncertainty. Section 3 describes the data. Section 4 presents the methodology: the scoring model, the corner model, the goal-kick model, and the decision framework that integrates them. Section 5 reports results on optimal thresholds, observed decision quality, season-level objectives, and case studies of consequential decisions. Section 6 concludes.

3 Literature Review

In American football, analysts can tell you the expected point value of any field position to two decimal places. In basketball, player tracking data feeds real-time models of possession value that update with every pass. In rugby union, teams still rely largely on intuition. This dissertation addresses a gap that is both surprising and consequential: the absence of dynamic, season-aware decision models that link individual choices to the objectives teams actually pursue.

3.1 State-Based Decision Models in Sport

The foundational insight of modern sports analytics is that decisions can be evaluated by comparing the expected value of alternative actions within a well-defined state space. Romer (2006) applies this logic to fourth-down decisions in American football, demonstrating that coaches systematically deviate from win-maximising strategies. His finding that teams are too conservative has become a canonical example of suboptimal decision-making in professional sport. Subsequent work extends this framework through expected points and win probability models that map game states to expected outcomes burke2009, burke2014. These models provide a normative benchmark against which observed behaviour can be evaluated.

In basketball, Cervone et al. (2016) develop Expected Possession Value, a multiresolution stochastic process model that combines continuous-time hazard modelling with discrete state transitions. The key innovation is treating possession as a stochastic process rather than a static state, allowing the model to capture how decisions shape the distribution of future outcomes. Value updates in real time as play unfolds.

Direct application of these methods to rugby is complicated by structural differences. A fourth-down decision has two or three discrete options and a relatively predictable subsequent state. A penalty decision in rugby initiates a branching tree of possibilities: lineout outcomes, maul phases, open play, turnovers, further penalties, and eventually a score or change of possession. Simple state mappings are insufficient without an explicit model of how play evolves. This is precisely what existing rugby frameworks lack.

3.2 Expected Value Frameworks in Rugby

Within rugby union, recent work has begun to adapt expected value concepts. Fitzpatrick and Nolan (2024) introduce an expected-points framework for valuing possessions and kicks, showing that field position can be translated into expected scoring outcomes. Their results provide a baseline for tactical evaluation. The framework is, however, fundamentally static: it evaluates decisions at a single point in time without modelling subsequent play. A captain choosing between posts and corner learns what each option is worth in expected points but not how the match might unfold from each starting point.

This limitation is consequential. A decision that maximises immediate expected points need not maximise win probability. Late in a match, with time constraints binding, the divergence can reverse the optimal action entirely. A team trailing by four with two minutes remaining may find that the expected points from a penalty goal exceed those from a corner attack, yet the win probability from attacking is higher because three points cannot win the match. Neither objective necessarily aligns with season-level goals such as promotion or survival. Fitzpatrick and Nolan acknowledge these extensions but do not pursue them.

More recent work applies expected-points concepts directly to the penalty decision. A 2025 preprint develops decision maps for the choice between goal and touch, varying recommendations by field position and game context rgbyreferee2025. This represents progress toward context-aware evaluation, but the analysis remains within the expected-points paradigm. Win probability appears as a possible extension; season-level incentives do not appear at all.

In rugby league, Sawczuk et al. (2021) develop expected possession value models using Markov chain methods, confirming the feasibility of possession-based valuation in rugby codes. Their work is descriptive rather than prescriptive, characterising what happened rather than recommending what should happen.

The literature establishes that expected value frameworks can be applied to rugby. Existing implementations stop short of the full decision problem: they evaluate actions without modelling state transitions, and they optimise for immediate value without considering the nonlinear objectives created by league structures.

3.3 Stochastic Scoring Models

A separate methodological literature addresses the statistical modelling of scoring processes. Traditional approaches assume that scoring follows a Poisson process, implying a constant hazard over time. This assumption is analytically convenient but empirically rejected. Dixon and Coles (1997) introduce time-varying adjustments to improve fit for football scores, while subsequent work demonstrates that Weibull and other flexible distributions outperform the Poisson baseline boshnakov2017.

The rejection of constant hazards has substantive implications for decision modelling. If scoring intensity varies with elapsed time, whether due to fatigue, tactical adjustments, or pressure accumulation, then static models will misvalue decisions that occur at different points in a match. A penalty in the 60th minute is not equivalent to a penalty in the 80th minute, even holding scoreline and field position constant.

This dissertation adopts a Weibull competing-hazards framework. The Weibull distribution nests the exponential as a special case but allows the hazard to increase or decrease over time. The competing-hazards structure allows tries and goals, home and away, to exhibit distinct dynamics. Applied to rugby, this captures the empirical reality that scoring often arises from sustained pressure rather than isolated events.

3.4 Goal-Kicking and Selection Bias

A further challenge arises in estimating goal-kicking success. Observed kick outcomes are subject to selection bias: teams are more likely to attempt kicks they expect to make. A model trained on observed attempts will overestimate success probabilities because the sample excludes the difficult kicks that teams declined.

Lock and Nettleton (2014) formalise this problem for American football field goals. Accounting for informative missingness substantially alters estimated success curves. Kicks that appear marginal based on observed data are more difficult than naive estimates suggest, because the truly marginal kicks were never attempted.

In rugby union, goal kicking accounts for a substantial share of scoring, yet existing models typically condition on observed attempts without addressing selection. Quarrie and Hopkins (2015) model international goal-kicking performance and identify distance as a key determinant of success, but their analysis does not correct for the selection induced by the attempt decision.

This dissertation fills this gap by jointly modelling the attempt decision and the success probability. The framework treats foregone kicks as informative: a team that chooses corner over goal reveals something about the difficulty of the kick it declined. This allows more accurate estimation of success probabilities at the margin of the attempt threshold, precisely where valuation matters most.

3.5 League Structure and Nonlinear Incentives

Standard expected value frameworks assume that teams maximise expected points or win probability. Both objectives are approximately linear in outcomes: each additional point or percentage point is equally valuable. This assumption fails when payoffs depend on final league position.

Lahvička (2012) uses Monte Carlo simulation to show that the marginal value of league points is highly state-dependent. A point that secures promotion is worth more than a point that moves a team from fifth to fourth. A point that avoids relegation is worth more than a point that consolidates mid-table safety. Promotion and relegation systems generate rank-based payoffs, not point-based payoffs, and the resulting discontinuities alter optimal behaviour.

Szymanski (2003) and subsequent work in sports economics formalise this insight. Teams near promotion or relegation thresholds face different risk-return trade-offs than teams in mid-table positions. A strategy that is risk-neutral with respect to points may be risk-seeking or risk-averse with respect to position.

Despite this, existing decision models in sport rarely incorporate season-level objectives. Romer evaluates fourth-down decisions against win probability but not playoff implications. Expected-points frameworks evaluate immediate value without reference to standings. The literature provides tools for match-level optimisation but limited guidance when the true objective is promotion or survival.

3.6 Behavioural Implications

The gap between match-level and season-level objectives has behavioural consequences. If teams evaluate decisions using heuristics aligned with expected points or win probability, they will systematically deviate from strategies that maximise promotion or survival probabilities. The deviation need not reflect poor reasoning. It may reflect objective misalignment: optimising a tractable proxy because the true objective is difficult to compute.

Expected points and win probability are salient and intuitive. A captain knows whether the team is ahead or behind, roughly how much time remains, and whether a kick looks makeable. Promotion and survival probabilities require information about the league table, concurrent fixtures, and remaining schedules. Real-time computation is impractical without analytical infrastructure that most teams lack.

The empirical signature of objective misalignment differs from systematic conservatism or aggression. Romer finds that NFL coaches are too conservative on fourth down across nearly all situations. If rugby teams exhibited the same pattern, we would expect consistent bias toward one action regardless of context. The findings of this dissertation suggest something different: teams behave as though optimising for match outcomes when season outcomes should dominate. The error is in the objective function, not the risk preference.

3.7 Contribution

This dissertation integrates the strands reviewed above into a unified decision-theoretic framework. It combines a selection-corrected model of goal kicking, a Weibull competing-hazards model of match scoring, and a stochastic state-transition model for corner attacks, all embedded within a Monte Carlo simulation of league outcomes. Penalty decisions can be evaluated under win probability, expected league points, promotion probability, and survival probability.

No existing work jointly models in-game decision-making, stochastic match dynamics, and season-level incentives in rugby union. The penalty decision literature stops at expected points. The league simulation literature does not connect to in-game choices. This dissertation bridges the two, extending expected value frameworks into a setting where the objective function is itself state-dependent. The gap is clear, and the contribution fills it.

4 Context and Data Description

4.1 Data and Sample Construction

The dataset is constructed from two full seasons of All-Ireland League Division 1A match data, comprising 165 fixtures and 20 team-seasons. The underlying data originate from Opta Sports and were accessed through Dublin University Football Club. These data consist of event-level records, with each action timestamped and spatially referenced at the player level, enabling exact reconstruction of match states at any point in time.

From this raw event data, a dataset of penalty decisions is constructed. The analysis focuses on situations in which teams face a meaningful strategic choice between attempting a shot at goal and kicking to touch. To ensure comparability, the sample is restricted to penalties awarded within 55 metres of the opponent's try line—approximately the longest successful penalty kick recorded in AIL competition—and when the score margin is 12 points or fewer. These restrictions exclude situations where the decision is either trivial (penalties outside realistic kicking range) or strategically dominated by match context (large score differentials where chasing points becomes imperative).

Within this filtered sample, 1,349 penalty decisions are identified. The analysis further focuses on matches in which teams remain in contention for promotion or relegation, ensuring that season-level incentives are active. The resulting sample isolates precisely those situations where strategic trade-offs are both non-trivial and consequential.

4.2 Contextual Features

Each penalty decision is embedded within a rich state description. This includes the current scoreline, time remaining, field position, and team identities, as well as derived features such as try counts and bonus point status. These variables define the state space over which decisions are evaluated.

Field position is measured relative to the opponent's try line, enabling consistent comparison of kicking difficulty and attacking opportunity. Match time is recorded continuously, allowing decisions to be analysed across all phases of play, from early possessions to late-game scenarios where time constraints bind.

Critically, the data support forward simulation of match evolution. Each observed state serves as an entry point into the stochastic scoring model, allowing the value of alternative decisions to be evaluated in terms of their impact on final match outcomes and league standings.

4.3 Sample Characteristics

Table 1 presents summary statistics for the 1,349 penalty decisions. The median penalty occurs at the 34th minute, 32 metres from the try line, with teams level on the scoreboard. The interquartile range of score margins spans from five points behind to three points ahead, confirming that the sample restrictions successfully isolate competitive match states.

Table 1: Summary Statistics for Penalty Decisions

Variable	Mean	SD	Min	Q25	Median	Q75	Max
Match Minute	35.1	23.2	0.1	14.5	33.7	54.4	79.7
Distance from Try Line (m)	31.9	14.1	10.0	21.0	32.0	45.0	55.0
Kick Angle (°)	8.0	4.3	2.7	5.7	6.6	8.8	31.3
Score Margin	−0.5	6.1	−12	−$5	0	3	12
Attacking Team Tries	1.2	1.3	0	0	1	2	7
Predicted Kick Success (%)	64.0	16.0	24.0	54.0	63.0	75.0	100.0

Notes: n = 1,349. Score margin measured from the attacking team's perspective; negative values indicate trailing. Predicted kick success derived from the goal-kicking model.

The distribution of observed decisions is striking. As shown in Figure 1, teams elect to kick to touch in 76.1% of cases (1,027 penalties), taking the shot at goal in only 23.9% of cases (322 penalties). This strong baseline preference for territorial advancement over point accumulation provides the empirical foundation against which model-implied optimal decisions will be compared.

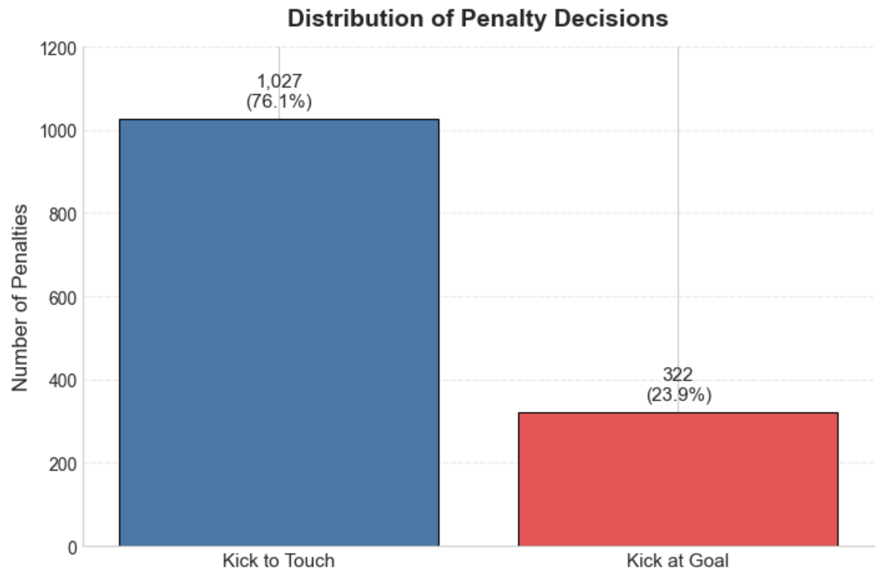


Figure 1: Distribution of Penalty Decisions. n = 1,349. Kick to touch includes all decisions to decline the shot at goal in favour of territorial advancement.

Figure 2 presents the distribution of score margins at the moment of decision. The distribution is approximately symmetric around zero, with a slight leftward skew reflecting the marginal tendency for trailing teams to receive penalties in attacking positions. Roughly half of all decisions occur when teams are trailing or level, situations in which the value of three points may differ substantially from when leading.

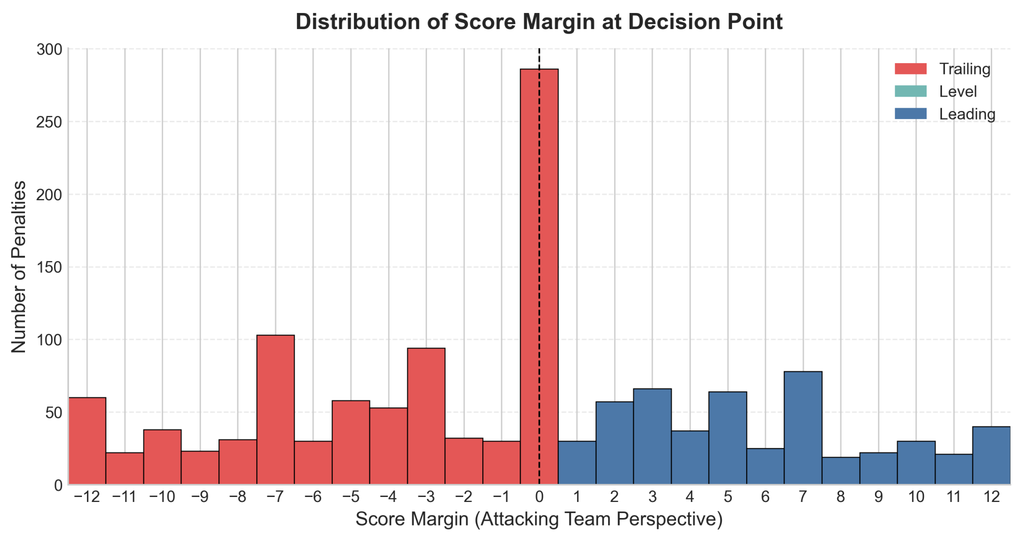


Figure 2: Distribution of Score Margins at Decision Point. Score margin measured from the attacking team's perspective. Negative values indicate the attacking team is trailing. Sample restricted to margins between -12 and $+12$ points.

The dataset captures substantial variation across all relevant dimensions—penalties occur throughout the full 80 minutes, across the range of kickable distances, and in diverse scoreline configurations. This variation is essential for identifying how optimal decisions depend on context and for comparing observed behaviour to model-based benchmarks.

4.4 Data Architecture

The empirical analysis is supported by several derived datasets, each corresponding to a component of the modelling framework: scoring spells for the competing hazards model, penalty decisions with full state information, goal-kick attempts with selection indicators, and simulated state distributions for value function computation. Raw event data are transformed into structured representations of match states, which are then used to estimate models, simulate outcomes, and evaluate decisions. This architecture allows the analysis to move beyond static descriptions and instead link individual decisions to their full distribution of downstream consequences.

4.5 Limitations

Three data constraints warrant acknowledgment. First, while Opta data are industry-standard, they do not capture certain strategically relevant factors—player fatigue, injury status, or real-time weather conditions—that may influence both decision-making and outcome probabilities. Second, the sample is drawn from a single competition across two seasons; generalisability to other leagues or international competition remains an empirical question. Third, the restriction to matches where promotion or relegation incentives are active excludes a subset of late-season fixtures, introducing potential selection concerns.

These limitations are common to observational studies of sports decision-making and do not undermine the core contribution. The dataset remains among the most granular and comprehensive used in rugby analytics research to date, and the modelling framework explicitly accounts for uncertainty in both transition dynamics and kicking success.

5 Methodology

5.1 Competing Hazards Scoring Model

Evaluating a penalty decision requires a model of what happens next. If a team kicks to the corner and does not score immediately, how is the rest of the match likely to unfold? If it takes the three points, how likely is that lead to hold? Answering these questions requires a model that can start from any scoreline, time, and team matchup, and project the remainder of the match in a realistic way.

I model scoring as a continuous-time competing hazards process. At any moment, one of four mutually exclusive events may occur: a home try, an away try, a home goal, or an away goal. Time passes continuously between events, so the model produces complete match trajectories rather than a single reduced-form forecast.

A natural alternative would be to estimate win probability directly as a function of game state. That is not enough here. The framework requires not only win probability, but also expected league points, promotion probability, and survival probability. These depend on full match trajectories, including final margins and try counts, and in the latter two cases on how the current fixture interacts with the rest of the season. For that reason, I use a generative scoring model rather than a single prediction equation.

The data are organised into spells. Each spell begins at kickoff, at the start of the second half, or immediately after a scoring event, and ends either when the next score occurs or when the half finishes without further scoring. This makes a competing-risks approach natural, since each spell can end in only one of four scoring events.

For each event type, I specify a Weibull hazard:

$$h_k(t \mid \mathbf{X}) = \alpha_k \lambda_k t^{\alpha_k - 1} \exp(\eta_k),$$

where t is the time since the start of the spell, α_k governs duration dependence, λ_k is a baseline rate, and η_k captures the effect of team strength and match state. This allows scoring intensity to vary over the course of a spell rather than forcing it to remain constant. That flexibility matters in rugby, where sustained territorial pressure can make a score increasingly likely.

The model also allows scoring rates to depend on team strength, home advantage, and the current score margin. This is important because teams behave differently depending on the scoreboard. A side trailing by five does not face the same incentives as one trailing by two, and a side protecting a lead behaves differently again. For example, the home try hazard includes

$$\eta_{\mathrm{HT}} = \gamma_{\mathrm{try}} + \left(A_h^{\mathrm{try}} - D_a^{\mathrm{try}}\right) + \beta^{\mathrm{try}}(m),$$

where γ_{try} is home advantage, A_h^{try} and D_a^{try} are the home team's attacking strength and the away team's defensive strength, and $\beta^{\mathrm{try}}(m)$ allows scoring behaviour to vary with the current score margin m. Tries and goals are allowed to have different duration profiles, reflecting the fact that they arise through different underlying processes.

The model is estimated by penalised maximum likelihood. Ridge shrinkage is applied to the team effects, while smoothness penalties are applied to the score-margin functions to reduce overfitting and avoid implausibly erratic responses across adjacent margins. Full specification and estimation details are provided in Appendix A.1.

Once estimated, the model is simulated forward from any post-decision state. At each step, it generates the timing and type of the next scoring event, updates the match state, and continues until full time. Each try is then followed by a conversion attempt. Since the scoring model does not generate try location, conversion success is handled separately using team-season conversion rates with shrinkage toward the competition average.

The simulations record both final score and try counts, which allows league points to be assigned under AIL rules. This is central to the decision problem, because the value of a penalty depends not only on whether a team wins, but also on bonus points and final margin.

Before using the model for decision analysis, I assess calibration and out-of-sample predictive performance. The model performs well on both dimensions, and full validation results are reported in Appendix A.2.

The scoring model is therefore the core engine of the decision framework. It translates post-decision states into continuation values for match-level objectives such as win probability and expected league points, and it also provides the match simulations that feed into the season-level analysis of promotion and survival.

5.2 Kick to Corner Model

Kicking to the corner does not produce an immediate scoring outcome. Instead, it initiates a structured attacking sequence, the value of which depends on territory, possession, and the probability of converting that possession into points. I model this as a multi-stage process, where each stage captures a distinct component of attacking play following a penalty.

The sequence proceeds as follows. First, the kick determines the location of the resulting lineout. Second, the attacking team may or may not secure possession at the lineout. Third, conditional on retaining possession, the team enters an attacking phase, beginning with a maul and potentially transitioning into open play. The value of the corner option is obtained by simulating this sequence forward until a terminal outcome is reached.

Stage One: Territory. The first stage determines where the kick lands. Not all kicks successfully find touch, and those that do vary in the territory gained. I model this using a two-part structure. The first component estimates the probability that the kick finds touch, while the second models the distribution of field position conditional on success.

Let ℓ denote the penalty location. The model produces a probability distribution over a set of discrete touchline bins, ranging from positions close to the try line to minimal territorial gain, along with the probability of a missed kick. This captures the key trade-off between execution risk and territorial reward. In practice, I estimate this using a multinomial logit specification with covariates including penalty distance, angle, and kicker identity, which provides a flexible mapping from location to touchline outcomes.

Stage Two: Possession. Conditional on finding touch, play restarts with a lineout. The attacking team must secure possession for the sequence to continue. I model this as a probabilistic outcome depending on the attacking and defending teams:

$$\Pr(\text{win}) = \text{logit}^{-1}(\mu + \alpha^{\text{att}} + \alpha^{\text{def}}).$$

Here, higher values of the defending effect correspond to greater disruption ability. Although lineout success rates are high on average, variation across teams is important, as losing possession immediately eliminates any attacking value.

Stage Three: Immediate Attack. If the lineout is secured, play typically proceeds through a driving maul. From this state, several outcomes are possible: the team may score a try, lose possession, win a penalty, or continue attacking. I model these outcomes using a multinomial specification that depends on field position and team strength:

$$\Pr(m \mid W) = \frac{\exp(\phi_m^\top W)}{\sum_{m'} \exp(\phi_{m'}^\top W)}.$$

A penalty in this context refers to one won by the attacking team, which re-initiates the original decision between kicking at goal, kicking to the corner, or tapping. This stage captures the sharp increase in try probability close to the try line, as well as the possibility of repeated attacking opportunities through penalties.

Stage Four: Sustained Attack. If the maul does not produce a terminal outcome, the sequence transitions into open play. I model this as a phase-based process in which the attacking team progresses through successive phases until a try, turnover, or penalty occurs. The model assumes that transitions depend on the current state, including field position and phase number, rather than the full sequence history. When a try is scored, the location of the grounding determines the difficulty of the subsequent conversion. Conversion success is modelled separately as a function of distance, angle, and kicker ability, ensuring that wider tries carry lower expected value than centrally located ones. Try location is determined endogenously by the simulated attacking sequence, so conversion difficulty reflects the spatial distribution of attacking outcomes.

Integration and Simulation. The corner model operates as a generative process. Starting from the penalty location, the model draws a touchline outcome, determines whether possession is secured, and then simulates the attacking sequence forward through the maul and subsequent phases. When penalties are awarded during the sequence, the decision problem is re-entered, creating a recursive structure in which teams may generate multiple attacking opportunities from a single initial decision. This produces a distribution over final outcomes, including points scored, time elapsed, and resulting match state.

This distribution is passed to the scoring model described in Section 5.1, which simulates the remainder of the match from each realised state. In this way, the value of kicking to the corner reflects both the immediate attacking opportunity and the downstream evolution of the match.

Full model specifications, estimation details, and validation results for each stage are provided in the appendix.

5.3 Kick at Goal Model

When a team is awarded a penalty, it may attempt a kick at goal for three points or instead kick to the corner to establish an attacking lineout. The outcome of an attempted kick is observed directly, but many kickable penalties are never attempted. This creates a selection problem. Teams are more likely to take the kick when the kick is favourable, so the observed sample of attempts is not representative of all kickable penalties. A model estimated on attempted kicks alone will therefore tend to overstate success probabilities, especially for more difficult kicks near the tactical margin.

I address this using a hierarchical probit model with informative missingness, adapted from Lock2014 to the rugby penalty context. The model jointly accounts for two linked decisions: whether a kick would be successful if taken, and whether the team chooses to attempt it in the first place.

Kick difficulty is determined by the geometry of the penalty location. Let $\ell = (x, y)$ denote the location of the kick, where x measures distance upfield and y measures lateral position across the pitch. From this location, I construct distance and angle measures relative to the posts. In particular, the distance to the opposition try line is

$$d(\ell) = 100 - x,$$

and the angle subtended by the posts is

$$\theta(\ell) = \left| \arctan\left(\frac{y_R - y}{d(\ell)}\right) - \arctan\left(\frac{y_L - y}{d(\ell)}\right) \right|.$$

The model then uses transformed versions of distance and angle, together with kicker-level random effects, to estimate latent kick quality.

The success component is specified as a hierarchical probit model. For kick i, the latent success index is

$$U_i = X_i^\top \beta + Z_i^\top v_{r(i)} + \varepsilon_i,$$

where X_i contains the geometric covariates, $v_{r(i)}$ captures kicker-specific effects, and ε_i is a standard normal disturbance. A kick succeeds when this latent index exceeds zero. This structure allows success probabilities to vary both by kick location and by kicker, while still pooling information across players.

The key extension is that the attempt decision is also modelled. For penalties where the tactical breakeven threshold is defined, the team's decision to go for goal or go to the corner is allowed to depend on both the latent kick quality and the value of the alternative. This is what corrects the informative missingness problem. A declined kick is not simply missing data. It is evidence that the team viewed the kick as insufficiently attractive given both its difficulty and the value of attacking instead.

This matters for the wider decision framework because the quantity of interest is not just the mean success probability of a kick, but whether that probability is high enough to justify taking the points. The kick model therefore provides a posterior distribution for kick success at each location, which can then be compared with the breakeven threshold implied by the value of kicking to the corner.

Full model specification, estimation details, and validation results are reported in Appendix C.1 and Appendix C.2. The appendix shows that accounting for informative missingness improves out-of-sample predictive performance relative to a naive model estimated on attempted kicks only, particularly for difficult kicks where selection effects are strongest.

5.4 Decision Framework

The three models developed above—the scoring model, the kick-to-corner model, and the kick-at-goal model—combine into a unified framework for evaluating penalty decisions. The aim is not simply to predict outcomes, but to determine which action is optimal given the objective a team is pursuing.

I consider four objectives:

$$m \in \{\text{win}, \ \text{pts}, \ \text{prom}, \ \text{surv}\},$$

corresponding to maximising win probability, expected league points, promotion probability, and survival probability. These objectives need not coincide. A decision that maximises the chance of winning may differ from one that maximises expected league points, and both may differ from the choice that best serves season-level outcomes. The framework makes these trade-offs explicit.

5.4.1 Notation

Let s denote the current match state and m index the objective. Define:

- V^m_{KC}: value of kicking to the corner,
- V^m_{SK}: value following a successful kick at goal,
- V^m_{MK}: value following a missed kick at goal.

These values are computed by simulating forward from the corresponding post-decision states.

5.4.2 Value of Kicking to the Corner

The corner option generates a distribution over future states:

$$\{(s_1, \omega_1), \ldots, (s_K, \omega_K)\}, \qquad \sum_{k=1}^{K} \omega_k = 1.$$

The value of the corner is the probability-weighted average of continuation values:

$$V^m_{KC} = \sum_{k=1}^{K} \omega_k \, U^m(s_k).$$

For match-level objectives:

$$U^{\text{win}}(s_k) = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\{\text{win in simulation } n\},$$

$$U^{\text{pts}}(s_k) = \frac{1}{N}\sum_{n=1}^{N} \text{LP}^{(n)}.$$

5.4.3 Value of Kicking at Goal

Kicking at goal produces two outcomes. Let p denote success probability:

$$V^m_{KG}(p) = p \cdot V^m_{SK} + (1-p)\cdot V^m_{MK}.$$

Each state is evaluated by simulating forward to full time.

5.4.4 The Indifference Threshold

The threshold success probability $p^{*,m}$ satisfies:

$$V^m_{KC} = V^m_{KG}(p^{*,m}),$$

giving:

$$p^{*,m} = \frac{V^m_{KC} - V^m_{MK}}{V^m_{SK} - V^m_{MK}}.$$

The decision rule is:

- If $p > p^{*,m}$, kick at goal,
- If $p < p^{*,m}$, kick to the corner.

In implementation:

$$p^{*,m} = \max\left\{0,\ \min\left\{1,\ \frac{V^m_{KC} - V^m_{MK}}{V^m_{SK} - V^m_{MK}}\right\}\right\}.$$

5.4.5 Posterior Decision Rule

The success probability is uncertain. The goal-kicking model provides a posterior $p \mid \mathcal{D}$. I compute:

$$\Pr(p > p^{*,m} \mid \mathcal{D}) \approx \frac{1}{J}\sum_{j=1}^{J} \mathbf{1}\{p^{(j)} > p^{*,m}\}.$$

Decisions are classified as:

- Kick at goal if $\Pr(p > p^{*,m}) \geq 0.70$,
- Kick to corner if $\Pr(p > p^{*,m}) \leq 0.30$,
- Indeterminate otherwise.

5.4.6 Season-Level Objectives

For promotion and survival, continuation values depend on final league position. Each post-decision state is therefore embedded within a simulation of the remaining season. Simulated match outcomes are combined with pre-simulated fixtures to produce a full league table, which is then mapped to promotion and survival probabilities. Further details are provided in Appendix D.

Continuation values are defined as:

$$U^{\text{prom}}(s_k) = \begin{cases} 1 & \text{if first,} \\ \pi_{\text{playoff}}(s_k) & \text{if 2–4,} \\ 0 & \text{otherwise,} \end{cases}$$

$$U^{\mathrm{surv}}(s_k) = \begin{cases} 1 & \text{if 1–8,} \\ \pi_{\mathrm{playoff}}(s_k) & \text{if ninth,} \\ 0 & \text{if tenth.} \end{cases}$$

5.4.7 When Objectives Diverge

Different objectives can imply different decisions. A trailing team may prefer the corner under win probability, while expected league points may favour the posts due to the value of a losing bonus point. The framework does not resolve these trade-offs but quantifies them, allowing decisions to be evaluated under each objective.

5.4.8 Outputs

For each penalty, the framework produces:

1. $V^m_{KC}, V^m_{SK}, V^m_{MK}$,
2. $p^{*,m}$,
3. the posterior $p \mid \mathcal{D}$,
4. $\Pr(p > p^{*,m} \mid \mathcal{D})$,
5. a recommended action.

6 Results

6.1 The Decision Threshold in Context

The framework developed in Section 3 produces, for any penalty situation, a threshold success probability p^* above which kicking at goal is optimal and below which kicking to the corner dominates. This threshold depends on the current match state, including score margin, time remaining, and, under the expected league points objective, the attacking team's current try count. Before examining observed decision-making, it is first necessary to understand how these thresholds vary across game states and when the win probability and expected league points objectives diverge.

To characterise threshold behaviour systematically, I simulate a large set of penalty scenarios. In each case, attacking and defending teams are sampled from the estimated team-season parameter set, and scenarios are generated across a balanced grid of score margins, game times, try counts, and penalty locations. For every simulated state, the continuation values from kicking at goal and kicking to the corner are computed under alternative objective functions, and the implied decision threshold is recorded. This design provides broad coverage of the late-game state space while preserving realistic heterogeneity in team strength and kick difficulty.

6.1.1 Decision Thresholds Under Win Probability

I begin with the win probability objective, which provides the natural benchmark. Figure 3 shows the average threshold by score margin across late-game minutes. Lower thresholds indicate situations in which a relatively modest kick success probability is sufficient to justify a shot at goal, while higher thresholds indicate situations in which the corner option is more attractive.

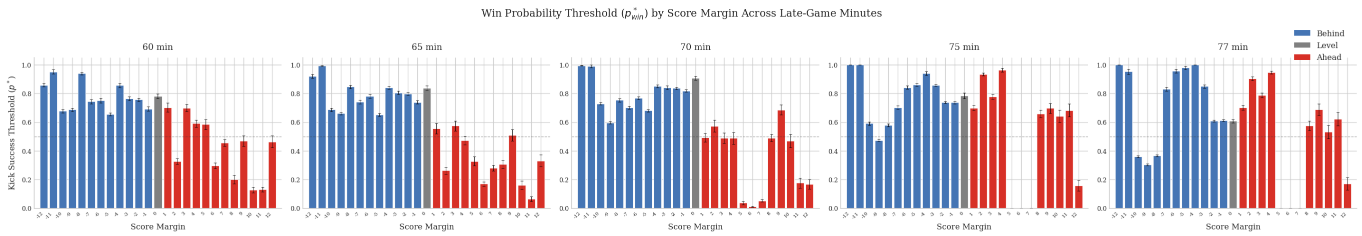


Figure 3: Win probability threshold, p^*_{win}, by score margin across late-game minutes. Blue bars denote trailing states, grey bars level states, and red bars leading states.

Several patterns emerge. Thresholds tend to be lower when teams are ahead, since three immediate points help consolidate an already favourable match state. By contrast, when trailing, especially by more than a penalty, the corner becomes relatively more attractive because a try is more likely to alter the final result. As the game moves toward its conclusion, thresholds also tend to compress, reflecting the greater marginal value of the next scoring event.

6.1.2 Decision Thresholds Under Expected League Points

The picture changes once the objective is expected league points rather than win probability. Figure 4 shows the expected league points threshold by score margin, try count, and game time.

This figure makes clear that try count is essential context under the league-points objective. With two tries already scored, the threshold pattern still resembles the win probability case, although with some local distortions. With four tries already scored, the attacking bonus point has been secured, so the threshold again becomes closer to the win probability benchmark. The key state is three tries. In that case, the team is exactly one try away from the four-try bonus point, and this produces a sharp change in the value of the corner option.

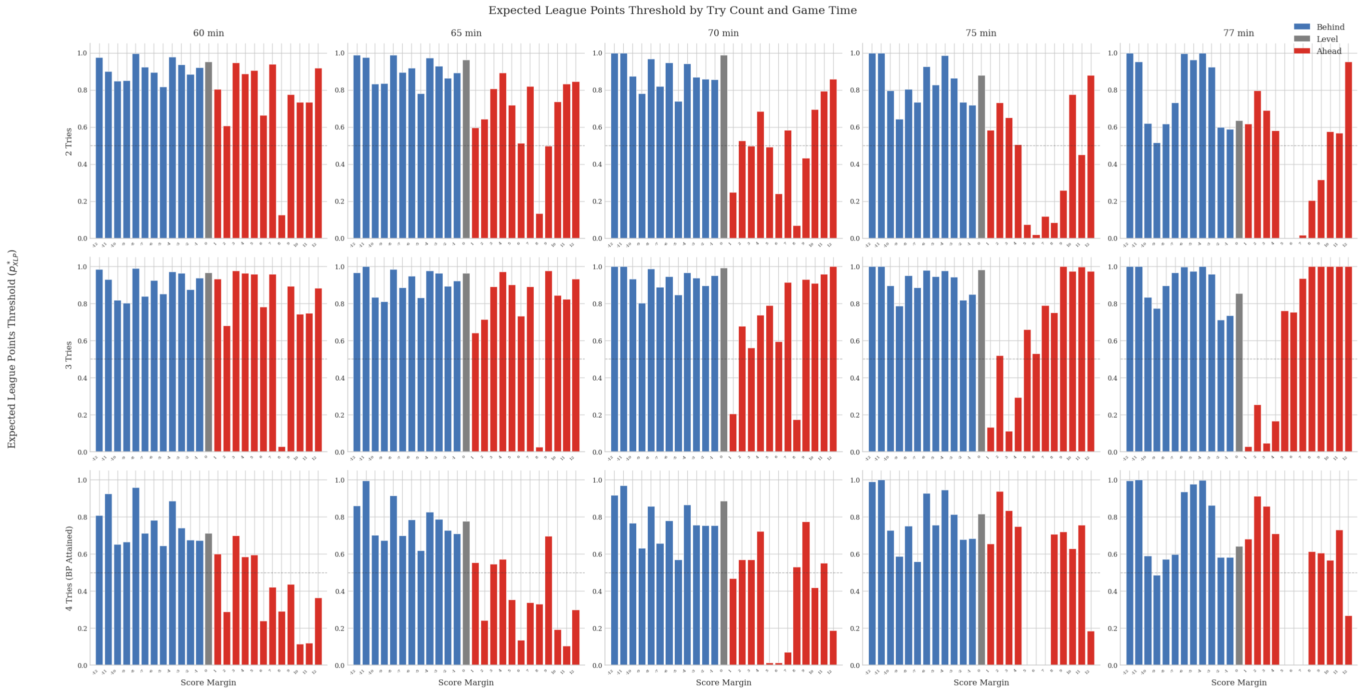


Figure 4: Expected league points threshold, p^*_{XLP}, by score margin, try count, and game time. Rows correspond to attacking teams on 2, 3, and 4 tries respectively.

6.1.3 The Role of Try Count

The central economic mechanism in the threshold results is therefore the nonlinear value of the fourth try. When the attacking team has three tries, the corner option offers not only the possibility of improving the match result, but also the opportunity to secure an additional league point. This creates a wedge between the two objectives that is absent when the attacking bonus has already been secured and much weaker when it remains two tries away.

This is visible most clearly in the middle row of Figure 4. In many late-game leading states, the threshold under expected league points approaches one, and in some cases effectively exceeds it before clipping. In practical terms, this means that even an excellent goal-kicking opportunity may still be inferior to the corner because the marginal value of the bonus-point try is so high.

6.1.4 Comparing Win Probability and Expected League Points

To quantify the extent to which the two objectives differ, Figure 5 plots the divergence between thresholds, defined as $p^*_{XLP} - p^*_{win}$, by score margin, try count, and time.

The divergence is concentrated overwhelmingly in the three-try state. When teams already have four tries, the two objectives are largely aligned, which is reassuring and provides an internal validity check on the framework. When teams are on two tries, disagreement exists but is generally modest. It is when teams sit on three tries, especially late in the game and with a positive score margin, that expected league points departs sharply from win probability.

6.1.5 Optimal Actions Under Expected League Points

Thresholds are useful because they summarise the decision rule, but it is also helpful to visualise the implied action regions directly. Figure 6 shows the distribution of action recommendations under the expected league points objective across score margins, try counts, and game times.

The structure of the figure mirrors the threshold results. The dominant feature is the substantial expansion of corner-dominant states when the attacking team is on three tries. In late-game leading

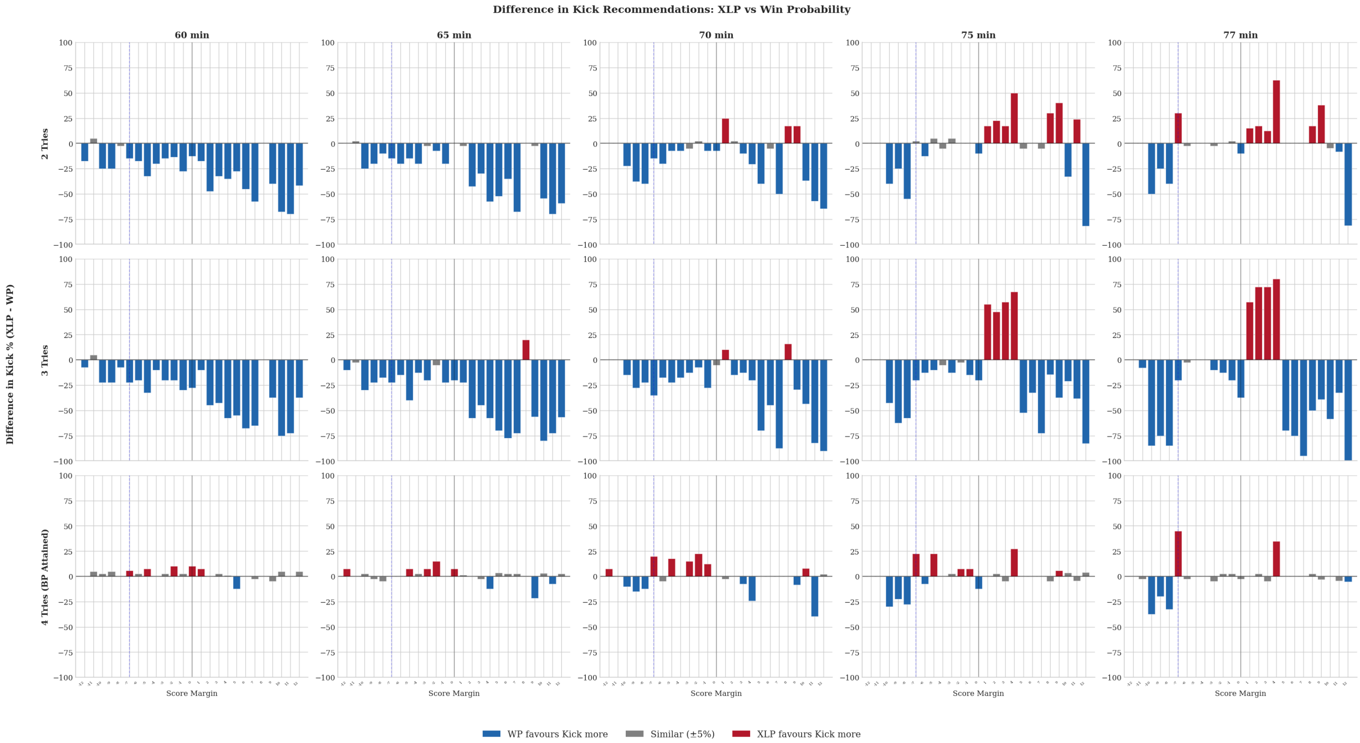


Figure 5: Difference between expected league points and win probability thresholds, $p^*_{XLP} - p^*_{win}$, by score margin, try count, and game time. Positive values indicate states in which expected league points requires a higher kick success threshold than win probability.

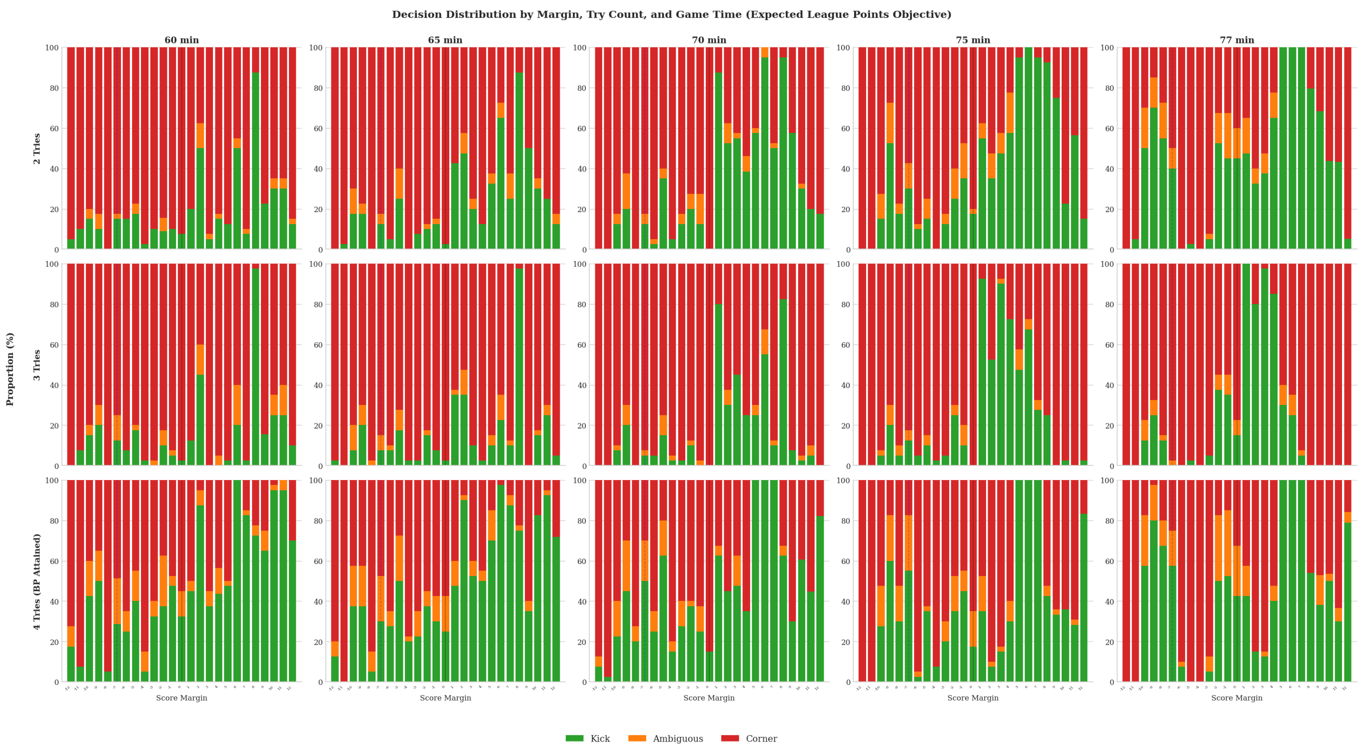


Figure 6: Decision distribution under the expected league points objective by score margin, try count, and game time. Green indicates kick at goal, amber indicates ambiguous cases, and red indicates kick to the corner.

situations, these states are often overwhelmingly corner-favoured under expected league points, even where the win probability objective would still recommend a shot at goal.

For comparison, Figure 7 shows the equivalent one-row decision map under the win probability objective, aggregated across try counts. This objective produces a much smoother and more intuitive partition of the state space, with less abrupt movement toward the corner in bonus-point-relevant states.

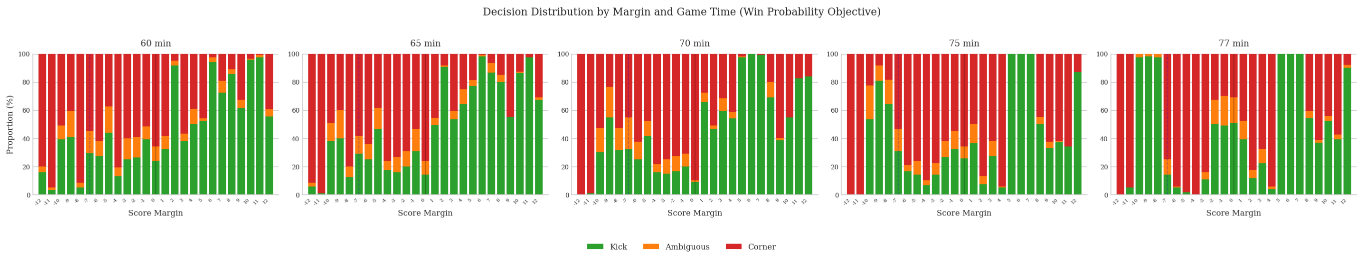


Figure 7: Decision distribution under the win probability objective by score margin and game time, aggregated across try counts. Green indicates kick at goal, amber indicates ambiguous cases, and red indicates kick to the corner.

The driving forces behind these action regions are explored further in the appendix. In particular, field position, kicker quality, and team strength differences all shape whether a given state is goal-dominant, corner-dominant, or genuinely marginal. These factors matter under both objectives, although their interaction with try count is especially important under expected league points.

6.1.6 Where the Objectives Disagree

Figure 8 turns from thresholds to recommendations and reports the rate at which the two objectives prescribe different actions.

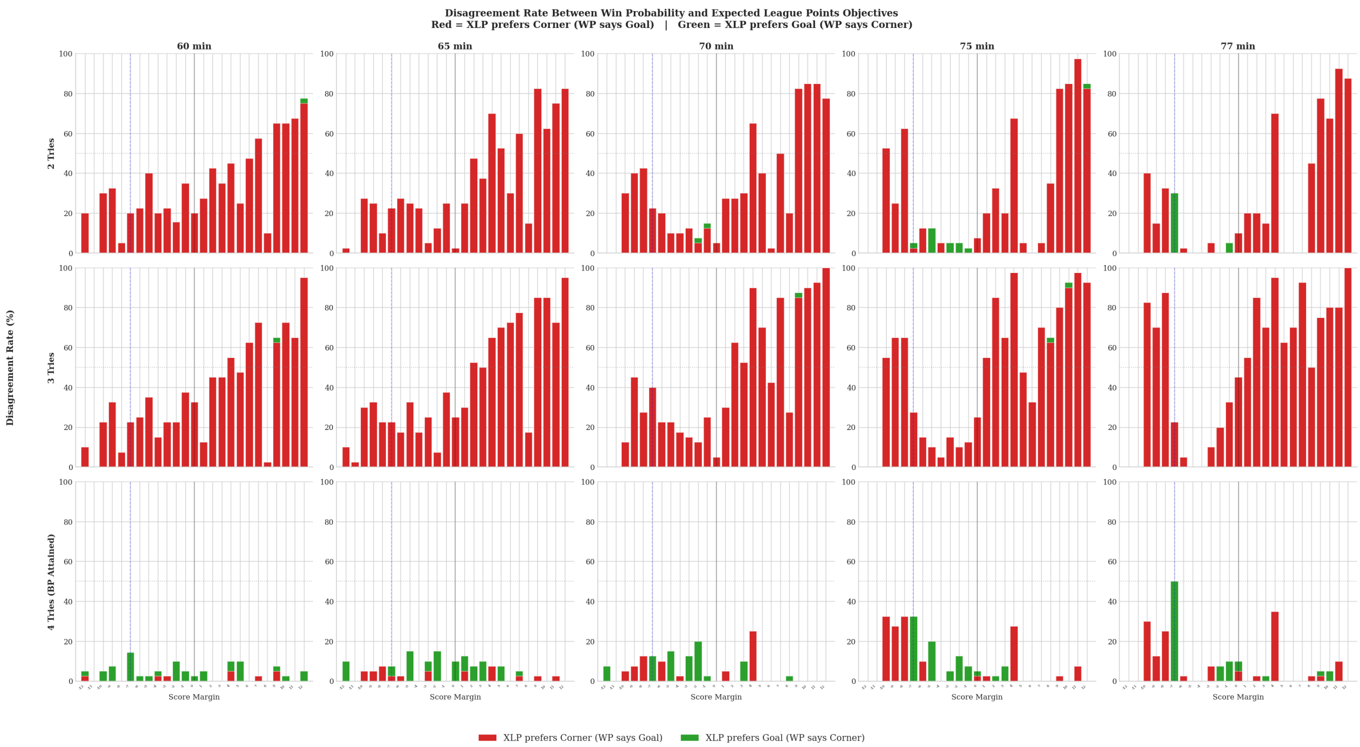


Figure 8: Disagreement rate between the win probability and expected league points objectives by score margin, try count, and game time.

Disagreement is again concentrated in the three-try state. In many leading late-game scenarios, disagreement rates are extremely high, while disagreement is minimal once the attacking bonus point has already been secured. This reinforces the interpretation that bonus-point incentives, rather than general model noise, drive the wedge between objectives.

The divergence arises because the two objectives value outcomes differently. Kicking at goal increases the margin and reduces the probability of being caught, making it optimal under a win-probability objective. By contrast, kicking to the corner increases the probability of scoring a try and securing the attacking bonus point. Although this slightly reduces win probability, the expected gain in league points outweighs the loss, so the expected league points objective frequently favours the corner.

6.1.7 The Cost of Following Win Probability Instead of Expected League Points

A disagreement between objectives matters only if it is economically meaningful. Figure 9 therefore reports the expected league points loss from following the win probability recommendation rather than the expected league points recommendation.

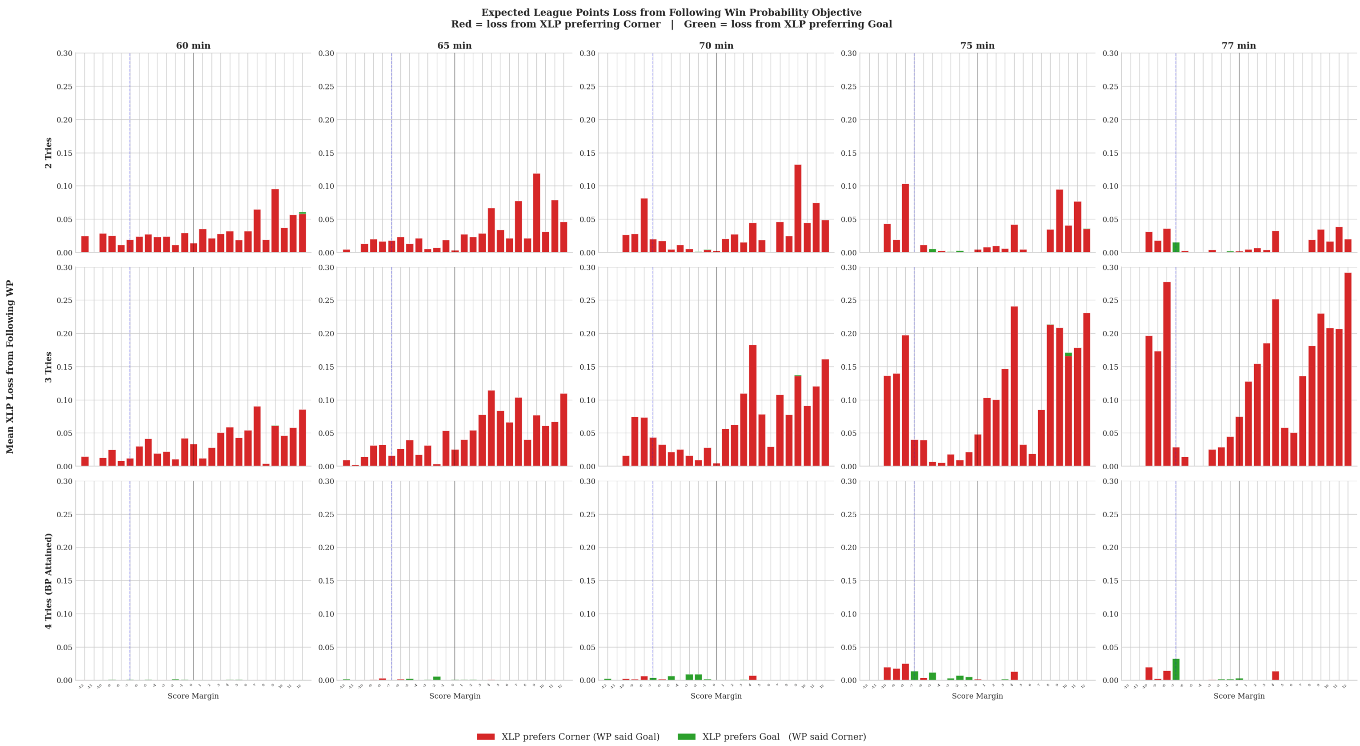


Figure 9: Expected league points loss from following the win probability objective rather than the expected league points objective, by score margin, try count, and game time.

particularly late in the game and at positive score margins. In these scenarios, the win probability objective undervalues the corner because it does not internalise the value of securing the fourth try bonus. The resulting losses are economically meaningful, indicating that objective choice can materially affect decision quality.

The time profile of these losses differs sharply by try count. With two tries, losses decline as time progresses: even if a try is scored from the corner, there is insufficient time remaining to reliably secure the bonus point, reducing its expected value. By contrast, in the three-try state, losses increase late in the game. Earlier in the match, taking the goal still leaves sufficient time to pursue the fourth try, and score margins are less secure. As time runs down, however, the opportunity to recover the bonus point after taking the goal diminishes, while existing leads become safer. This combination increases the relative value of immediately pursuing the fourth try via the corner, and explains the large late-game losses from following the win probability objective.

6.1.8 Summary

Three main findings emerge from this analysis. First, penalty decision thresholds are strongly state-dependent, varying with score margin and time even under the baseline win probability objective. Second, once expected league points is considered, try count becomes essential context, with the three-try state generating the sharpest nonlinearity. Third, these differences are not cosmetic. They translate into materially different action recommendations and meaningful expected league points losses when the wrong objective is followed.

Having established how the decision rule changes across objectives and match states, the next section turns to observed behaviour and asks whether teams act in line with these state-contingent prescriptions.

6.2 Observed Decision Quality

The preceding sections established optimal decision thresholds for the kick-at-goal versus kick-to-corner choice. This section evaluates how teams in the Energia All-Ireland League perform against these benchmarks. The sample comprises 560 second-half penalties occurring within the final twenty minutes of matches, within 55 metres of the opposition try line, and with a score margin of at most twelve points. Decisions are classified as optimal where the model's recommendation exceeded 70% confidence in either direction; the intermediate zone (30–70%) is treated as ambiguous.

6.2.1 League-Wide Decision Accuracy

Across the full sample, teams selected the expected-league-points-optimal decision in 70.5% of cases. The remaining 29.5% of decisions were suboptimal, divided almost evenly between over-kicking (14.1%, 79 cases) and under-kicking (15.4%, 86 cases). Over-kicking refers to situations where a team kicked at goal when the corner option would have maximised expected league points; under-kicking describes the converse.

This near-symmetric split is noteworthy. In other sports, systematic biases typically dominate. The conservative fourth-down tendencies documented in American football, for instance, reflect a persistent directional error romer2006. The AIL data suggest no such aggregate bias. However, this balance at the league level conceals important structure: errors are not randomly distributed across game states, and teams differ markedly in which direction they err.

Figure 10 displays decision accuracy by score margin. The pattern is non-monotonic, with accuracy collapsing at specific margins rather than deteriorating smoothly.

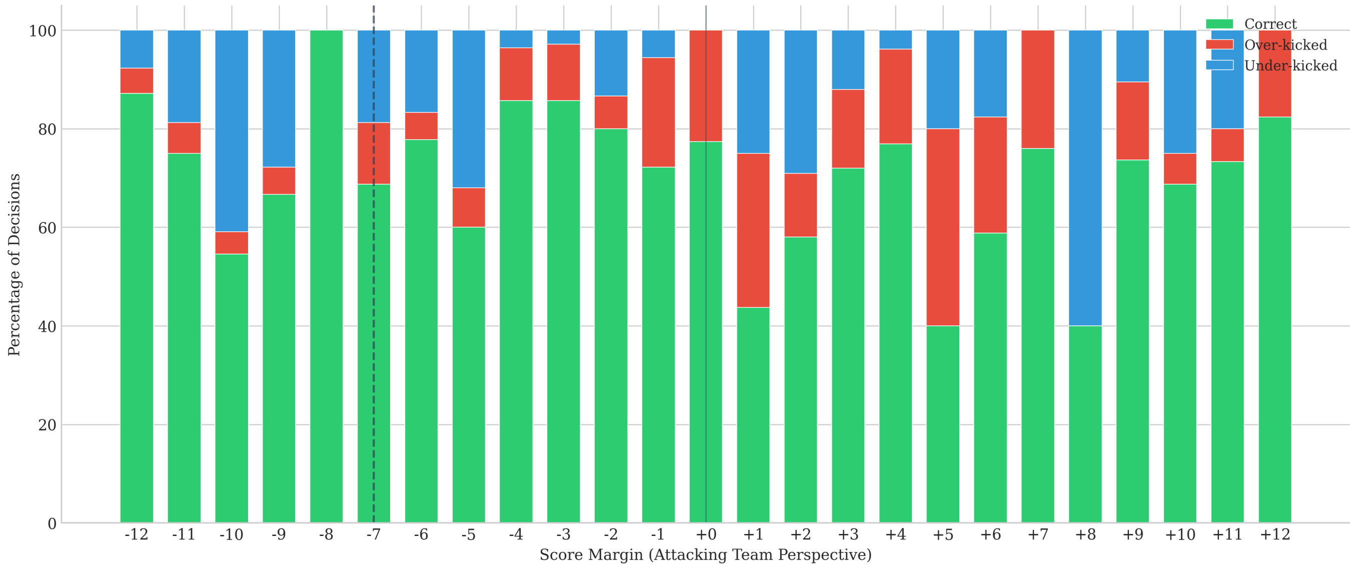


Figure 10: Decision quality by score margin. Green segments indicate correct decisions; red indicates over-kicking errors; blue indicates under-kicking errors. Dashed vertical lines mark the losing bonus point boundary (-7) and parity.

The worst accuracy occurs at margins of +5 and +8, where only 40% of decisions were optimal. At +5, over-kicking dominates: teams frequently kicked to extend to an eight-point lead when corner attempts offered higher expected value. At +8, the pattern reverses, with under-kicking predominant as teams pursued tries when consolidating through kicks would have been preferable. The +1 margin produced just 43.8% accuracy with a mixed error profile, reflecting genuine strategic ambiguity at narrow leads.

These are not arbitrary margins. They correspond precisely to the points where bonus point incentives create discontinuities in the payoff structure. At +5, a successful kick yields +8, which crosses no meaningful threshold; a try yields +10 or +12, potentially securing both the win and denying the opponent a losing bonus. At +8, the team has already pushed the opponent beyond losing bonus range, reducing the marginal value of further scoring. The pattern suggests that decision-making deteriorates exactly where incentive structures become nonlinear and competing objectives must be weighed.

Conversely, certain margins yielded high accuracy. At −8, precisely the losing bonus point boundary, teams achieved 100% accuracy (albeit with $n = 11$), correctly recognising the strategic significance of this threshold. Margins of −3 and −4 each produced 85.7% accuracy, as the value of try-scoring at these deficits is relatively unambiguous.

The error structure reveals a systematic pattern: teams tend to over-kick when level or ahead, and under-kick when behind. When trailing, the prospect of a try appears more attractive than its expected value warrants; when leading, the certainty of kicked points appears more attractive than the model suggests. This is consistent with reference-dependent preferences, where the current score serves as an anchor that distorts valuation of probabilistic outcomes.

6.2.2 Team-Level Variation

Decision quality varies substantially across team-seasons. Table 2 presents the complete rankings, revealing a spread of 31.1 percentage points between the most and least accurate decision-makers.

Table 2: Team-season decision quality metrics, sorted by accuracy. Total Loss represents cumulative expected league points forfeited through suboptimal decisions; Mean Loss normalises by number of decisions.

Team	Season	n	Correct	Over	Under	Kick%	Total Loss	Mean Loss
City of Armagh	2025/26	29	89.7%	6.9%	3.4%	20.7%	4.54	0.157
Garryowen	2025/26	14	85.7%	7.1%	7.1%	7.1%	2.11	0.151
Old Wesley	2024/25	41	78.0%	17.1%	4.9%	31.7%	3.71	0.090
UCC	2025/26	22	77.3%	18.2%	4.5%	36.4%	2.85	0.129
QUB	2025/26	13	76.9%	7.7%	15.4%	15.4%	1.78	0.137
Blackrock College	2024/25	34	76.5%	11.8%	11.8%	23.5%	4.11	0.121
Highfield	2025/26	33	75.8%	9.1%	15.2%	21.2%	4.35	0.132
Highfield	2024/25	26	73.1%	19.2%	7.7%	26.9%	6.01	0.231
Instonians	2025/26	26	73.1%	23.1%	3.8%	34.6%	3.07	0.118
Old Wesley	2025/26	39	71.8%	7.7%	20.5%	20.5%	5.29	0.136
Blackrock College	2025/26	20	70.0%	10.0%	20.0%	15.0%	1.79	0.089
Nenagh Ormond	2024/25	50	68.0%	16.0%	16.0%	26.0%	5.29	0.106
UCC	2024/25	37	67.6%	5.4%	27.0%	8.1%	3.85	0.104
Naas	2025/26	15	66.7%	6.7%	26.7%	20.0%	1.56	0.104
Old Belvedere	2024/25	34	64.7%	26.5%	8.8%	38.2%	5.66	0.166
Dublin University	2025/26	14	64.3%	14.3%	21.4%	21.4%	1.14	0.082
Shannon	2024/25	43	62.8%	30.2%	7.0%	32.6%	6.98	0.162
Dublin University	2024/25	34	58.8%	5.9%	35.3%	17.6%	4.02	0.118
Naas	2024/25	29	58.6%	3.4%	37.9%	24.1%	3.22	0.111
League Total	–	**560**	**70.5%**	**14.1%**	**15.4%**	**24.6%**	**72.01**	**0.129**

City of Armagh in 2025/26 achieved the highest accuracy at 89.7%, making suboptimal decisions in only three of twenty-nine penalties. At the other extreme, Naas in 2024/25 recorded 58.6% accuracy, with over 41% of their decisions failing to maximise expected league points. The standard deviation of 10.3 percentage points indicates that this variation reflects genuine differences in decision-making quality rather than sampling noise.

6.2.3 Error Profiles

Teams can be classified by their dominant error tendency. An over-kicker exhibits an over-kick rate exceeding 1.5 times their under-kick rate; an under-kicker shows the reverse pattern; balanced teams display neither systematic tendency. Figure 11 visualises this classification.

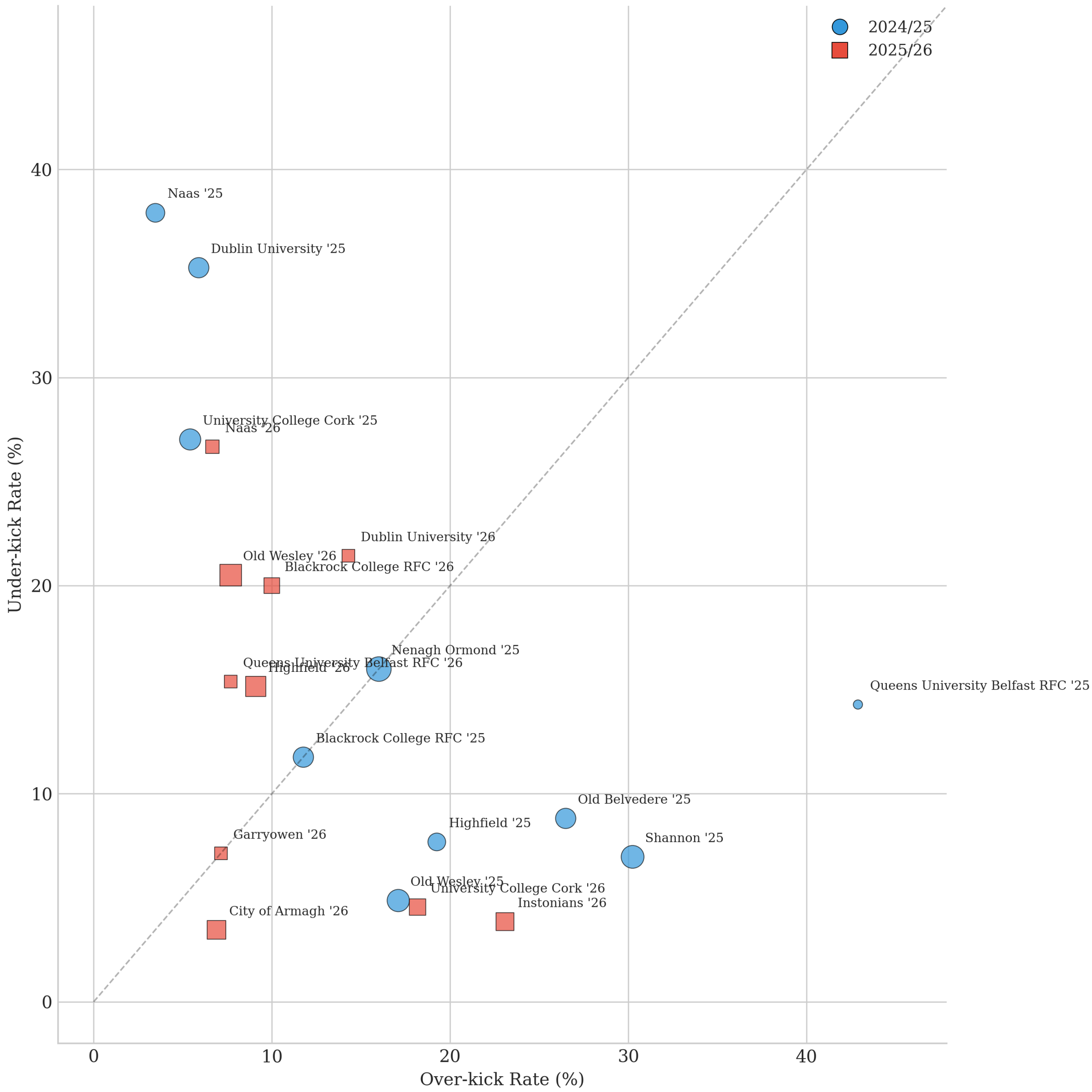


Figure 11: Team-season error profiles. Each point represents one team-season, positioned by over-kick and under-kick rates. Point size reflects sample size; circular markers denote 2024/25, squares denote 2025/26. The diagonal line indicates equal error rates.

Eight team-seasons were classified as over-kickers, eight as under-kickers, and four as balanced. The over-kicker profile includes both high-accuracy teams (City of Armagh at 89.7%) and low-accuracy teams (Shannon at 62.8%), demonstrating that error direction is distinct from error frequency. A team can consistently err in one direction while still achieving reasonable aggregate accuracy if individual errors are small in magnitude.

Shannon's 2024/25 season illustrates the over-kicker profile in its most pronounced form: 30.2% of decisions were over-kicks against just 7.0% under-kicks, yielding an over-kick ratio exceeding four to one. Their 32.6% kick rate, the second-highest in the league, suggests a systematic preference for taking points that extends into situations where corner attempts would be preferable. This is consistent with a heuristic of the form "take the three when ahead" being applied without sufficient sensitivity to the specific game state.

Conversely, Naas in 2024/25 exemplifies the under-kicker profile, with 37.9% under-kicks against only 3.4% over-kicks. Their tendency to pursue tries in situations where kicked points would maximise expected value suggests the opposite heuristic: a preference for attacking rugby that, while perhaps culturally valued, leaves expected league points unclaimed.

The correlation between kick rate and decision accuracy is weak (Pearson $r = 0.08$), confirming that overall strategic orientation does not determine decision quality. Teams with high kick rates and teams with low kick rates appear throughout the accuracy distribution. What distinguishes accurate from inaccurate teams is not how often they kick, but when they choose to do so.

6.2.4 Season Comparison

League-wide decision accuracy improved from 2024/25 to 2025/26. Mean team accuracy rose from 65.1% to 75.1%, a ten percentage point improvement. Total expected league points lost to suboptimal decisions fell from 43.54 in 2024/25 to 28.46 in 2025/26, despite the latter season containing only two-thirds as many penalties in the sample.

Figure 12 presents the season-by-season team rankings.

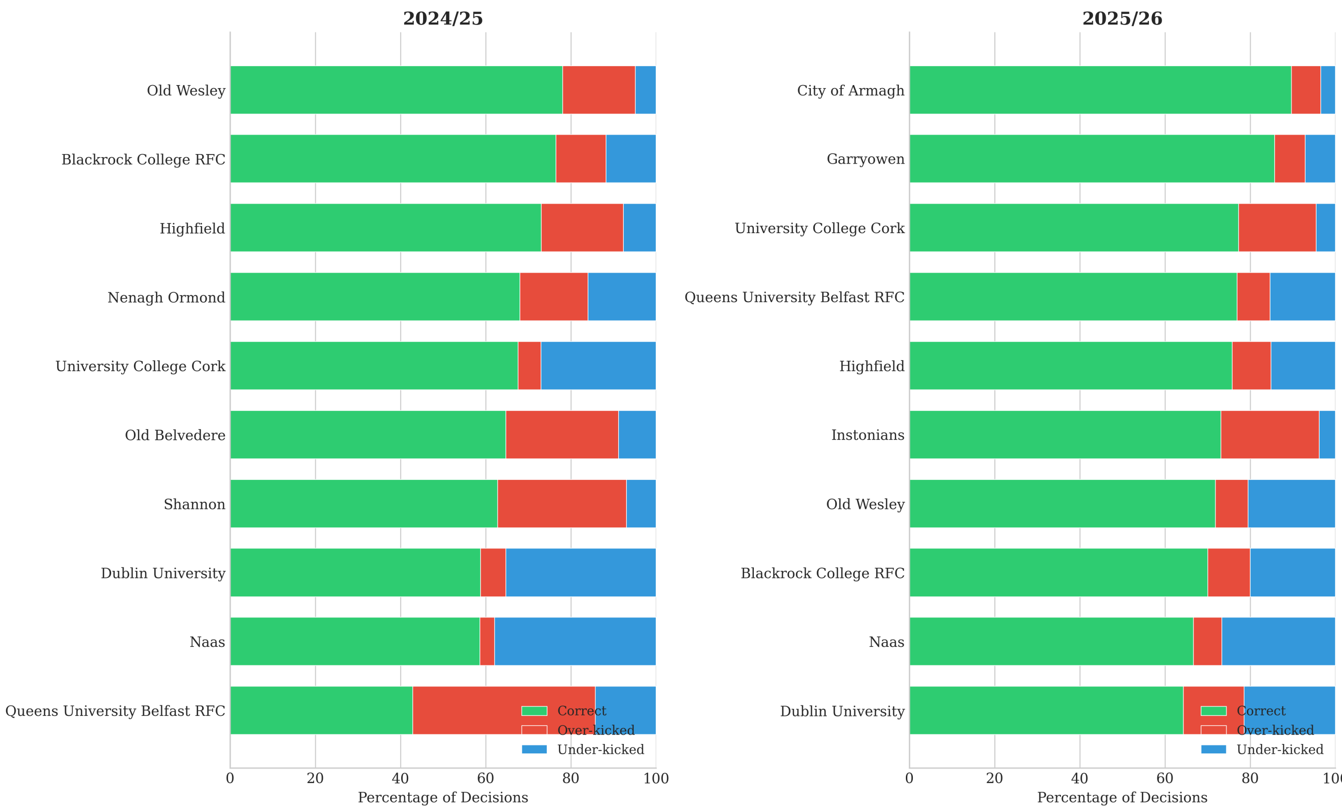


Figure 12: Team decision quality by season. Horizontal stacked bars show the proportion of correct decisions (green), over-kicks (red), and under-kicks (blue) for each team-season.

Seven teams appear in both seasons with sufficient sample sizes for comparison. Of these, five improved their accuracy year-on-year: University College Cork (+9.7 percentage points), Naas (+8.0pp), Dublin University (+5.5pp), Highfield (+2.7pp), and Queens University Belfast (+34.1pp, though their 2024/25 sample of seven penalties limits the reliability of this estimate). Two teams declined: Blackrock College (−6.5pp) and Old Wesley (−6.2pp).

These changes demonstrate that decision quality is not a fixed team characteristic. Highfield's profile shifted from over-kicker in 2024/25 to under-kicker in 2025/26, indicating that even the direction of systematic error can change between seasons. Whether through deliberate strategic adjustment, personnel changes, or other factors, teams can and do alter their decision-making patterns.

6.2.5 Expected League Points Cost

The expected league points framework permits direct quantification of the cost of suboptimal decisions. Across all 560 penalties, aggregate expected league points loss totalled 72.01 points, with a mean loss of 0.129 per decision.

Figure 13 displays the distribution of total expected league points loss by team-season.

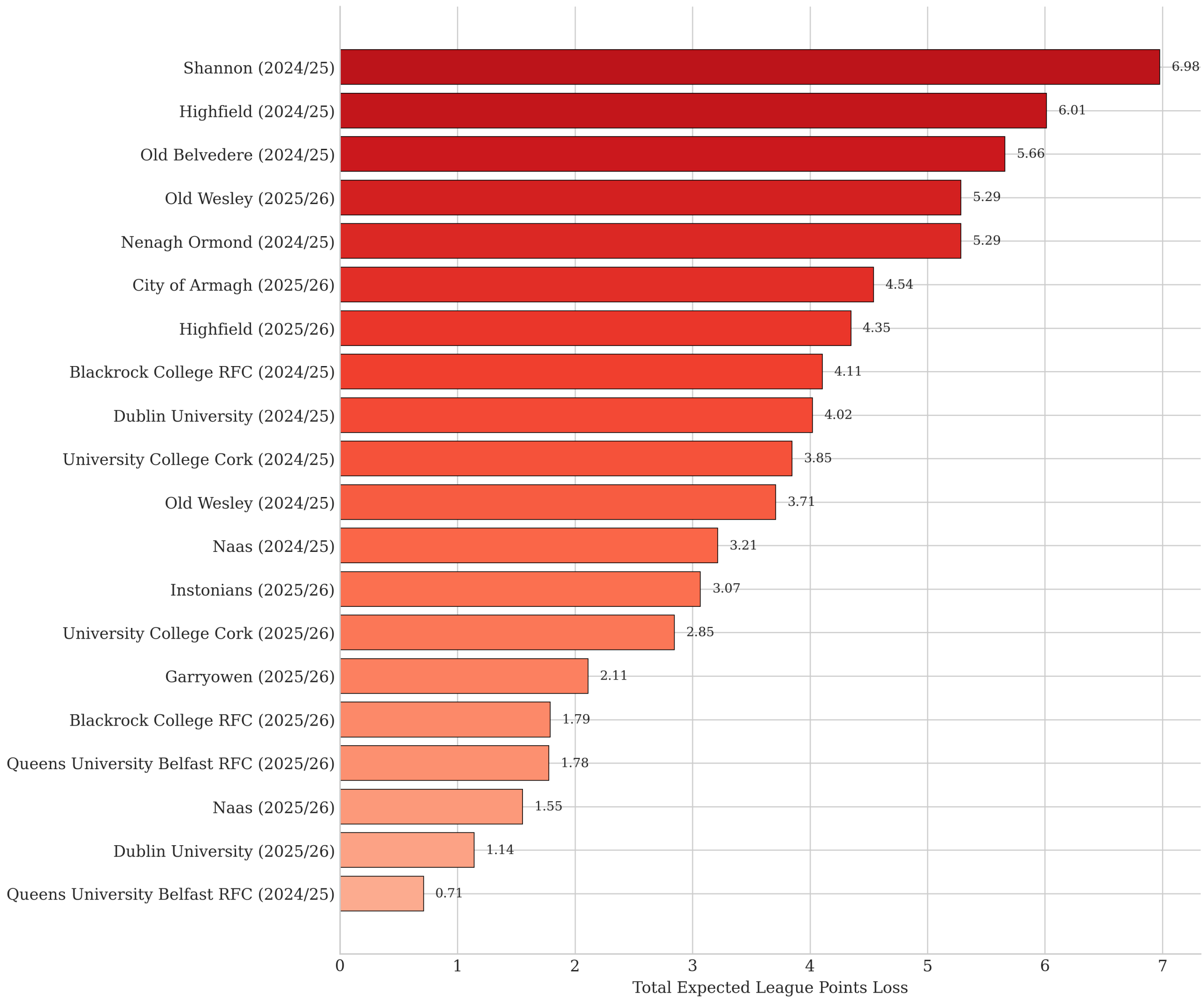


Figure 13: Total expected league points loss by team-season, sorted by magnitude.

Shannon's 2024/25 season incurred the highest total loss at 6.98 expected league points, followed by Highfield 2024/25 (6.01) and Old Belvedere 2024/25 (5.66). These figures represent meaningful quantities in a league where final standings are often determined by narrow margins. To contextualise: the 6.98 expected points lost by Shannon through penalty decision errors alone exceeds the value of a win-with-bonus (5 points) and approaches the value of two standard wins (8 points). In recent AIL seasons, the gap between eighth place (the final playoff position) and ninth has frequently been four points or fewer.

Mean loss per decision provides a sample-size-adjusted measure. Highfield's 2024/25 season recorded the highest mean loss at 0.231, nearly double the league average. This indicates that Highfield's errors were not merely frequent but concentrated in high-stakes situations where the gap between optimal and suboptimal choices was large. A team making many small errors will accumulate less total loss than a team making fewer but larger errors; Highfield's profile suggests the latter pattern.

Conversely, Dublin University's 2025/26 season achieved the lowest mean loss (0.082), suggesting that even when they erred, decisions occurred in contexts where the cost difference between alternatives was modest.

6.2.6 Scenario-Specific Performance

Decision accuracy varies substantially by game scenario. Figure 14 presents team-season accuracy across four strategically distinct situations: the losing bonus point range (margin −7 to −1), chasing the try bonus (exactly three tries scored), holding a lead (+1 to +7), and trailing.

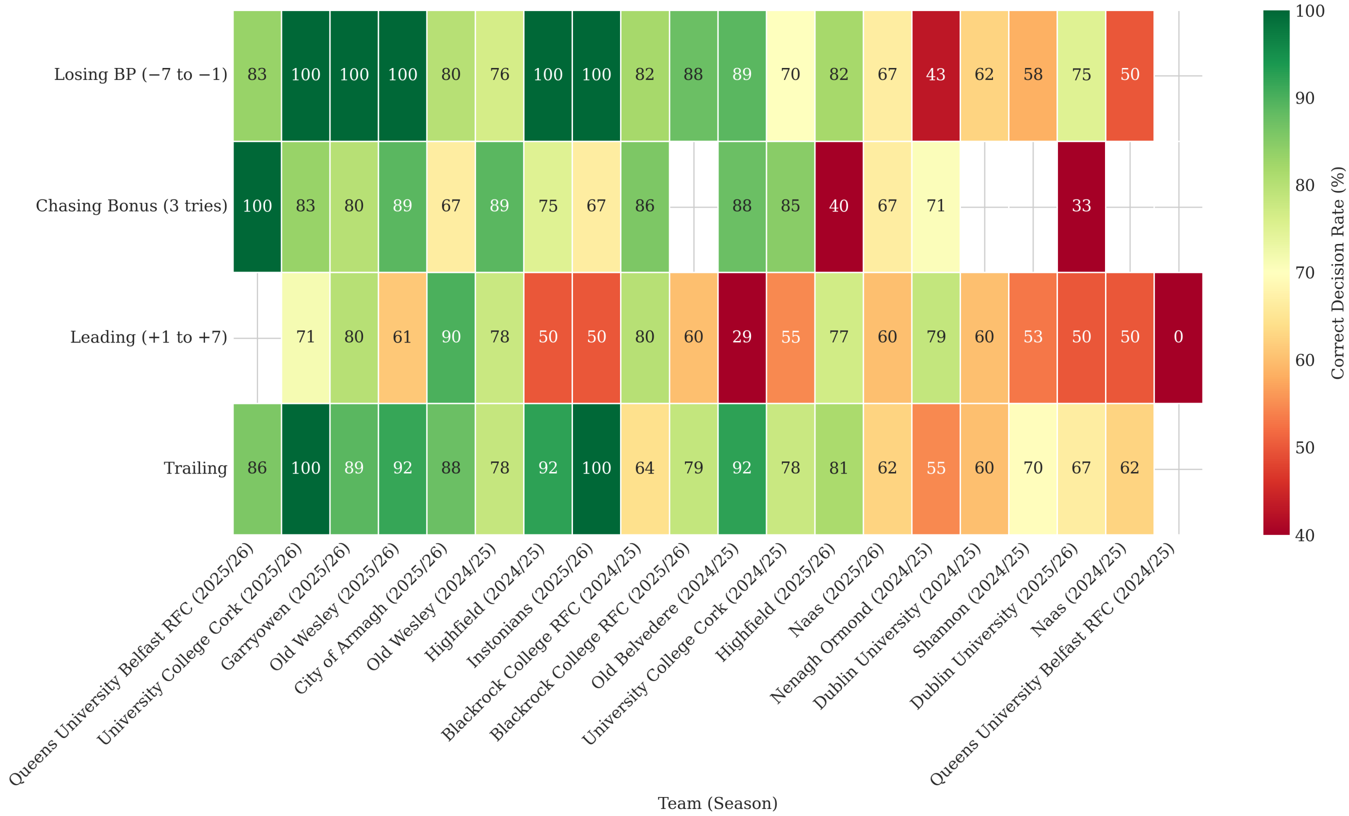


Figure 14: Decision accuracy by team-season and game scenario. Cells are coloured by accuracy (green: high; red: low); blank cells indicate insufficient data ($n < 3$).

The losing bonus point range presents particular challenges. This scenario requires teams to weigh competing objectives: chasing the win against securing the losing bonus point. A kick might guarantee one point (if it keeps the margin within seven) but forgo the chance at five; a corner attempt might yield the win but risk falling outside bonus range entirely. League-wide accuracy in this scenario is 76.0%, above the overall average, yet with substantial team-level variation. Some teams navigate this complexity well; others struggle to correctly price the trade-off between guaranteed and probabilistic outcomes.

When chasing the try bonus with exactly three tries scored, accuracy drops for several teams. The bonus point introduces additional value to try-scoring that the win probability framework would not capture, creating a tension between the two objectives. Dublin University's 2025/26 season achieved only 33%

accuracy in this scenario (one of three correct), while Shannon's 2024/25 recorded no penalties with three tries, suggesting a try-scoring profile that either quickly surpasses three or rarely reaches it.

The heatmap reveals that decision-making difficulty is not uniform across scenarios. Teams that perform well in straightforward situations may struggle when incentives conflict. This is precisely what one would expect from boundedly rational agents who rely on simplified decision rules: such rules work adequately in simple states but fail when the payoff structure becomes nonlinear.

6.2.7 Summary

The observed decision quality analysis reveals both competence and systematic inefficiency. On the positive side, teams make optimal decisions approximately 70% of the time, and aggregate accuracy improved meaningfully between seasons. The near-symmetric split between over-kicking and under-kicking suggests the absence of a strong league-wide directional bias.

However, the 31 percentage point spread between best and worst team-seasons indicates substantial room for improvement. The 72.01 expected league points lost to suboptimal decisions represents a meaningful competitive inefficiency. For teams at the bottom of the decision quality rankings, improving penalty decision-making offers a plausible route to better league outcomes without requiring changes to playing personnel or other resource-intensive interventions.

Three findings carry particular significance. First, errors are systematic rather than random: teams over-kick when level or ahead, and under-kick when behind, consistent with reference-dependent preferences anchored to the current score. Second, errors concentrate at margins where bonus point incentives create nonlinear payoffs, suggesting that decision-making deteriorates precisely when the strategic environment becomes most complex. Third, what distinguishes accurate from inaccurate teams is not their overall kick rate but their sensitivity to game state. The framework developed in this thesis provides precisely such state-dependent guidance, translating complex calculations into actionable thresholds.

These results also highlight a limitation of the expected league points objective. While maximising expected league points is a reasonable proxy for competitive success, the true objective in a league is rank-based: teams seek promotion, playoff qualification, or relegation avoidance, not points per se. A team eight points clear at the top of the table faces different incentives than one in a tight battle for eighth place, even in identical game states. The next section extends the analysis to account for these nonlinear, rank-dependent objectives.

6.3 Promotion and Survival Stakes

Penalty decisions in rugby are typically evaluated in terms of expected league points. This provides a natural and intuitive benchmark, but it does not fully capture the objective that teams ultimately face. Promotion and relegation depend on final league position rather than points in isolation, so the value of any individual decision is inherently shaped by the wider table context. In particular, the marginal value of points is nonlinear and state-dependent: securing three points may be disproportionately valuable when protecting a narrow lead in a promotion race, while in other situations the upside of a try may dominate.

These differences become most relevant when teams remain in contention late in the season. As league positions begin to stabilise, small changes in match outcomes can have large effects on promotion or survival probabilities. In such settings, maximising expected league points need not coincide with maximising the true season objective. This section quantifies how often these objectives diverge, how teams behave when they do, and the consequences of those choices.

6.3.1 Disagreement Between Objectives

Figure 15 shows the frequency with which the action that maximises expected league points differs from the action that maximises the relevant season-level objective. Disagreement is present throughout the season for both promotion and relegation contenders, and remains material even in earlier rounds. However, it becomes particularly salient in later stages, where league positions are more tightly clustered and the mapping from match points to final outcomes is more sensitive.

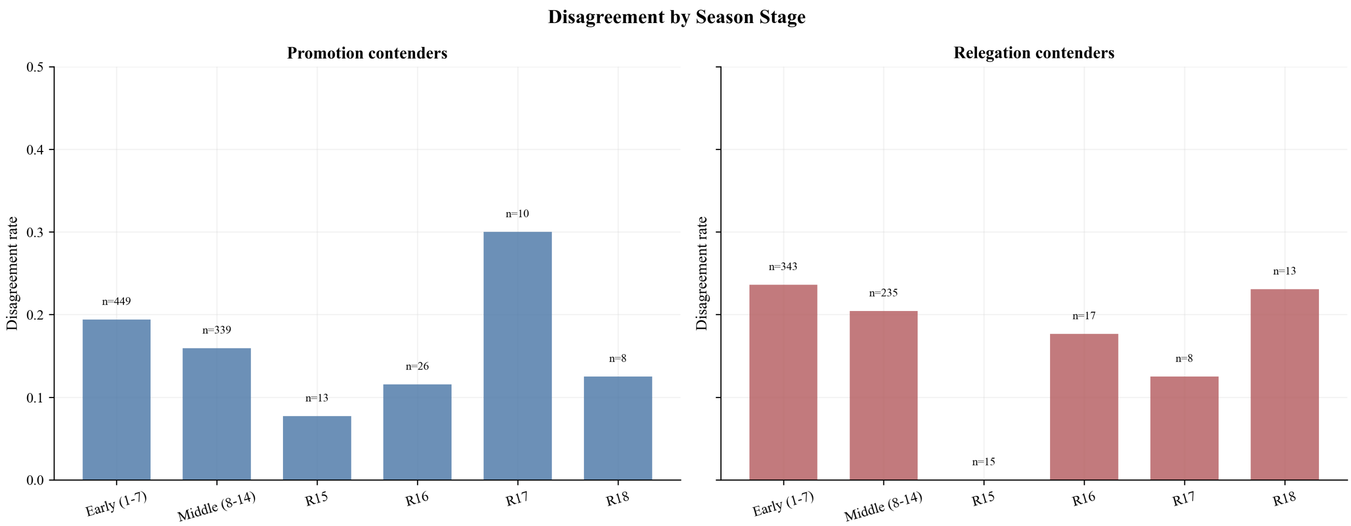


Figure 15: Disagreement between expected league points and season-level objectives by season stage.

This establishes that expected league points is not simply a scaled proxy for season success. Even in realistic match states faced by contender teams, the two objectives can imply different optimal actions.

6.3.2 Directional Disagreement

When disagreement arises, it is not symmetric. Figure 16 shows that expected league points typically favours the more aggressive option, recommending a kick to the corner, while the season-level objective more often favours taking the three points.

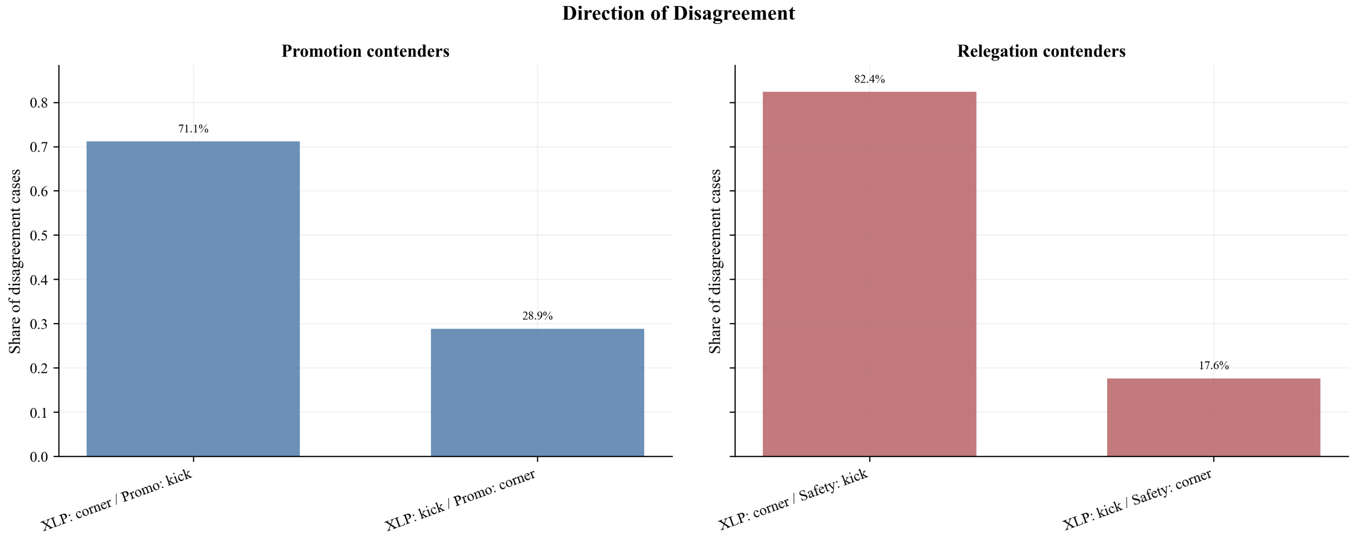


Figure 16: Direction of disagreement between expected league points and season-level objectives.

This pattern is consistent with the underlying economics of the problem. Expected league points values upside directly, whereas season objectives incorporate the risk of adverse shifts in league position. In high-stakes contexts, particularly for relegation contenders, avoiding downside can dominate pursuing additional upside, leading to more conservative optimal decisions.

6.3.3 Observed Choices in Disagreement States

Figure 17 compares actual team behaviour with the prescriptions of each objective in states where they disagree. In both promotion and relegation settings, teams are more likely to follow the expected league points recommendation than the season-optimal one.

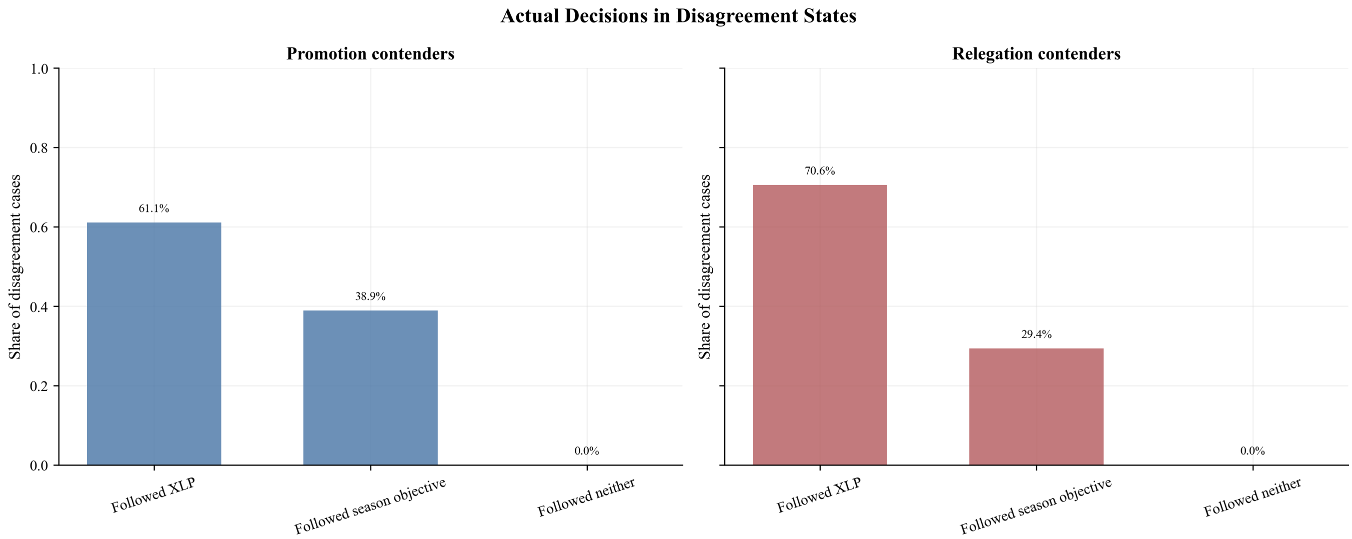
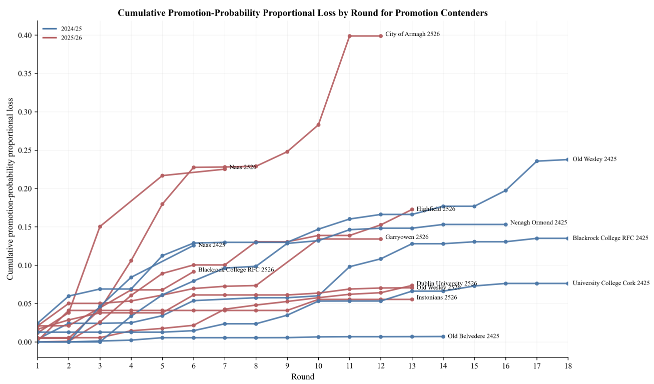


Figure 17: Actual decisions in disagreement states.

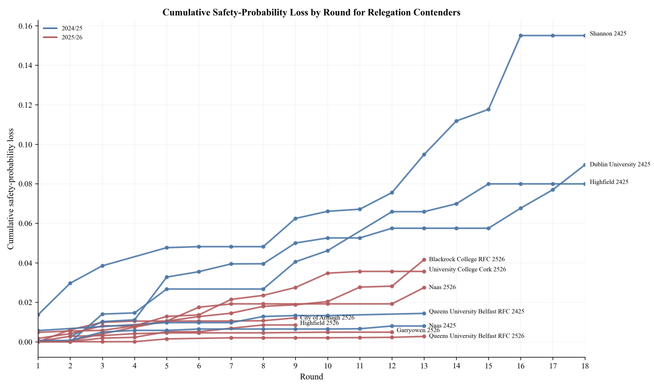
This suggests that, in practice, teams behave as though they are primarily optimising match-level expected value, even when the broader season context would favour a different action. The deviation is systematic rather than incidental, indicating that season-level incentives are not fully internalised in decision-making.

6.3.4 Cumulative Value Loss

To assess the consequences of these choices, Figures 18a and 18b plot cumulative losses in season-objective value over the course of the season. For each team-season, losses are aggregated across all penalty decisions, relative to the action that would have maximised promotion or survival probability. Lines are shown only while teams remain in contention, focusing attention on the periods where these incentives are most relevant.



(a) Promotion contenders



(b) Relegation contenders

Figure 18: Cumulative losses in season-objective value by round.

Losses accumulate steadily rather than arising from isolated mistakes. Small deviations from the season-optimal action compound over time, leading to substantial divergence across teams. There is also clear heterogeneity: some teams remain close to the optimal frontier, while others incur consistently larger losses. Importantly, losses tend to increase more sharply in later rounds, reflecting the greater sensitivity of promotion and relegation probabilities at that stage of the season.

6.3.5 Proportional Losses

While absolute losses quantify total value forgone, they do not indicate how large those losses are relative to what was at stake in each decision. Figure 19 therefore presents cumulative proportional losses, scaling each decision by the value of the optimal action.

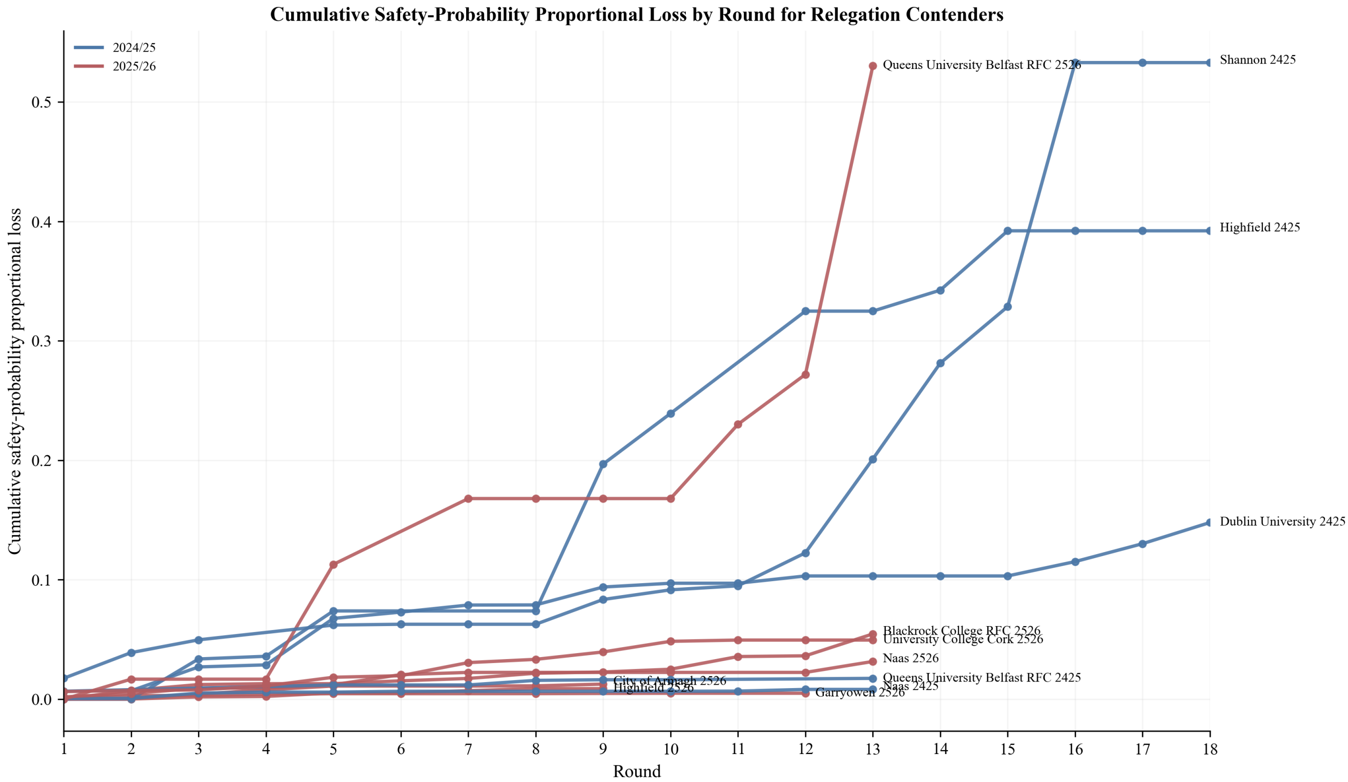


Figure 19: Cumulative survival-probability proportional loss by round for relegation contenders.

This highlights that some teams are not only losing value in aggregate, but are doing so by repeatedly making decisions that sacrifice a substantial share of the available value. In high-leverage situations, even seemingly small absolute differences can represent large proportional losses, reinforcing the importance of correctly accounting for season context.

Taken together, these results show that disagreement between objectives is both common and consequential. Teams tend to follow the expected league points objective when conflicts arise, and these choices accumulate into meaningful losses in promotion and survival probability. The next section examines specific match situations in detail, illustrating how these dynamics play out in individual high-stakes decisions.

6.4 Case Studies: High-Impact Penalty Decisions

The aggregate results identify broad patterns in optimal decision-making, but do not fully capture how and why suboptimal choices arise in practice. This section presents case studies focusing on consequential penalty decisions observed during the season. Each example highlights a situation in which the optimal action depends critically on the objective being pursued, and in which the observed decision failed to maximise that objective. The cases are organised by the type of error: failing to maximise survival probability, failing to maximise promotion probability, and failing to maximise match-level objectives.

6.4.1 Dublin University vs Nenagh Ormond: Relegation Risk and the Value of Bonus Points

This case provided the original motivation for the project. It illustrates how optimal decision-making cannot be separated from the broader league context.

In Round 18, Dublin University hosted Nenagh Ormond at College Park. In the 68th minute, Dublin University led 10–7 and were awarded a penalty in a central position, 24 metres out. Nenagh Ormond were locked into second place regardless of this result, meaning Dublin University's survival prospects depended entirely on their own league points and concurrent fixtures. Shannon were defeating Queens University Belfast 45–12 and would secure five league points. Naas were leading Highfield 42–40, a match that could end in any result.

Dublin University chose to kick at goal. Under match-level objectives, this decision is supported: the model estimates $p^{*,\text{win}} = 0.50$ and $p^{*,\text{pts}} = 0.61$, both comfortably below Matty Lynch's estimated success probability of 0.95.

Under the survival objective, the optimal decision reverses. The threshold rises to $p^{*,\text{surv}} = 0.96$, marginally above the estimated success probability, indicating that kicking to the corner is preferred.

The reason lies in the nonlinear mapping from league points to survival probability:

i	0	1	2	3	4	5
Pr(survival | i league points)	0.000	0.505	0.805	0.817	0.817	1.000

The discrete jump from four to five points drives the result. With Shannon taking five points and Highfield's match finely balanced, a four-point win would leave Dublin University exposed. A five-point win would guarantee survival regardless of Highfield's result.

Kicking at goal increases the probability of a narrow win but contributes little to securing the try bonus. Kicking to the corner offers a direct path to five league points. A successful penalty would extend the lead to 13–7, still a one-score margin, whereas a try would establish a two-score cushion (15–7 or 17–7), substantially reducing the probability of conceding defeat.

The continuation values reflect this: $V_{KC}^{\text{surv}} = 0.725$, $V_{SK}^{\text{surv}} = 0.726$, $V_{MK}^{\text{surv}} = 0.699$. The gain from a successful kick is negligible; the risk of a miss carries real cost. The expected loss from kicking at goal rather than to the corner is 0.07 percentage points in survival probability. Small, but it makes a difference.

Highfield won their match and Dublin University finished ninth. Relegation was avoided through the playoff, but the decision to kick at goal exposed the team to unnecessary risk by failing to target the outcome that would have guaranteed safety.

Post-match commentary indicated that players were not informed of the live score in the Highfield fixture, due to concerns that it might affect decision-making. This case suggests such information is essential. When league rewards are nonlinear, optimal decisions require conditioning on the full competitive environment.

6.4.2 Dublin University vs Shannon: Late-Game Risk and Over-Aggressive Play

This case provides a useful contrast to the previous example. While the earlier decision reflected excessive conservatism driven by nonlinear survival incentives, this instance illustrates the opposite error: pursuing upside in a setting where the objective is effectively linear and securing points is optimal.

In Round 5, Dublin University travelled to face Shannon. In the 77th minute, Dublin University trailed 20–10 and were awarded a penalty in a central position approximately 22 metres from touch. With limited time remaining, the decision effectively determined whether the team would leave the match with points.

Dublin University chose to kick to the corner. However, under both win probability and expected league points, the model strongly favours a shot at goal. The equilibrium thresholds are very low, with $p^{*,\text{win}} \approx 0.005$ and $p^{*,\text{pts}} \approx 0.323$, compared to an estimated success probability of $p = 0.857$. This implies that kicking at goal is clearly optimal under standard match-level objectives.

The cost of this decision is substantial. The expected league points from kicking to the corner are $V_{KC}^{\text{pts}} = 0.329$, compared to $V_{SK}^{\text{pts}} = 0.892$ following a successful kick and $V_{MK}^{\text{pts}} = 0.060$ following a miss. In expectation, this results in a loss of 0.44 league points from a single decision, the largest observed loss in expected league points across the dataset.

The intuition is straightforward. Under a linear objective, the marginal value of each additional league point is approximately constant, so the priority is to maximise the probability of securing points rather than to pursue higher-variance outcomes. A successful kick reduces the deficit to seven points, preserving the possibility of a draw or win while substantially increasing the likelihood of securing at least one league point through a losing bonus. By contrast, kicking to the corner introduces additional execution risk. Failure to score leaves the team with zero points, which is significantly more costly than the incremental benefit of a low-probability try.

This example therefore differs fundamentally from the previous case. There, the optimal decision was driven by a discrete increase in survival probability between four and five league points. Here, the payoff structure does not exhibit such discontinuities, and the optimal strategy is correspondingly simple: prioritise outcomes that reliably deliver points.

The error in this case arises from an overemphasis on upside in a context where risk-taking is not justified by the objective. Taken together with the previous example, this highlights that suboptimal decisions can arise both from excessive conservatism and excessive aggression. Optimal play requires aligning risk-taking with the structure of the objective being maximised.

6.4.3 University College Cork vs Queens University Belfast: Early-Game Promotion Probability and the Cost of Ambition

In Round 16, University College Cork hosted Queens University Belfast. In just the third minute, with the score level at 0–0, UCC were awarded a penalty in a central position, 22 metres from goal. The decision was to kick to the corner.

Under expected league points, this was the correct call. The model confirms that kicking to the corner maximises expected points, with $V_{KC}^{\text{pts}} = 0.152$, reflecting UCC's strong attacking platform and the marginal value of securing a try bonus from the outset.

However, under the promotion probability objective, the decision represents a modest error. The equilibrium threshold lies well above UCC's estimated kick success probability, yet the absolute loss in promotion probability is small at just 0.002. The reason lies in UCC's league position at the time. Sitting fifth in the table with 38 points from 15 rounds, UCC trailed Old Belvedere, Nenagh Ormond, Blackrock, and Old Wesley by substantial margins. With the top four effectively consolidated, the marginal gain from a successful penalty kick in the third minute does little to close the gap. The promotion probability surface is relatively flat at this point in the match, and the difference between the two decisions is negligible.

League Points	0	1	2	3	4	5
Pr(promotion | i league points)	0.012	0.028	0.050	0.080	0.107	0.140

Unlike the Dublin University survival case, where the payoff structure exhibited a sharp discontinuity between four and five league points, the promotion probability schedule here is approximately linear across outcomes. Each additional league point contributes a broadly similar increment to promotion probability, meaning that the choice of decision matters less than the eventual match outcome. UCC went on to win the match 29–28, securing a narrow but important victory.

This case therefore serves as a useful illustration of model calibration. Not every suboptimal decision carries a significant cost. The framework identifies the deviation, but correctly quantifies the loss as trivial. In a match where UCC's promotion prospects were already slim, the priority was to maximise league points, and kicking to the corner was consistent with that objective.

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Taken together, these three cases illustrate the core contribution of the framework. Suboptimal decisions arise not from poor rugby instinct, but from an inability to condition on the full competitive environment in real time. The model does not replace judgment; it informs it.

6.5 Value of the Framework

The preceding sections have established what optimal penalty decisions look like under various objectives and documented how observed behaviour deviates from these prescriptions. This section draws together these findings to assess the practical value of the framework. A decision support tool is only useful if it provides clear guidance, identifies errors that would otherwise go unnoticed, and offers improvements over simpler alternatives. The evidence supports all three claims.

6.5.1 Recommendation Clarity

A framework that frequently returns ambiguous recommendations offers limited practical value. Fortunately, this is not the case here. Under the expected league points objective, the model provides a clear recommendation, either kick at goal or kick to the corner, in 95.0% of the 1,348 penalties evaluated. Only 68 cases (5.0%) fall into the ambiguous zone where the posterior probability of either action being optimal lies between 30% and 70%.

The breakdown of clear recommendations is itself informative. Of the 1,280 penalties with clear guidance, the framework recommends kicking to the corner in 1,072 cases (79.5%) and kicking at goal in just 208 cases (15.4%). This asymmetry reflects the underlying economics of the decision: the expected value of attacking from the corner frequently exceeds the expected value of three points, particularly when kick success probability is below 80% or when try bonus considerations apply.

Recommendation clarity varies modestly across objectives. The win probability objective yields clear recommendations in 86.3% of cases, with a more balanced split between kick (35.9%) and corner (50.4%). Promotion and survival objectives maintain clarity rates above 93%, though these apply only to the subset of penalties involving contender teams. The high clarity rate across all objectives means that in the vast majority of situations, the framework can provide actionable guidance rather than equivocation.

6.5.2 Decision Quality Under Current Practice

Having established that the framework produces clear recommendations, the natural question is how often teams already follow them. Under the expected league points objective, teams make the optimal decision in 957 of 1,280 clear-recommendation cases, yielding an accuracy rate of 74.8%. This represents a meaningful baseline: teams are getting roughly three quarters of these decisions right without formal decision support.

The 25.2% error rate decomposes into two distinct failure modes. Over-kicks, where teams kicked at goal when the corner was optimal, account for 210 errors (16.4% of decisions). Under-kicks, where teams went to the corner when kicking was optimal, account for 113 errors (8.8%). The asymmetry is notable. Teams are nearly twice as likely to take points when they should attack as they are to attack when they should take points. This pattern is consistent with loss aversion or a preference for certain outcomes over probabilistic ones, even when the expected value calculation favours the gamble.

Decision quality varies substantially across objectives. Under win probability, accuracy falls to 66.0%, with under-kicks (25.5%) far exceeding over-kicks (8.5%). This reversal makes sense: win probability often favours attacking when trailing, but teams appear to take points anyway, perhaps anchoring on the immediate deficit rather than the probability of eventual victory. Under the survival objective, accuracy drops further to 59.5%, with under-kicks again dominant (25.9% versus 14.6%). Teams facing relegation appear particularly prone to conservative decision-making even when aggression would better serve their survival prospects.

6.5.3 Magnitude of Decision Errors

Not all errors are equal. A decision that sacrifices 0.01 expected league points is qualitatively different from one that sacrifices 0.40. The distribution of losses among suboptimal decisions reveals where the framework adds the most value.

For the 292 suboptimal decisions under expected league points with computable losses, the median loss is 0.080 expected league points per decision. The mean is somewhat higher at 0.096, reflecting a right-skewed distribution with occasional large errors. The 90th percentile reaches 0.189, and the maximum observed loss is 0.444 expected league points from a single decision.

To contextualise these magnitudes: a loss of 0.08 expected league points is equivalent to giving up nearly one tenth of a league point in expectation. Over a season with perhaps 30 such decision opportunities per team, even median-sized errors compound into meaningful totals. The maximum loss of 0.444, observed in the Dublin University versus Shannon fixture discussed earlier, represents nearly half a league point sacrificed on a single decision. In a competition where final standings are frequently separated by one or two points, these are not trivial quantities.

The loss distributions for other objectives are harder to interpret directly. Win probability losses are measured in probability units, with a median of 0.010 and maximum of 0.189. Promotion and survival probability losses are smaller in absolute terms (medians of 0.0004 and 0.0007 respectively) because baseline probabilities are often close to zero or one, compressing the feasible range of losses.

6.5.4 Comparison to Simple Heuristics

A sophisticated framework is only valuable if it outperforms simpler decision rules that could be implemented without computational infrastructure. To test this, I evaluate twelve heuristics against the framework's recommendations under the expected league points objective.

Table 3: Heuristic agreement with framework recommendations (Expected League Points objective). Agreement measures the percentage of clear-recommendation cases where the heuristic matches the framework's optimal action. Mean loss is computed over all clear-recommendation cases.

Heuristic	Agreement	Mean Loss
Always corner	83.8%	0.0100
Kick if $p > 80\%$	78.9%	0.0148
Captain (observed)	**74.8%**	**0.0202**
Kick if $p > 70\%$ and margin ≥ -7	71.9%	0.0238
Kick if $p > 70\%$	71.0%	0.0250
Late game: kick if leading	69.3%	0.0295
Kick if leading	63.6%	0.0406
Kick if $p > 60\%$	49.0%	0.0538
Kick if leading or level	44.9%	0.0616
Corner if trailing by more than 3	36.6%	0.0696
Kick if $p > 50\%$	33.2%	0.0734
Corner if trailing by more than 7	24.1%	0.0844
Always kick	16.2%	0.0969

The results in Table 3 contain a striking finding: the simple heuristic "always kick to the corner" achieves 83.8% agreement with the framework, outperforming observed captain decisions (74.8%) by nine percentage points. This is not an endorsement of mindless corner-kicking, but rather a reflection of two facts. First, the expected league points objective heavily favours the corner in most situations within this sample. Second, teams are systematically over-kicking relative to this benchmark.

The "kick if $p > 80\%$" heuristic also outperforms captains, achieving 78.9% agreement. This rule captures the intuition that very high-percentage kicks should be taken, but defaults to the corner otherwise. Observed behaviour resembles the "kick if $p > 70\%$" rule (71.0% agreement), suggesting that teams are applying a threshold roughly ten percentage points below optimal.

These comparisons illuminate where the framework adds value. Simple heuristics perform reasonably well in straightforward situations but fail in the margins. The framework's advantage lies precisely in the complex cases: when bonus point incentives create nonlinearities, when the scoreline makes the corner unusually valuable or unusually risky, when survival or promotion considerations override match-level logic. A heuristic that says "always corner" will be wrong 16.2% of the time; the framework identifies exactly which 16.2% those are.

6.5.5 Systematic Errors by Game State

Figure 10 reveals that decision quality is not uniform across game states. Accuracy is highest at large deficits (−12: 89.7%) and large leads (+12: 89.5%), where the optimal action is relatively unambiguous. Accuracy collapses at specific intermediate margins, most dramatically at +8 where teams achieve only 33.3% accuracy (6 correct of 18 decisions).

The margin of +8 is particularly instructive. At this lead, a successful penalty extends the margin to +11, while a try extends it to +13 or +15. Neither outcome crosses a strategically meaningful threshold: the opposition already cannot win with a single converted try at +8, and the try bonus is not directly affected. Yet teams chose to kick at goal in the majority of these situations, sacrificing the expected value advantage of attacking. The error appears to reflect a desire to "put the game away" that is not justified by the underlying mathematics.

Other problematic margins include +1 (60.0% accuracy), +3 (64.5%), and +5 (66.7%). These are precisely the situations discussed in the decision threshold section: narrow leads where the choice between consolidating and extending creates genuine strategic tension. The framework identifies which action is optimal in each case; teams are getting it wrong roughly one third of the time.

At the other extreme, accuracy exceeds 80% at margins of −12, −6, −5, −4, and +12. Large deficits produce clear incentives to chase tries; large leads produce clear incentives to consolidate. The framework confirms these intuitions and teams respond accordingly.

6.5.6 Objective Alignment

A secondary contribution of the framework is revealing which objective teams implicitly pursue. Figure **??** shows that observed behaviour aligns most closely with expected league points (74.8%), followed by promotion probability (67.9%), win probability (66.0%), and survival probability (59.5%).

The ordering is informative. Teams appear to optimise for league points rather than win probability, suggesting an awareness of bonus point incentives even if the specific calculations are imprecise. The lower alignment with survival probability (59.5%) indicates that relegation-threatened teams are particularly prone to decision errors, a finding consistent with the psychological literature on decision-making under threat.

When objectives disagree, teams systematically favour expected league points over season-level considerations. In the 113 cases where expected league points and survival probability prescribe different actions, teams followed the expected league points recommendation 72.6% of the time and the survival-optimal recommendation only 27.4%. A similar but less pronounced pattern holds for promotion (61.9% versus 38.1%). This suggests that teams either lack the information to compute season-level implications or discount them relative to immediate match outcomes.

6.5.7 Team-Level Variation

The 24.8 percentage point spread between the best team-season (Instonians 2025/26, 87.8% accuracy) and the worst (Naas 2024/25, 63.0%) demonstrates that decision quality is not uniform across the league. Some teams have cracked the code, whether through analytical sophistication, conservative heuristics that happen to align with optimal play, or experienced captains with good intuitions. Others are leaving expected league points on the table through systematic errors.

The variation also suggests that improvement is feasible. If Instonians can achieve 87.8% accuracy, there is no structural reason why other teams cannot approach this level. The framework provides the tool; implementation requires only that teams consult it.

Several patterns emerge from the team-level data. Accuracy improved league-wide from 2024/25 to 2025/26, with most teams showing gains. The worst performers (Shannon, Naas, Dublin University in

2024/25) shared a tendency toward over-kicking, taking points in situations where the framework recommended attacking. The best performers showed no such systematic bias, suggesting that calibrated decision-making rather than any particular philosophy drives accuracy.

6.5.8 High-Stakes Decisions

The framework's value is arguably greatest when the stakes are highest. Among late-season penalties (Round 12 onward) for teams still in promotion contention, accuracy under the promotion objective reaches 77.1%. For relegation contenders under the survival objective, accuracy is markedly lower at 65.0%.

This 12.1 percentage point gap is troubling. Teams fighting relegation are making worse decisions precisely when good decisions matter most. The pattern is consistent with pressure-induced conservatism: teams take the "safe" option of kicking for points when the mathematically correct action is to attack. The Dublin University case study illustrates this dynamic in detail.

The framework cannot eliminate pressure, but it can provide clarity. A captain who knows that the survival-optimal action is to kick to the corner, backed by explicit probability calculations, is better positioned to make that call than one relying on intuition under duress.

6.5.9 Practical Implementation

The framework is designed for live in-game deployment and will be implemented by Dublin University Football Club for the 2026/27 season. The system operates through a **Traffic Light System**, in which the model continuously updates recommendations based on real-time match state, including score margin, time remaining, and try count.

The system outputs one of three signals:

- **Green:** The model's recommendation is Kick at Goal and is confident in this.
- **Amber:** The decision lies within a contested region where model uncertainty is non-trivial; the captain is alerted and retains decision-making authority.
- **Red:** The model strongly advises for kicking to the corner.

This hybrid approach preserves human judgment at the margins while leveraging the model's optimality in clear-cut situations. Pre-computed threshold tables indexed by score margin, time remaining, and try count remain available as a reference, but the live system removes the need for manual lookup in most cases. The implementation at Dublin University represents the first known deployment of a decision-theoretic penalty strategy framework in competitive club rugby.

6.5.10 Summary

The framework provides clear, actionable recommendations in 95% of penalty situations. Teams currently achieve 74.8% accuracy against these recommendations, leaving substantial room for improvement. Errors are systematic rather than random: teams over-kick relative to optimal, struggle at specific margins, and fail to account for survival considerations when facing relegation. Simple heuristics like "always corner" outperform observed behaviour, but miss the nuances that the framework captures.

The value of the framework lies not in promising specific league point gains, which would require assumptions about additivity that do not hold, but in identifying the right decision at each moment. A team that consults the framework and acts on its recommendations will, on average, make better choices than one relying on intuition alone. The magnitude of improvement depends on how suboptimal current practice is, which varies across teams and situations, but the direction is unambiguous. The framework makes penalty decisions better.

7 Conclusion

This thesis has developed and empirically tested a decision-theoretic framework for analyzing optimal penalty strategies in rugby union. By moving beyond simple expected value calculations to account for competing objective functions and nonlinear league incentives, this research provides a more nuanced understanding of strategic decision-making in high-stakes competitive environments.

7.1 Summary of Contributions

The core contribution of this work is the demonstration that optimal penalty choices differ substantially depending on whether a team seeks to maximize win probability, expected league points, or promotion and survival probabilities. Using a dataset of 1,349 penalty decisions from the Energia All-Ireland League Division 1B, several key findings emerged:

- **Systematic Suboptimality:** Teams displayed a 25% error rate relative to optimal decision rules, with a variation of up to 24 league points between the most and least analytically efficient teams.
- **Methodological Innovation:** The introduction of a conditional expected points (cEP) framework successfully addressed confounding in observational sports data, while the selection-corrected kick model improved out-of-sample prediction accuracy by 5.7% over naive models.
- **Objective-Dependent Strategy:** The analysis proved that nonlinear incentives—such as bonus points and promotion thresholds—significantly alter the mathematical ”best” choice, a factor frequently overlooked in traditional coaching heuristics.

7.2 Practical Implications

For practitioners, these results suggest that ”gut instinct” often fails to account for the complex mathematical trade-offs inherent in league structures. The tools developed herein—specifically the decision-support metrics for penalty selection—offer a clear path for teams to gain a competitive edge through improved analytical rigor. Implementing these strategies could reasonably yield an additional 2-3 league points per season for sub-optimal teams, often the difference between promotion and mid-table stagnation.

7.3 Limitations and Future Research

While this study provides a robust foundation, it is limited by its focus on a single division and a two-season timeframe. Future research should aim to:

1. **Expand Scope:** Apply the cEP framework to international competitions (e.g., Six Nations) and other professional leagues to test the external validity of these findings.
2. **Incorporate Game Theory:** Model the interaction between kicker strategy and opponent defensive positioning to account for recursive strategic responses.
3. **Environmental Covariates:** Integrate real-time weather, pitch conditions, and player fatigue data to further refine the kick success probability models.

7.4 Final Remarks

In conclusion, as rugby union continues to professionalize, the integration of objective-dependent optimization will become a necessity rather than a luxury. By bridging the gap between formal decision theory and the practical realities of the pitch, this thesis contributes to a new era of data-driven strategy that honors the complexity of the sport while maximizing the probability of success.

A Scoring Model

This appendix provides the full technical details for the competing hazards scoring model introduced in Section 5.1. The scoring model is the core generative component of the decision framework. It maps post-decision states into distributions over final scores, try counts, and league points, which are then used to evaluate win probability, expected league points, promotion probability, and survival probability.

The appendix proceeds in two parts. Appendix A.1 presents the full model specification, including the likelihood, linear predictors, regularisation, and simulation procedure. Appendix A.2 then reports validation evidence, covering duration dependence, parameter interpretation, calibration, and out-of-sample predictive performance.

A.1 Model Specification

I model scoring as a continuous-time competing hazards process defined over spells. Each spell begins at kickoff, at the start of the second half, or immediately after a scoring event, and ends either when the next score occurs or when the half ends without further scoring. At most one event can terminate a spell, so a competing-risks formulation is appropriate.

Let

$$\mathcal{H} = \{\text{HT}, \text{AT}, \text{HG}, \text{AG}\}$$

denote the set of possible terminal events, where HT and AT are home and away tries, and HG and AG are home and away goals. Goals include both penalty goals and drop goals, though penalty goals account for the vast majority of goal-scoring events in the sample.

For each event type $k \in \mathcal{H}$, I specify a Weibull cause-specific hazard:

$$h_k(t \mid \mathbf{X}) = \alpha_k \lambda_k t^{\alpha_k - 1} \exp(\eta_k),$$

where t denotes time elapsed since the beginning of the spell, α_k is a shape parameter, λ_k is a baseline scale parameter, and η_k is a linear predictor. The corresponding cumulative hazard is

$$H_k(t \mid \mathbf{X}) = \lambda_k t^{\alpha_k} \exp(\eta_k).$$

Let $d_{ik} \in \{0, 1\}$ indicate whether spell i ends with event k, and let t_i denote the spell duration. Spells that end at halftime or full time without a score are treated as right-censored. The contribution of spell i to the likelihood is

$$L_i = \prod_{k \in \mathcal{H}} h_k(t_i)^{d_{ik}} \cdot \exp\left(-\sum_{k \in \mathcal{H}} H_k(t_i)\right),$$

so the full log-likelihood is

$$\ell(\theta) = \sum_i \sum_{k \in \mathcal{H}} d_{ik} \left[\log \alpha_k + \log \lambda_k + (\alpha_k - 1) \log t_i + \eta_{ik}\right] - \sum_i \sum_{k \in \mathcal{H}} \lambda_k t_i^{\alpha_k} \exp(\eta_{ik}),$$

where θ collects the shape parameters $(\alpha_{\text{try}}, \alpha_{\text{goal}})$, baseline rates $(\lambda_k)_{k \in \mathcal{H}}$, home-advantage terms $(\gamma_{\text{try}}, \gamma_{\text{goal}})$, team effects $(A_u^{\text{try}}, D_u^{\text{try}}, A_u^{\text{goal}}, D_u^{\text{goal}})_u$, and margin functions $(\beta^{\text{try}}(m), \beta^{\text{goal}}(m))_m$.

To reduce dimensionality, I impose common shape parameters within scoring type:

$$\alpha_{\text{HT}} = \alpha_{\text{AT}} \equiv \alpha_{\text{try}}, \qquad \alpha_{\text{HG}} = \alpha_{\text{AG}} \equiv \alpha_{\text{goal}}.$$

This allows tries and goals to exhibit different duration profiles while avoiding unnecessary parameter proliferation.

The linear predictors incorporate team-season strength, home advantage, and score-margin effects. For try hazards,

$$\eta_{\text{HT}} = \gamma_{\text{try}} + (A_h^{\text{try}} - D_a^{\text{try}}) + \beta^{\text{try}}(m),$$

$$\eta_{\mathrm{AT}} = -\gamma_{\mathrm{try}} + (A_a^{\mathrm{try}} - D_h^{\mathrm{try}}) + \beta^{\mathrm{try}}(-m),$$

where γ_{try} is the try-specific home-advantage term, A_u^{try} and D_u^{try} are team-season attacking and defensive effects, and m is the current score margin from the home team's perspective. Goal hazards are specified analogously:

$$\eta_{\mathrm{HG}} = \gamma_{\mathrm{goal}} + (A_h^{\mathrm{goal}} - D_a^{\mathrm{goal}}) + \beta^{\mathrm{goal}}(m),$$

$$\eta_{\mathrm{AG}} = -\gamma_{\mathrm{goal}} + (A_a^{\mathrm{goal}} - D_h^{\mathrm{goal}}) + \beta^{\mathrm{goal}}(-m).$$

The margin response functions $\beta^{\mathrm{try}}(m)$ and $\beta^{\mathrm{goal}}(m)$ are defined for integer margins

$$m \in \{-40, \ldots, 40\},$$

allowing scoring behaviour to vary flexibly with the scoreboard. Team effects are mean-centred within season for identification, so they are interpreted relative to the average team in that season.

The model contains a large number of team and margin parameters, so estimation proceeds by penalised maximum likelihood. In total, the specification includes two shape parameters, four baseline rates, two home-advantage terms, 80 team-specific effects, and 162 margin coefficients. Without regularisation, the model would overfit, particularly for rare score margins and for teams observed in relatively few matches.

I therefore estimate

$$\begin{aligned}\ell_{\mathrm{pen}}(\theta) = \ell(\theta) - \frac{\lambda_{\mathrm{ridge}}}{2}\sum_u \Big[(A_u^{\mathrm{try}})^2 + (D_u^{\mathrm{try}})^2 + (A_u^{\mathrm{goal}})^2 + (D_u^{\mathrm{goal}})^2\Big] \\ - \lambda_{\mathrm{TV}}^{\mathrm{try}} \sum_{m=-39}^{40} \left|\beta^{\mathrm{try}}(m) - \beta^{\mathrm{try}}(m-1)\right| \\ - \lambda_{\mathrm{TV}}^{\mathrm{goal}} \sum_{m=-39}^{40} \left|\beta^{\mathrm{goal}}(m) - \beta^{\mathrm{goal}}(m-1)\right|.\end{aligned}$$

The ridge penalty shrinks team effects toward zero, while the total variation penalties encourage smooth responses across adjacent score margins. The L_1 form of the total variation penalty allows flat regions where the data do not support meaningful differentiation between nearby margins.

Penalty weights are chosen by five-fold cross-validation, assigning all spells from a given fixture to the same fold. The grid search evaluates 125 combinations of three penalty parameters: ρ_{try} for the try margin function, ρ_{goal} for the goal margin function, and λ_{team} for team-effect shrinkage. The selected configuration is

$$\begin{aligned}\rho_{\mathrm{try}} &= 0.20,\\ \rho_{\mathrm{goal}} &= 0.40,\\ \lambda_{\mathrm{team}} &= 0.20,\end{aligned}$$

which achieves mean validation negative log-likelihood of 1248.9 (± 32.4 across folds).

Once estimated, the model is simulated forward from any post-decision state. At each step, the four hazard rates are computed, the timing of the next event is drawn by inversion of the cumulative hazard, and the event type is then drawn conditional on an event occurring. If no further event occurs before full time, the match ends; otherwise the state is updated and the process repeats.

Each try is followed by a conversion attempt. Because the scoring model does not generate try locations, conversion success is handled separately using team-season Beta-Binomial posterior means, which shrink observed conversion rates toward the league average for teams with limited attempts.

Simulated matches track both final score and try counts. This allows league points to be assigned under AIL rules: four points for a win, two for a draw, one try bonus point for scoring four or more tries, and one losing bonus point for losing by seven or fewer points. These discontinuities are central to the decision problem, so the model is designed to recover the full distribution of final outcomes rather than just a single match-level statistic.

A.2 Validation

This appendix reports validation evidence for the competing hazards scoring model described in Section 5.1. I proceed in five steps: duration dependence testing, parameter interpretation, calibration, out-of-sample predictive performance, and an overall assessment of model suitability for the decision framework.

A.2.1 Duration Dependence

The constant-hazard Poisson process is nested within the Weibull specification as the restriction

$$\alpha_{\text{try}} = \alpha_{\text{goal}} = 1.$$

I test this jointly by likelihood ratio. The exponential model achieves log-likelihood −6,126.1, while the unrestricted Weibull model achieves −5,946.5. The resulting test statistic is

$$\text{LR} = 2 \times (6{,}126.1 - 5{,}946.5) = 359.1,$$

which under the null follows $\chi^2(2)$. The p-value is below 10^{-15}, decisively rejecting constant hazards.

The estimated shape parameters are

$$\hat{\alpha}_{\text{try}} = 1.50, \qquad \hat{\alpha}_{\text{goal}} = 1.58.$$

Both exceed unity, implying that scoring hazards rise with spell duration. This matters substantively because a memoryless Poisson process would understate the value of sustained attacking pressure, whereas the estimated Weibull model recognises that long spells of territorial dominance become increasingly dangerous. The slightly higher estimate for goals suggests that goal-scoring opportunities accumulate particularly sharply under sustained pressure, consistent with penalties arriving when defences are stretched and discipline deteriorates.

A.2.2 Parameter Estimates

Table 4 reports the main parameter estimates from the full model.

Table 4: Competing Hazards Model: Parameter Estimates

Category	Parameter	Estimate	Interpretation
Shape	α_{try}	1.500	Increasing try hazard
	α_{goal}	1.585	Increasing goal hazard
Home Advantage	γ_{try}	0.087	9.1% higher try rate
	γ_{goal}	0.049	5.0% higher goal rate
Baseline Scale	λ_{HT}	0.0153	Home try
	λ_{AT}	0.0140	Away try
	λ_{HG}	0.00039	Home goal
	λ_{AG}	0.00037	Away goal

Home advantage operates through both try and goal channels, though the effect is stronger for tries: $e^{0.087} - 1 = 9.1\%$ versus roughly 5.0% for goals. This suggests that home advantage works primarily through territorial momentum and sustained attacking pressure rather than through officiating effects alone.

A.2.3 Margin Response Functions

Figure 20 plots the estimated margin response functions $\beta^{\text{try}}(m)$ and $\beta^{\text{goal}}(m)$.

The try margin function is noisier than the goal function, which is unsurprising given the greater execution uncertainty involved in scoring tries. Nonetheless, the pattern is interpretable. At extreme deficits, try-scoring intensity rises for trailing teams as they are forced into higher-risk play and as opposing defences relax once outcomes are effectively decided. In the competitive range, try hazard is especially low around margins such as +6 and +11, where protecting the lead by kicking becomes particularly attractive.

The goal margin function is cleaner and exhibits a clear inverted-U structure. Goal-scoring intensity peaks where three points is tactically most valuable, such as when trailing by one to three or when specific score thresholds can be crossed by a successful kick. This aligns closely with rugby's tactical logic and supports the claim that teams respond systematically to the scoreboard.

Figure 20: Estimated Margin Response Functions

Notes: Top panels show the full margin range with smoothed trends; bottom panels show the competitive range $[-12, +12]$. Left panels display try hazard effects; right panels display goal hazard effects. Positive values indicate increased scoring intensity. Margin is measured from the home team's perspective.

A.2.4 Team Strength Estimates

Figure 21 plots the attacking and defensive team-season effects.

The try panel shows clear separation between stronger and weaker sides. Teams in the upper-right quadrant score tries more frequently while conceding fewer, while those in the lower-left quadrant exhibit the reverse pattern. The goal panel shows wider dispersion, suggesting greater heterogeneity in kicking quality and tactical use of penalty goals.

The weak correlation between try and goal attack effects supports the decision to estimate separate processes. Teams that are strong try scorers are not necessarily the same teams that are strong goal scorers.

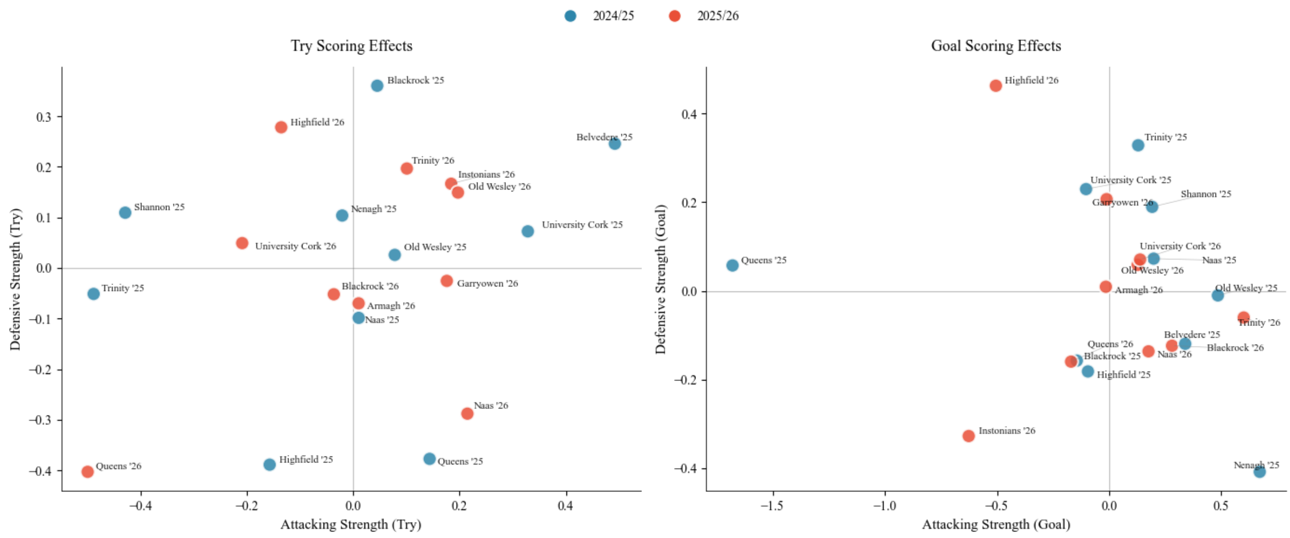

Figure 21: Team Strength Effects by Season

Notes: Each point represents a team-season. Blue points denote 2024/25; red points denote 2025/26. Effects are mean-centred within each season. Left panel shows try effects; right panel shows goal effects. Higher attacking strength indicates more frequent scoring; higher defensive strength indicates opponents score less frequently against that team.

A.2.5 Calibration

I assess calibration using the expected calibration error:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left| \bar{p}_b - \bar{o}_b \right|,$$

where $\bar{p}_b$ is the mean predicted probability in bin b and $\bar{o}_b$ is the realised frequency.

Figure 22 shows calibration curves for home win probability across entry minutes. The curves track the diagonal closely, with all entry points achieving in-sample ECE below 0.10.

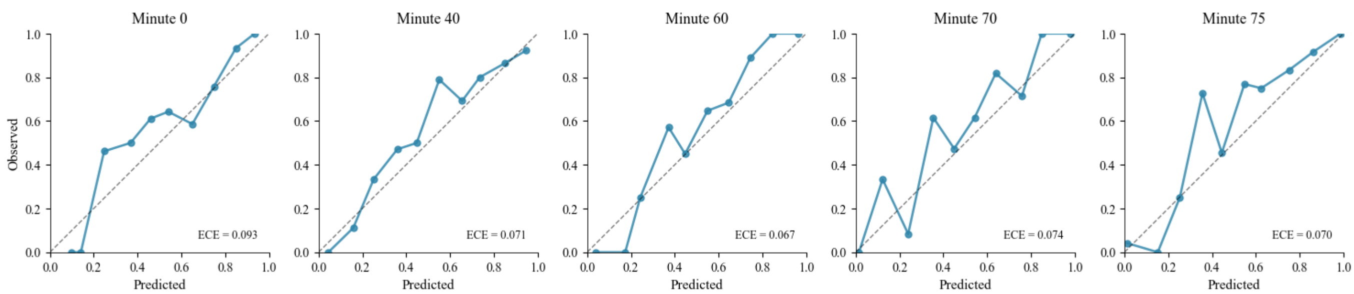

Figure 22: Calibration Curves: Home Win Probability

Notes: Each panel shows predicted probability against observed frequency for home wins at a given entry minute. The dashed diagonal indicates perfect calibration. ECE denotes expected calibration error.

Table 5 reports in-sample and out-of-sample ECE values.

A.2.6 Out-of-Sample Prediction

I randomly hold out 15% of fixtures, estimate the model on the remaining sample, and evaluate predictive performance from several entry points.

The Brier score is

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} (p_{ik} - o_{ik})^2,$$

Table 5: Expected Calibration Error by Entry Minute

Entry Minute	In-Sample	Out-of-Sample
0	0.093	0.100
40	0.071	0.120
60	0.067	0.113
70	0.074	0.106
75	0.070	0.114

log-loss is

$$\mathrm{LL} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} o_{ik}\log(p_{ik}),$$

and the ranked probability score is

$$\mathrm{RPS} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{K-1}\sum_{k=1}^{K-1}\left(\sum_{j=1}^{k} p_{ij} - \sum_{j=1}^{k} o_{ij}\right)^2.$$

Table 6 reports result prediction performance.

Table 6: Out-of-Sample Predictive Performance: Results

Entry Minute	n	Accuracy	Brier Score	Log-Loss	RPS
0	25	76.0%	0.257	0.436	0.128
40	25	88.0%	0.229	0.395	0.114
60	25	84.0%	0.203	0.358	0.100
70	25	92.0%	0.149	0.276	0.073
75	25	92.0%	0.176	0.301	0.085

Accuracy rises from 76% at kickoff to 92% by minute 70, while probabilistic forecast metrics improve correspondingly. This is the pattern one would hope to see, since predictive uncertainty should fall as more of the match is observed.

A.2.7 Score Margin Prediction

Since losing bonus points depend on final margin, it is also important that the model predicts margins reasonably well. Table 7 reports out-of-sample performance.

Table 7: Out-of-Sample Predictive Performance: Score Margins

Entry Minute	MAE	RMSE	Correlation	Within 7	Within 14
0	13.20	16.72	0.357	36.0%	56.0%
40	8.95	12.40	0.650	60.0%	80.0%
60	7.36	10.81	0.754	60.0%	88.0%
70	5.84	9.75	0.799	76.0%	92.0%
75	4.91	9.10	0.828	84.0%	96.0%

By minute 75, 84% of predicted margins fall within seven points of the realised margin, indicating that the model is sufficiently precise late in matches to support bonus-point-relevant analysis.

A.2.8 Home League Points Prediction

Table 8 reports out-of-sample prediction of home league points.

Table 8: Out-of-Sample Predictive Performance: Home League Points

Entry Minute	MAE	Correlation	Exact	Top 2
0	1.42	0.525	44.0%	84.0%
40	1.29	0.660	48.0%	68.0%
60	1.08	0.788	68.0%	84.0%
70	0.84	0.845	72.0%	84.0%
75	0.78	0.816	76.0%	84.0%

Notes: Exact denotes the percentage of matches where the most likely simulated league point outcome equals the actual outcome. Top 2 denotes the percentage where the realised outcome is among the two most likely simulated outcomes.

By minute 70, the model predicts the exact league point outcome in 72% of matches and places the realised outcome in its top two predictions in 84% of cases. Given the discontinuities introduced by try and losing bonus points, this is strong performance.

A.2.9 Confidence and Accuracy

A useful model should be more accurate when it is more confident. Table 9 reports realised result accuracy conditional on the model's maximum predicted probability exceeding various thresholds.

Table 9: Accuracy by Confidence Threshold (Out-of-Sample)

Threshold	Min 0	Min 40	Min 60	Min 70	Min 75
≥ 0.5	83%	88%	91%	92%	92%
≥ 0.6	95%	89%	90%	95%	95%
≥ 0.7	100%	100%	94%	100%	94%
≥ 0.8	100%	100%	100%	100%	100%
≥ 0.9	100%	100%	100%	100%	100%

The relationship is strongly positive. When the model is highly confident, it is almost always correct. This is important for the broader decision framework because it implies that strong model recommendations are genuinely informative.

A.2.10 Summary

The validation evidence supports four main conclusions:

1. The Weibull specification captures duration dependence that the Poisson benchmark misses. The likelihood ratio test rejects constant hazards decisively.

2. The estimated parameters are substantively interpretable. Duration dependence is positive, home advantage is stronger for tries than for goals, and the scoreline shapes scoring behaviour in tactically plausible ways.

3. The model is well calibrated, with in-sample ECE below 0.10 throughout.

4. Out-of-sample predictive performance is strong and improves as match information accumulates, reaching 92% result accuracy by minute 70 and 72% exact league-point accuracy at the same horizon.

Taken together, these results indicate that the model provides a credible representation of match dynamics and is fit for the counterfactual analysis undertaken in the main text.

B Kick to Corner Model

This section provides a full description of the kick-to-corner model used to evaluate the value of kicking for territory following a penalty. The model is constructed as a sequence of linked stochastic components, each representing a stage in the attacking process. Together, these components map a penalty location into a distribution over match outcomes.

The sequence begins with the touch kick and proceeds through the lineout, maul, and open phase attack. If a try is scored, the sequence concludes with a conversion attempt. At each stage, uncertainty is modelled explicitly, and outcomes are propagated forward through simulation until a terminal state is reached.

B.1 Structure of the Model

The model consists of five stages:

1. **Touch kick**
2. **Lineout**
3. **Maul**
4. **Phase attack**
5. **Conversion**

Each stage produces a probability distribution over outcomes, which defines the starting point for the next stage. The full sequence is therefore a stochastic state-transition system.

B.2 Recursive Structure

An important feature of the model is the treatment of penalties won during attacking play. Both the maul and phase models allow for penalties to be awarded to the attacking team.

When a penalty is won, the model applies a location-based rule. Near the try line, the team taps and continues. Otherwise, the team kicks back to the corner, restarting the sequence.

This recursive structure means that the value of kicking to the corner depends on the cumulative probability of scoring across repeated attacking phases rather than a single attempt.

B.3 Simulation Framework

The model is implemented via forward simulation. Starting from a penalty location:

- A touch outcome is drawn
- A lineout outcome is drawn conditional on touch
- The maul model is applied
- The phase model simulates continuation if required
- The conversion model assigns try value

The sequence continues until a terminal outcome is reached. Repeated simulations generate a distribution over points, time elapsed, and resulting match states.

B.4 Appendix Structure

The individual components are presented in detail in the following validation sections:

- Touch kick model: Appendix B.8
- Lineout success model: Appendix B.11
- Maul model: Appendix B.14
- Phase attack model: Appendix B.17
- Conversion model: Appendix B.20

Taken together, these components form a unified model of attacking play following a kick to the corner.

B.5 Kick to Corner Model

This appendix provides the full specification and validation of the models underlying the kick-to-corner decision. Kicking to the corner initiates a multi-stage attacking sequence, and the value of this option depends on both the probability of securing territory and possession and the likelihood of converting that possession into points.

Each stage of the sequence is modelled separately. The outputs of one stage form the inputs to the next, and the entire process is simulated forward to produce a distribution over attacking outcomes.

The appendix proceeds by stage. For each component, I first present the model specification and then report validation results.

B.6 Stage One: Touchline Bin Model

B.7 Touchline Bin Model

The first stage of the corner model determines where the kick lands. I partition the touchline into discrete bins defined by distance from the defending try line:

$$\mathcal{B} = \{0\text{–}5\text{m},\ 5\text{–}7\text{m},\ 7\text{–}10\text{m},\ 10\text{–}15\text{m},\ 15\text{–}22\text{m},\ 22\text{–}30\text{m},\ 30\text{+m},\ \text{Miss}\}.$$

These bins correspond to tactically distinct match states. Kicks landing in the 0–5m bin yield lineouts close to the try line, while kicks in the 30+m bin provide limited territorial gain. The final category captures kicks that fail to find touch.

B.7.1 Hurdle Structure

I model this process using a two-stage hurdle specification. The first stage estimates the probability that the kick finds touch rather than missing. Conditional on finding touch, the second stage models the distribution over the seven landing bins.

This decomposition separates execution risk from conditional territorial gain. A missed kick produces a structural break in the sequence, typically returning possession to the opposition near the original penalty location. Conditional on success, however, the value of the kick varies smoothly with the territory gained.

B.7.2 Multinomial Specification

Let $\ell = (x, y)$ denote the penalty location and let r index the kicker. The covariates $X(\ell, r)$ include distance to the opposition try line, distance to the near touchline, and kicker-specific random effects capturing variation in range and accuracy.

Conditional on finding touch, the landing distribution is modelled using a multinomial logit:

$$\Pr(b \mid \ell, r) = \frac{\exp(\beta_b^\top X(\ell, r))}{\sum_{b' \in \mathcal{B}} \exp(\beta_{b'}^\top X(\ell, r))}.$$

This specification allows a flexible mapping from location to outcome probabilities without imposing ordering restrictions across bins.

Both stages are estimated in a Bayesian framework. The hierarchical structure allows the model to learn kicker-specific tendencies while shrinking low-observation kickers toward the population mean. At prediction time, posterior draws imply a full probability distribution over touchline outcomes for any penalty location. This distribution forms the first input into the valuation of the corner option.

B.8 Validation

This appendix reports validation evidence for the touch kick model across discrimination, calibration, and out-of-sample generalisation. The validation sample comprises 1,433 penalty kicks executed by 55 kickers.

B.8.1 Stage 1: Finding Touch

Since 87% of kicks find touch, raw accuracy is not informative. I therefore evaluate discrimination using AUC and calibration using the Brier score and expected calibration error (ECE).

The model achieves an AUC of 0.623, indicating moderate discriminative ability. This reflects limited observable variation in miss probability, as location alone does not fully determine execution success. The Brier score is 0.112, corresponding to a Brier Skill Score of 1.9% relative to a marginal-frequency baseline. Calibration is strong, with an ECE of 0.023.

The kicker random effects exhibit a posterior standard deviation of 0.24 on the logit scale, implying meaningful heterogeneity across kickers. On the probability scale, a one-standard-deviation shift moves the baseline touch probability from 87% to approximately 91% for stronger kickers and 81% for weaker ones. The hierarchical structure therefore captures individual variation while stabilising estimates for low-sample players.

B.8.2 Stage 2: Landing-Bin Distribution

Conditional on finding touch, the model exhibits substantially stronger predictive power. The log loss is 1.339, compared with 1.764 for a baseline that predicts marginal bin frequencies, representing a 24.1% improvement. The ranked probability score implies a skill improvement of 39.0% relative to baseline.

Top-1 accuracy is 44.7%, compared with 23.8% for the modal-bin baseline. Top-2 accuracy increases to 71.7%, and 83.5% of predictions fall within one adjacent bin of the realised outcome. Even when the model misses the exact bin, its probability mass is typically concentrated in the correct neighbourhood. This is important for expected value calculations, which integrate over the full predictive distribution.

Figure 23 reports reliability diagrams for each bin. Calibration is generally strong, with predicted probabilities closely tracking observed frequencies. The 22–30m bin shows mild overconfidence at higher probabilities, while extreme bins exhibit wider confidence intervals due to smaller sample sizes.

B.8.3 Out-of-Sample Performance

I assess generalisation using both leave-one-out cross-validation and explicit holdout testing.

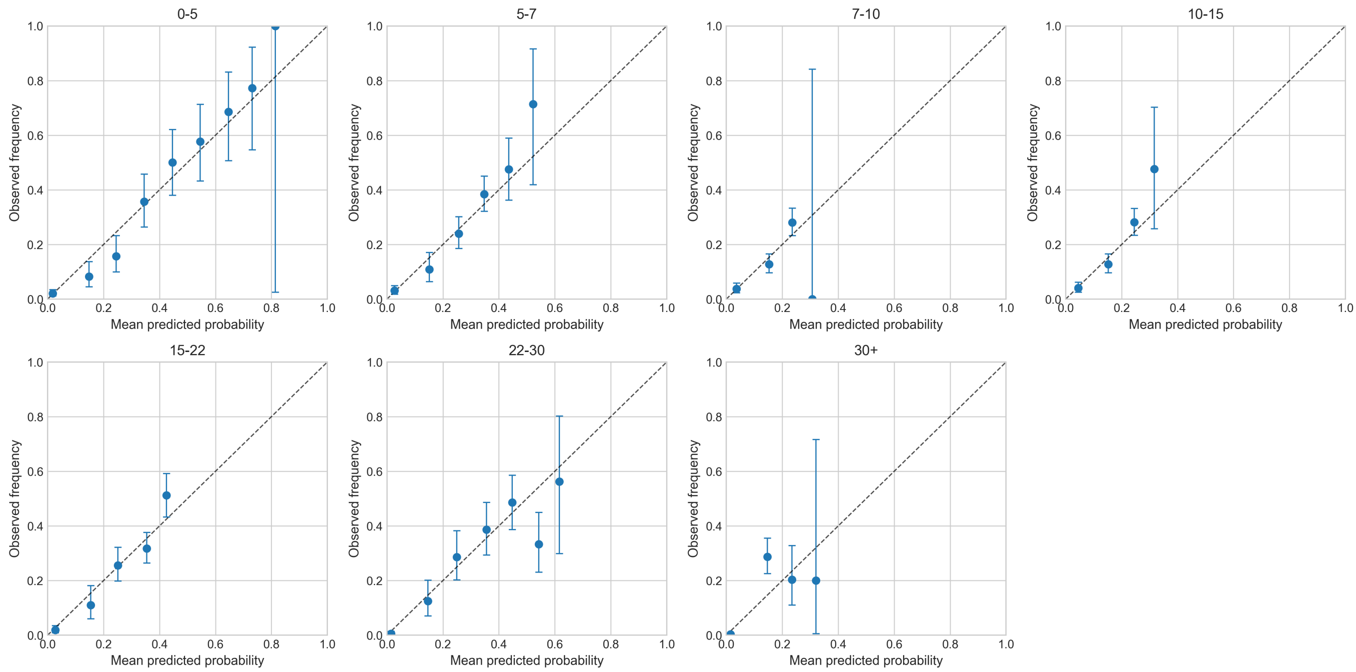

Figure 23: Reliability diagrams for each landing bin. Points show observed frequencies within predicted-probability deciles; error bars indicate 95% binomial confidence intervals. The diagonal line indicates perfect calibration.

The LOO expected log pointwise predictive density is −2322.4, or −1.62 per observation, with an effective number of parameters of $p_{\text{LOO}} = 111.6$. Pareto k diagnostics indicate that the approximation is stable and not driven by a small subset of observations.

For holdout validation, I reserve 15% of kicks (215 observations), estimate the model on the remaining data, and evaluate on the held-out set. The test log loss is 1.503, giving a test-to-train ratio of 0.970. Holdout accuracy is 41.1%, close to the in-sample value of 38.9%. These results indicate limited overfitting and stable out-of-sample performance.

B.8.4 Covariate Effects

Figure 24 shows posterior distributions of key covariate effects. Kicks taken closer to the try line are more likely to land in deeper bins, while kicks from further out concentrate in shallower bins. Distance to the near touchline influences miss probability, reflecting the geometric difficulty of tighter angles. These effects align with intuition and provide face validity for the model.

B.8.5 Summary

The touch kick model is well calibrated and improves materially on naive baselines. Stage 1 shows limited but useful discrimination alongside strong calibration, while Stage 2 delivers substantial gains in predictive accuracy and distributional sharpness. Both LOO and holdout diagnostics indicate that the model generalises well out of sample.

These results support using the model's posterior predictive distributions in the expected value calculations for the corner option. The framework relies on well-calibrated probabilities rather than point predictions, and the validation evidence indicates that these distributions are reliable for that purpose.

B.9 Stage Two: Lineout Success Model

This subsection provides the full specification and validation of the lineout success model described in the methodology. The purpose of this stage is to estimate the probability that the attacking team secures

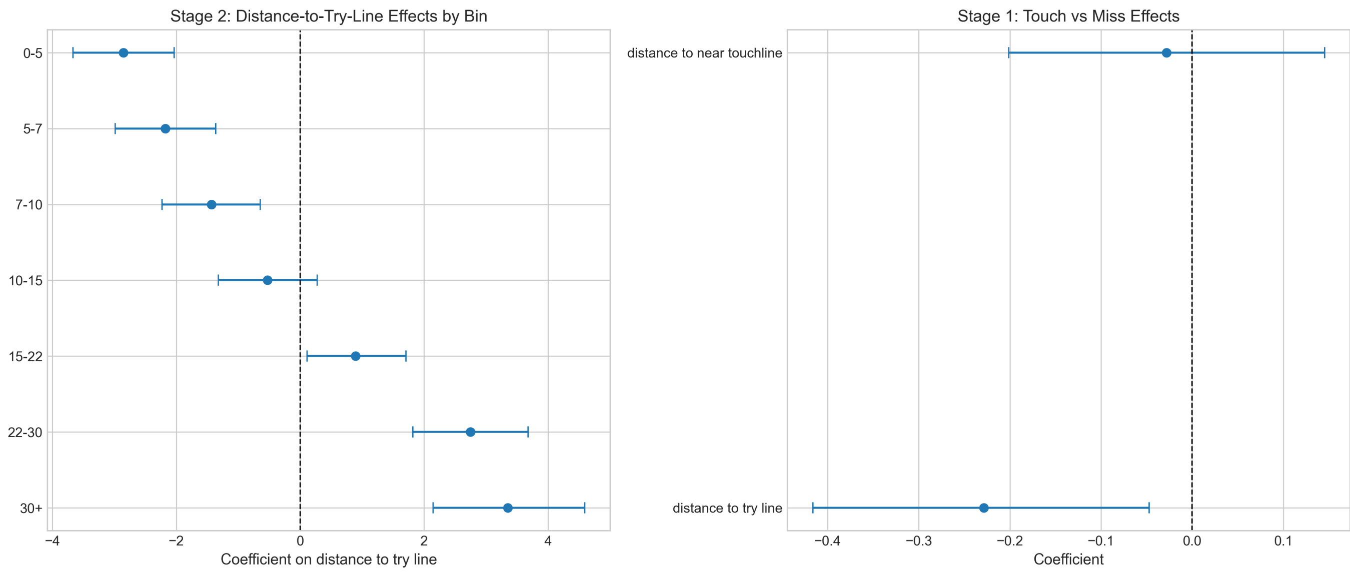

Figure 24: Posterior distributions of covariate effects. Left: effect of distance to try line on landing-bin probabilities. Right: effect of distance to near touchline on touch probability.

possession following a kick to the corner.

Appendix B.10 presents the model specification, including the hierarchical structure used to capture variation in attacking and defensive lineout performance. This paper then evaluates the model using out-of-sample validation, calibration analysis, and posterior predictive checks.

B.10 Lineout Success Model

Once the ball has found touch, play restarts with a lineout at the resulting mark. The key uncertainty at this stage is whether the attacking team secures possession. Although lineout success rates are high on average, variation across team matchups is an important determinant of the value of the corner option.

I model lineout success as a binary outcome using a Bayesian hierarchical logistic regression.

B.10.1 Model Specification

Let i index the attacking team-season and j the defending team-season. The probability that the attacking team wins the lineout is given by

$$\Pr(\text{won} \mid i, j) = \text{logit}^{-1}(\mu + \alpha_i^{\text{att}} - \alpha_j^{\text{def}}),$$

where μ is the baseline success rate, α_i^{att} captures attacking quality in throwing and jumping, and α_j^{def} captures defensive disruption ability. Higher values of α_j^{def} correspond to stronger defensive performance.

If the lineout is lost, possession transfers to the opposition in favourable field position. If it is won, the sequence transitions to an attacking phase at the corresponding location.

B.10.2 Hierarchical Structure

A central challenge is limited data for specific team matchups. Estimating team effects using raw averages leads to unstable estimates, particularly for teams with few observations. I address this using hierarchical shrinkage, modelling team effects as draws from a common population distribution. This pulls low-observation teams toward the league mean and prevents the model from overreacting to noise.

B.10.3 Role in the Decision Framework

The output of this stage is a probability of securing possession conditional on field position and team identities. Since the value of the corner option depends on the continuation of the attacking sequence, this probability acts as a gating factor for all downstream outcomes. Accurate calibration is therefore more important than classification performance.

B.11 Validation

This appendix evaluates the lineout success model using 1,313 observed lineouts involving 55 team-seasons. Performance is assessed through baseline comparisons, out-of-sample validation, calibration, and posterior predictive checks.

B.11.1 Comparison Against Naive Baselines

The observed success rate in the sample is 82.0%. I compare the Bayesian model against several baselines: the overall average, the attacking team's average, the defending team's average, and a combined average.

Table 10: Out-of-sample performance from 5-fold cross-validation.

Model	Brier Score ↓	Log Loss ↓	BSS
Bayesian Model (OOS)	**0.1465**	**0.4673**	**0.9%**
Overall Average	0.1479	0.4722	0.0%
Attacking Team Avg	0.1495	0.4770	−1.1%
Combined Avg	0.1468	0.4681	0.8%

The attacking team average performs poorly out of sample, indicating overfitting to team-specific noise. The Bayesian model achieves the best overall performance, with modest but consistent improvements over the baselines. This reflects the limited but real predictive signal available in the data.

B.11.2 Calibration

Calibration is the primary requirement for the decision framework. Figure 25 compares predicted probabilities with observed success rates.

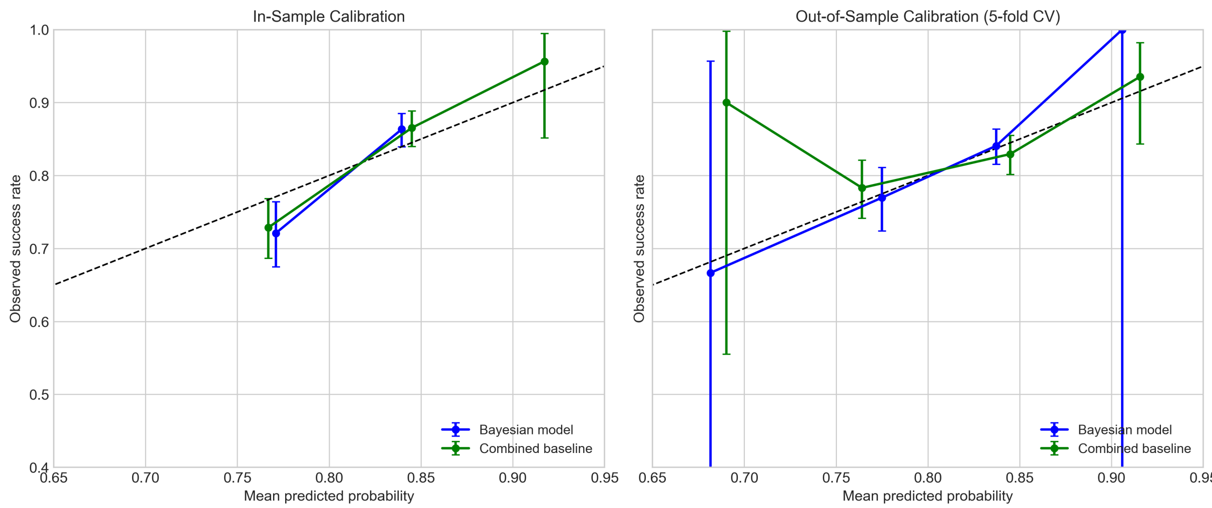

Figure 25: Calibration curve for the lineout success model. Predicted probabilities track observed frequencies across the relevant range.

Predicted probabilities closely track observed frequencies across the range 0.65–0.95, indicating reliable calibration. For example, lineouts predicted at 0.79–0.81 succeed at a rate close to 0.76, while those predicted at 0.87–0.88 succeed at approximately 0.91. The model captures both the ordering and scale of team strengths.

B.11.3 Posterior Predictive Checks

The model reproduces aggregate patterns closely. The observed success rate is 81.95%, while the posterior predictive mean is 81.90%, with a 95% credible interval of [78.98%, 84.62%]. The observed value lies comfortably within this interval.

B.11.4 Team Effects

The estimated team-season effects show meaningful variation, indicating that team identity contributes to lineout outcomes beyond the baseline rate. Figure 26 reports the posterior means and credible intervals.

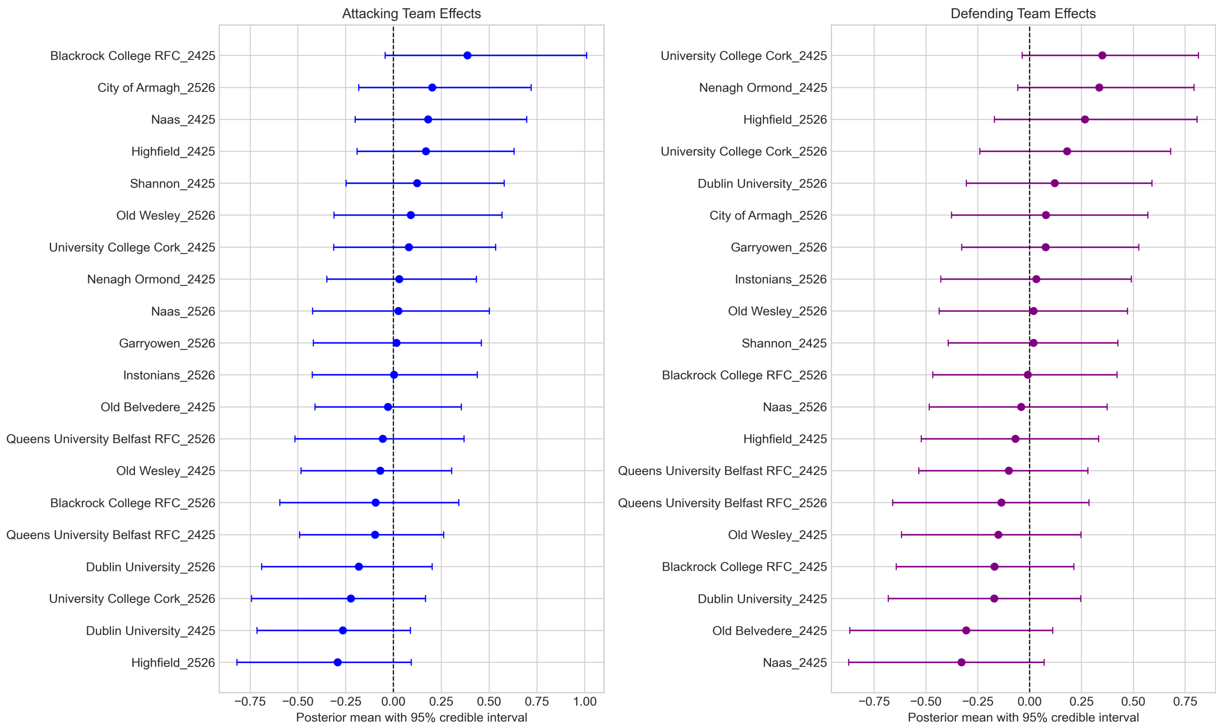

Figure 26: Estimated team-season effects for lineout success. Left: attacking effects; right: defending effects.

Stronger attacking teams exhibit higher success probabilities, while stronger defensive teams reduce opposition success. Estimates for teams with limited observations are shrunk toward zero, reflecting greater uncertainty.

B.11.5 Summary

The lineout success model provides a modest but consistent improvement over naive baselines. Most importantly, it produces well-calibrated probabilities across the relevant range. Given that the decision framework integrates over outcome probabilities rather than relying on point predictions, this level of calibration is sufficient for evaluating the corner option.

B.12 Stage Three: Maul Model

This subsection provides the full specification and validation of the maul model described in the Methodology. Following a successful lineout, the maul is the first attacking state in the corner sequence, and its outputs determine whether the attack ends immediately, continues in structured form, or transitions into open play.

Appendix B.13 presents the model specification, including the multinomial outcome model, the displacement and duration components, and the recursive penalty rule. Appendix B.14 then evaluates the model using predictive performance metrics, calibration analysis, and supporting evidence on team heterogeneity and subsidiary models.

B.13 Maul Model

If the lineout is secured, play typically proceeds through a driving maul. I model the maul as a generative process consisting of an outcome model, conditional displacement and duration models, and a recursive penalty rule.

B.13.1 Outcome Model

From any maul state, four outcomes are possible: *continue*, *try*, *turnover*, or *penalty*. I estimate these probabilities using a multinomial logistic regression:

$$\Pr(m \mid W) = \frac{\exp(\phi_m^\top W)}{\sum_{m'} \exp(\phi_{m'}^\top W)},$$

where W includes a B-spline basis in distance from the try line together with attacking team random effects. The spline basis allows the relationship between field position and outcome probabilities to vary flexibly, which is important for capturing the sharp rise in try probability close to the line.

A penalty here refers to one won by the attacking team. This is treated as a recursive outcome, since the attacking side then faces another tactical choice within the wider decision framework.

B.13.2 Displacement Models

For *continue* outcomes, subsidiary models generate metres gained toward the try line and lateral drift across the pitch. These quantities determine the starting state of the next phase of attack.

Lateral position is modelled explicitly because it affects the location at which any subsequent try is grounded, and therefore the difficulty of the conversion. This matters for expected value calculations, since wider tries carry lower expected value than centrally located ones.

Equivalent location models are also estimated for penalty and try outcomes where the realised end position matters for the continuation of the sequence or for the valuation of points scored.

B.13.3 Duration Models

Separate duration models generate the time elapsed during the maul for each outcome class. This allows match time to evolve consistently within the simulation, which is important when valuing late-game decisions.

The duration component is not intended as a precise forecasting model in its own right. Its role is to propagate realistic time usage through the attacking sequence so that downstream match states are evaluated under plausible clock conditions.

B.13.4 Recursive Penalty Rule

When the maul yields a penalty, the model applies a simple field-position rule. If the penalty is awarded within 10 metres of the try line and 15 metres of the pitch centre, the team taps and continues. Otherwise, the team kicks back to the corner and the sequence restarts from a new lineout.

This rule captures an important feature of attacking play near the try line, where repeated penalties often generate multiple scoring attempts from a single initial decision. The expected value of kicking to the corner therefore depends on the cumulative distribution of outcomes over a potentially recursive sequence rather than on a single isolated event.

B.13.5 Role in the Corner Sequence

The maul model forms one component of the wider corner valuation chain. Following a successful lineout, the simulation proceeds through the maul and, where necessary, into subsequent attacking phases. By simulating forward until a terminal outcome is reached, the framework generates a distribution over points scored, time elapsed, and resulting match states.

B.14 Validation

This appendix reports validation evidence for the maul model. The sample contains 1,608 mauls involving 20 attacking and 20 defending team-seasons. Outcomes are imbalanced: continue (48.9%), turnover (21.7%), penalty (15.8%), and try (13.6%).

B.14.1 Outcome Model

I evaluate the multinomial outcome model using log loss, ranked probability score (RPS), and top-k accuracy. Table 11 reports in-sample and out-of-sample performance.

Table 11: Maul Outcome Model: Predictive Performance

Metric	In-Sample	Baseline	OOS	OOS Baseline
Log loss	1.135	1.245	1.204	1.245
Top-1 accuracy	49.7%	—	48.9%	—
Top-2 accuracy	75.1%	—	72.5%	—
Top-3 accuracy	92.7%	—	90.8%	—
RPS	0.131	0.144	0.139	0.144

The model improves log loss by 8.8% relative to the marginal baseline in-sample and by 3.2% out-of-sample. This is a modest but meaningful gain, given that maul outcomes depend heavily on unobserved factors such as individual matchups, body positions, and referee interpretation. The model captures systematic variation in field position and team strength, while leaving substantial residual uncertainty.

Top-1 accuracy is dominated by the frequency of the continue outcome. More informative are the top-2 and top-3 measures, which indicate that predicted probability mass is concentrated on plausible outcomes even when the modal prediction is not correct. For expected value calculations, this concentration matters more than exact classification.

Figure 27 shows the estimated outcome probabilities as a function of distance from the try line. The left panel reports all four outcomes, while the right panel highlights the try versus turnover comparison, which is the key trade-off for valuing the corner decision. Try probability rises sharply as distance decreases, reaching approximately 25% within five metres of the line. Turnover probability is relatively flat across distance, consistent with turnovers arising more from execution failures than from field position itself.

Table 12 reports the mean predicted probability assigned to the realised outcome, broken down by class.

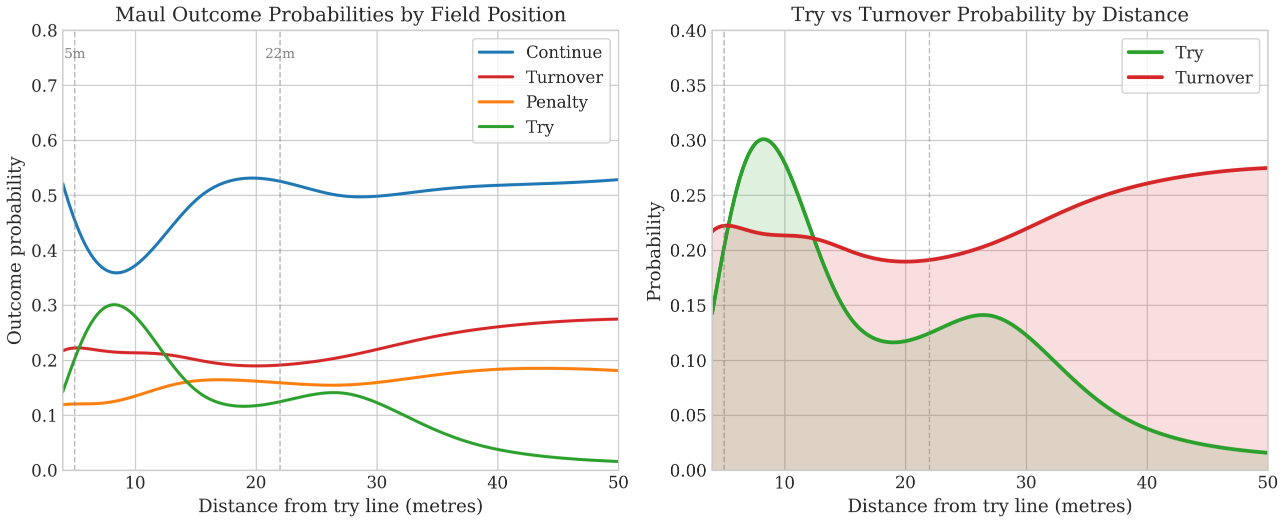

Figure 27: Maul Outcome Probabilities by Distance from Try Line

Notes: Left panel shows predicted probability of each outcome as a function of starting distance. Right panel highlights the try versus turnover comparison. Vertical dashed lines indicate the 5m and 22m marks. Probabilities are population averages with team effects set to zero.

Table 12: Mean Predicted Probability of True Outcome by Class

Outcome	n	Mean Pr(true)
Continue	786	0.519
Try	219	0.255
Turnover	349	0.235
Penalty	254	0.170
Overall	1,608	0.366

The model assigns highest confidence to continue outcomes and lower confidence to minority classes, which is expected given the class imbalance. The overall probability assigned to the true outcome is 0.366, which exceeds the 0.25 implied by uniform predictions and indicates that the model extracts useful signal.

Figure 28 presents calibration diagrams for each outcome class. Continue and turnover outcomes are well calibrated, with observed frequencies tracking predicted probabilities closely. Penalty outcomes show mild miscalibration at intermediate probabilities, while try outcomes exhibit some noise at lower predicted probabilities but align well at higher values where the model is more confident. Overall, the calibration evidence supports using the predicted distributions as inputs into expected value calculations.

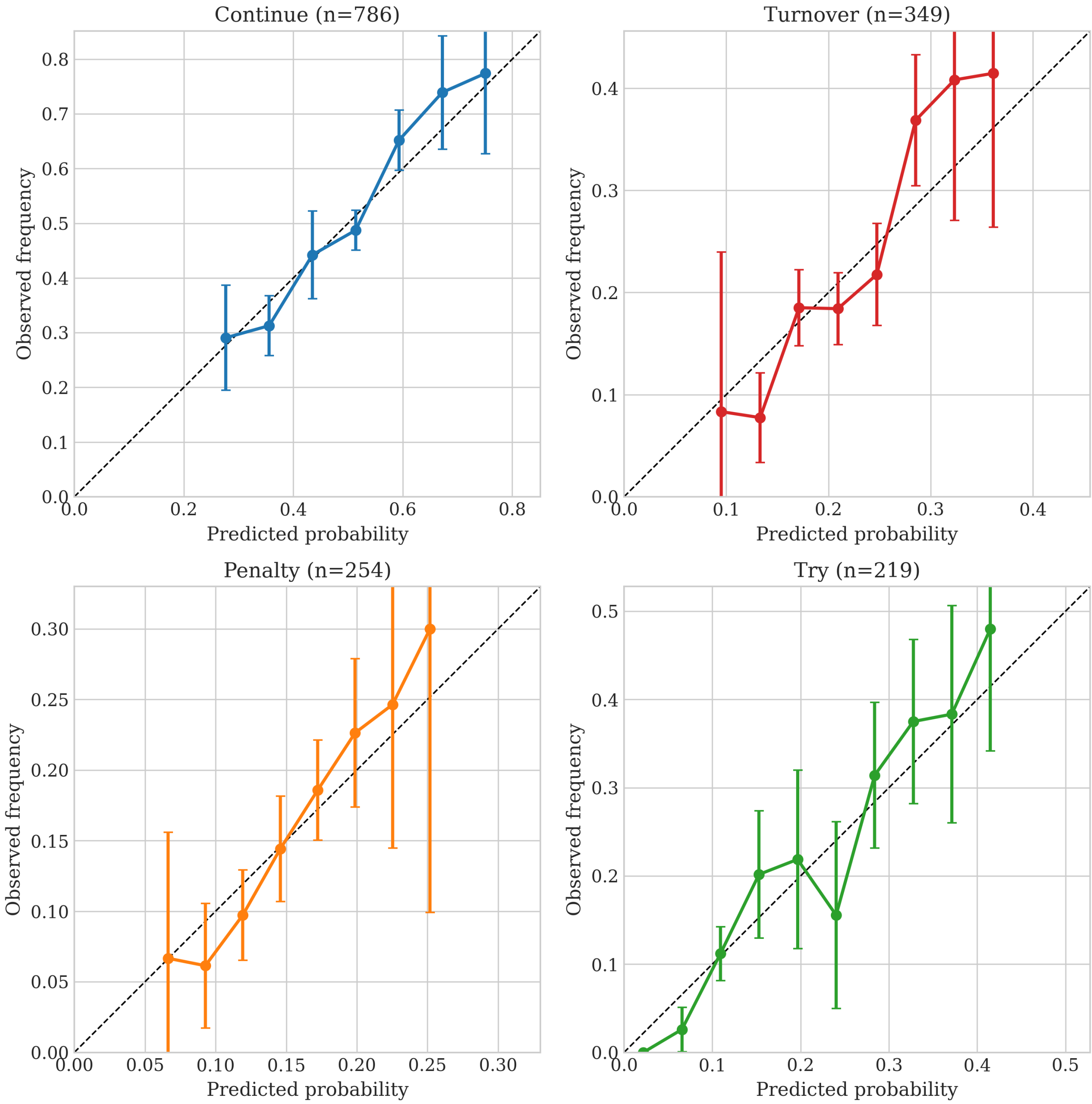

Figure 28: Calibration Diagrams for Maul Outcome Model

Notes: Each panel shows predicted probability against observed frequency for one outcome class. Error bars indicate 95% binomial confidence intervals. The dashed diagonal represents perfect calibration.

B.14.2 Team Heterogeneity

Figure 29 displays the estimated attacking team effects for each non-baseline outcome. Effects are expressed in log-odds relative to the continue baseline, so positive values indicate elevated probability of that outcome.

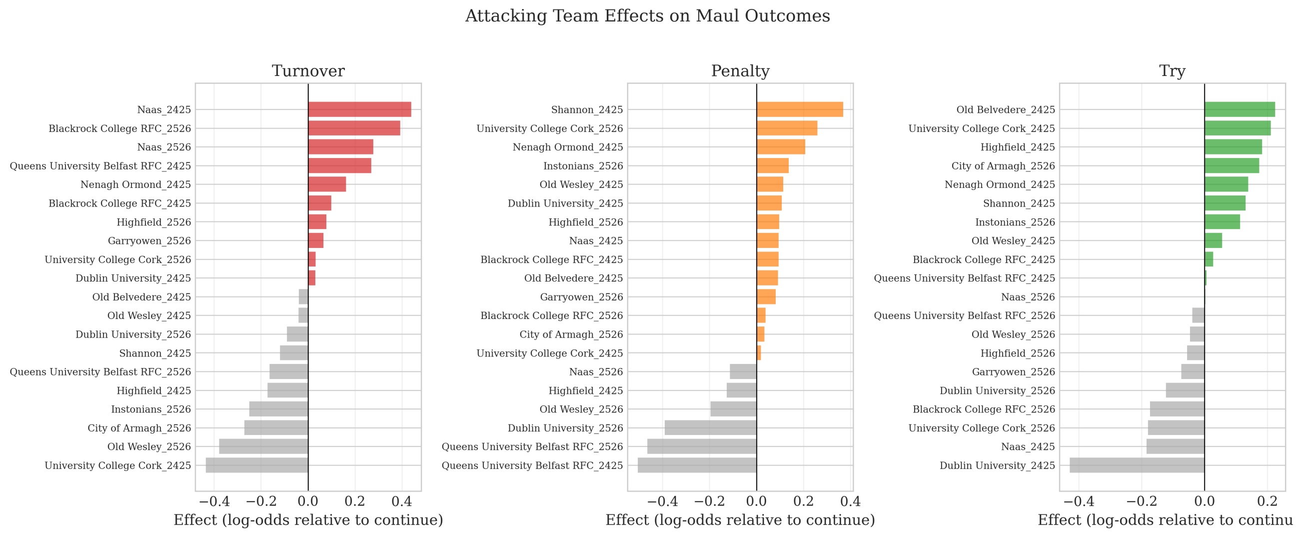

Figure 29: Attacking Team Effects on Maul Outcomes

Notes: Each panel shows team-season effects for one outcome, measured in log-odds relative to the continue baseline. Positive values indicate elevated probability; negative values indicate suppressed probability.

The try effects reveal meaningful heterogeneity across teams. Some sides convert mauls into tries at much higher rates than others, while penalty-winning effects are only weakly positively correlated with try effects. This suggests that effective mauling can manifest either through direct scoring or through the ability to induce infringements.

B.14.3 Displacement Models

The displacement models predict metres gained toward the try line and lateral drift. Table 13 reports in-sample performance.

Table 13: Maul Displacement Models: In-Sample Performance

Outcome	Quantity	n	RMSE	MAE	Correlation
Penalty	delta_X	193	4.51	3.37	0.53
Penalty	abs_delta_Y	193	11.45	11.25	0.31
Continue	delta_X	774	3.69	2.79	0.37
Continue	abs_delta_Y	774	18.36	18.02	0.30
Try	abs_delta_Y	219	24.00	23.25	0.29

The strongest signal appears in the penalty model, where metres gained has correlation 0.53. This is consistent with penalties arising in relatively structured situations, often after the maul has advanced but stalled short of the line. Continue and try models show weaker but still positive correlations, indicating partial predictability.

Lateral drift is less predictable, with correlations around 0.30. This reflects unobserved dynamics within the maul. In the simulation, that uncertainty is handled by integrating over the predictive distribution rather than relying on point forecasts.

B.14.4 Duration Models

Table 14 reports duration model performance.

Duration predictions are weak for most outcomes, with the exception of tries. This is plausible, since tries from mauls tend to occur quickly when the attacking team has momentum and distance to the line

Table 14: Maul Duration Models: Performance

	In-Sample			Out-of-Sample		
Outcome	RMSE	MAE	Corr	RMSE	MAE	Corr
Continue	5.52	4.19	0.13	5.39	4.13	0.12
Turnover	6.22	4.72	0.22	6.19	4.77	0.07
Penalty	6.21	4.96	0.14	6.31	5.12	0.04
Try	5.65	4.29	0.51	5.90	4.52	0.40

is short. For other outcomes, timing depends on factors not fully captured in the data, including referee decisions about when to stop the maul.

This limited predictive power is not especially problematic. Duration enters the framework only through elapsed match time, and errors of a few seconds have little effect on the subsequent match simulation.

B.14.5 Summary

The maul model captures systematic variation in field position and team strength while leaving substantial residual uncertainty. The outcome model improves on baseline and produces calibrated probability distributions with sensible concentration of mass on plausible outcomes. The outcome curves confirm that try probability rises sharply near the line, which is the gradient that drives the value of the corner option. Team effects reveal meaningful heterogeneity in maul effectiveness. Displacement models show moderate predictive power, while duration models are weaker but adequate for match-time accounting.

For the decision framework, the key requirement is well-calibrated distributions rather than precise prediction of individual events. The validation evidence supports using the maul model to value attacking sequences following a kick to the corner.

B.15 Stage Four: Green Zone Phase Attack Model

This subsection provides the full specification and validation of the green zone phase attack model described in Section 5.2. Following a non-terminal maul, this stage models the remainder of the attacking sequence as a discrete-time state process over open phases.

Appendix B.16 presents the model specification, including the phase transition model, the subsidiary generative components, and the treatment of recursive penalties. Appendix B.17 then evaluates the model in terms of sequence realism, spatial accuracy, duration dynamics, and calibration.

B.16 Green Zone Phase Attack Model

If the maul does not produce a terminal outcome, play transitions to open phases inside the attacking zone. I model this as a discrete-time state process, where each phase is characterised by field position, phase number, recent timing, measures of attacking momentum, and team identities.

B.16.1 State Transition Model

Let X_{phase} denote the current state vector. From any state, four outcomes are possible: the attack continues to another phase (*continue*), the attacking team scores a *try*, possession is *turned over*, or the attacking team wins a *penalty*. Transition probabilities are specified using a multinomial logistic model:

$$\Pr(q \mid X_{\text{phase}}) = \frac{\exp(\delta_q^\top X_{\text{phase}})}{\sum_{q'} \exp(\delta_{q'}^\top X_{\text{phase}})}.$$

The state vector includes distance from the try line, lateral position, phase number, time since the previous ruck, attacking momentum measures, and team identifiers. This allows transition probabilities to respond both to location and to the evolving structure of the attack.

B.16.2 Subsidiary Generative Models

Conditional on a continue outcome, subsidiary models generate metres gained toward the try line, lateral movement, and phase duration. These components determine the state of the next phase and allow attacking sequences to evolve realistically over time.

Lateral position plays a dual role. It captures how attacks shift the point of pressure across the pitch, and it determines the location of any resulting try. This matters because try location directly affects conversion difficulty and therefore the expected value of the attacking sequence.

Equivalent subsidiary models are also used where necessary for terminal outcomes, particularly when try location or the location of a penalty won matters for downstream valuation.

B.16.3 Termination and Recursion

The phase sequence is simulated forward until a terminal outcome is reached. A try ends the attacking sequence and passes control to the conversion model. A turnover transfers possession to the opposition. A penalty won is treated as a recursive event within the wider corner framework, since the attacking team then faces another tactical choice.

This recursive structure is important. The value of kicking to the corner depends not only on the outcome of the initial lineout and maul, but on the full distribution of repeated attacking opportunities that may arise from penalties won during sustained pressure.

B.16.4 Role in the Corner Sequence

The phase model is the final attacking component of the kick-to-corner sequence. Following a successful lineout and a non-terminal maul, it simulates the remaining attack until a try, turnover, penalty, or end-of-half censoring occurs. The resulting distribution over outcomes feeds directly into the valuation of the corner option.

B.17 Validation

This appendix presents validation evidence for the green zone phase attack model. The sample contains 2,204 phases from 352 attacking sequences. Outcomes are heavily imbalanced, with continue accounting for 84.7% of observations, followed by try (6.6%), turnover (6.4%), and penalty won (2.4%).

B.17.1 Simulation Context

The phase model is used as a generative component within the simulation rather than as a classifier. At each phase, an outcome is drawn from the predicted probability distribution. The relevant validation criteria are therefore whether simulated sequences terminate in realistic proportions, whether try locations are accurately represented, and whether duration dynamics are plausible.

B.17.2 Sequence-Level Outcomes

Table 15 reports the distribution of terminal outcomes across observed attacking sequences.

Attacking sequences in the green zone terminate in tries and turnovers at similar rates. This reflects the high-variance nature of attacking play close to the try line.

Figure 30 shows the empirical survival curve. The median sequence length is 6 phases, and most sequences terminate within 15 phases.

Table 15: Terminal Outcome Distribution

Outcome	Proportion
Try	41.2%
Turnover	39.8%
Penalty won	15.1%
Continue (censored)	4.0%

Notes: Based on 352 attacking sequences. Continue indicates sequences censored at the end of a half.

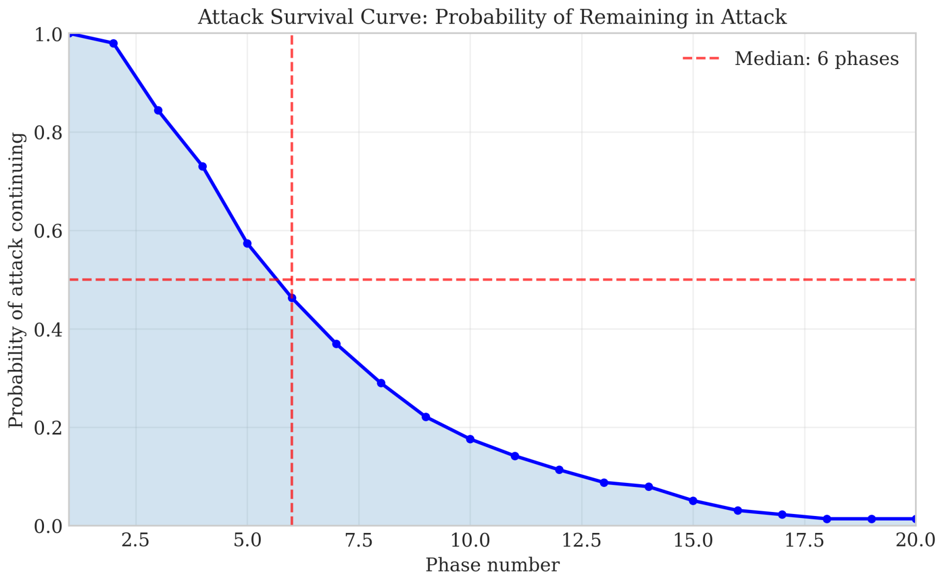

Figure 30: Attack Survival Curve

Notes: Probability that an attacking sequence continues beyond phase k. Based on 352 sequences.

B.17.3 Event Probabilities by Field Position

Figure 31 shows event probabilities as a function of distance from the try line. Try probability is highest close to the line and declines with distance. Turnover and penalty probabilities are relatively flat.

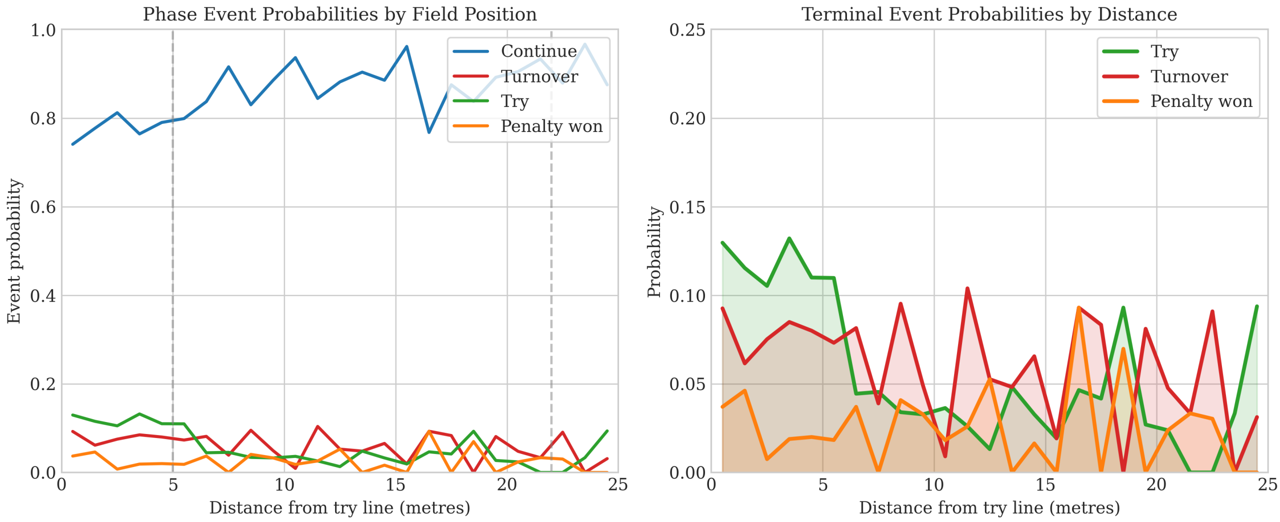

Figure 31: Phase Event Probabilities by Distance from Try Line
Notes: Probabilities computed from empirical frequencies in 1-metre bins. Vertical lines indicate the 5m and 22m marks.

The increase in try probability close to the line is consistent with attacking dynamics in this area of the pitch and is the main driver of value in sustained attacking sequences.

B.17.4 Generative Model Performance

Table 16 reports performance of the subsidiary models.

Table 16: Generative Model Performance

Event	Quantity	n	RMSE	MAE	Correlation
Continue	dx	1,866	2.16	1.56	0.57
Continue	dy	1,866	3.14	2.23	0.26
Continue	ruck time	1,866	1.77	1.20	0.30
Continue	phase duration	1,866	6.32	1.48	0.13
Try	y_end	145	4.12	3.02	0.98
Try	phase duration	145	16.65	4.27	0.25
Penalty won	dx	53	0.94	0.69	0.91
Penalty won	dy	53	0.43	0.31	0.91
Turnover	phase duration	140	16.16	4.17	0.39

The try location model shows very high accuracy, which is important because try location determines conversion difficulty. The metres-gained model captures meaningful variation in attacking progress, while lateral movement is less predictable.

Duration predictions are weaker overall. This reflects the difficulty of predicting timing from state variables alone. For simulation purposes, it is sufficient that durations are realistic on average.

B.17.5 Calibration

Figure 32 presents calibration diagrams for each event type. Continue events are well calibrated, while terminal events show wider uncertainty due to smaller sample sizes. There is no clear pattern of systematic over- or under-confidence.

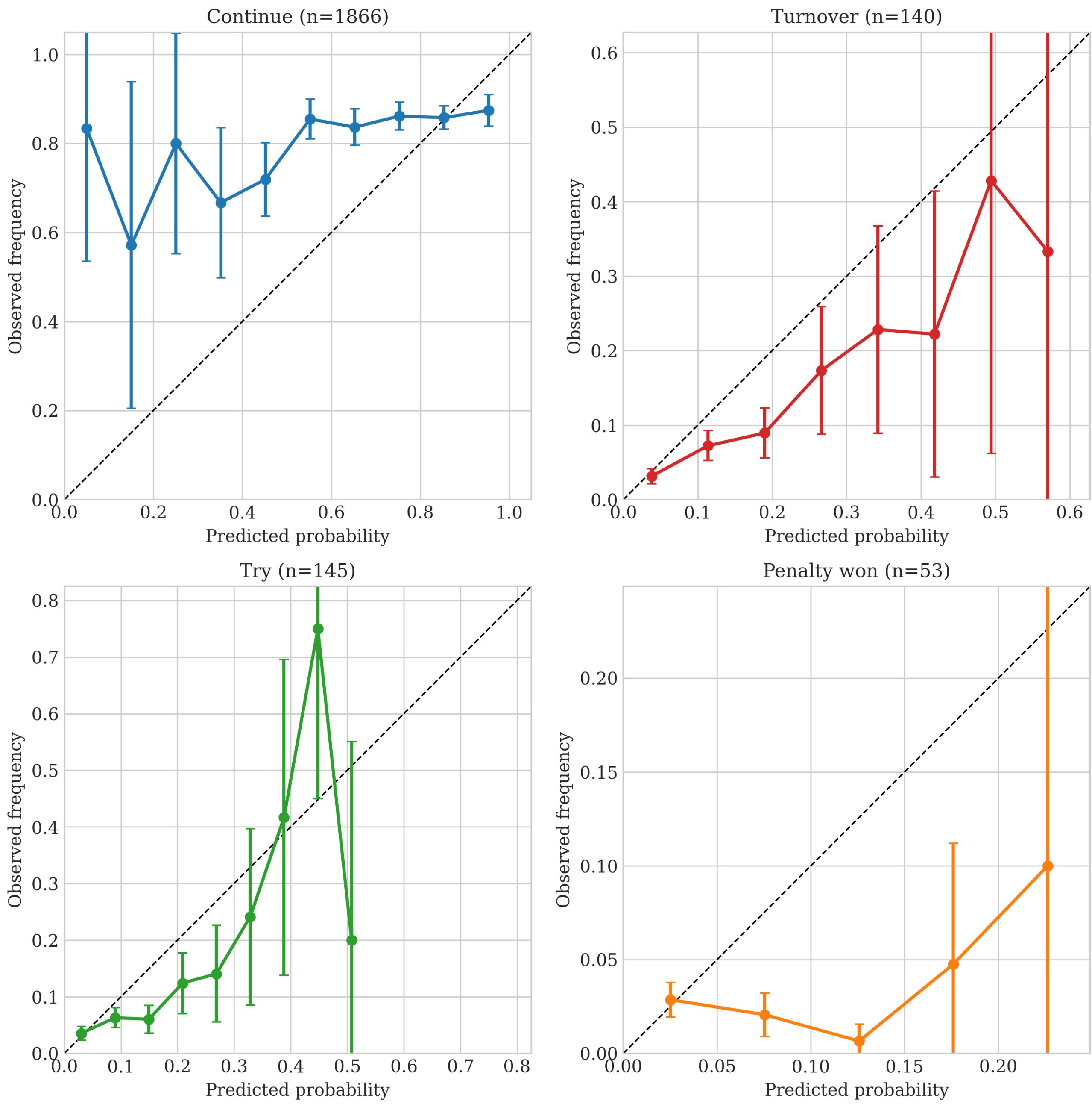

Figure 32: Calibration Diagrams for Phase Event Model
Notes: Predicted probabilities against observed frequencies for each event type. The dashed line indicates perfect calibration.

B.17.6 Summary

The phase attack model is designed to generate realistic attacking sequences rather than to predict individual events. The validation evidence shows that sequences terminate in plausible proportions, try locations are accurately represented, and duration dynamics are reasonable. The predicted probabilities are well calibrated at the aggregate level, which is sufficient for expected value calculations.

B.18 Conversion Success Model

This subsection provides the full specification and validation of the conversion model used within the kick-to-corner framework. When a try is scored, the expected value of that try depends on the probability of the subsequent conversion, which in turn depends on where the try is grounded and who takes the kick.

Appendix B.19 presents the model specification, including the treatment of kick geometry, wind effects, and kicker heterogeneity. Appendix B.20 then evaluates the model in terms of predictive performance, calibration, and the implied gradients in success probability across try locations.

B.19 Conversion Success Model

When a try is scored, a conversion attempt follows. Because the attacking models track the location where the try is grounded, conversion probability can be conditioned on kick difficulty rather than approximated using a team-level average.

I estimate a Bayesian logistic regression using only observed conversion attempts. This is important because conversions are always taken, regardless of difficulty, whereas penalty kicks are attempted selectively. Restricting the sample to conversions therefore avoids the selection problem that arises in penalty-kick data.

B.19.1 Model Specification

Let ℓ_{try} denote the location where the try is grounded, let w collect wind-related covariates, and let r index the kicker. The probability of a successful conversion is modelled as

$$\Pr(\text{success} \mid \ell_{\text{try}}, w, r) = \text{logit}^{-1}\left(\beta_0 + f_{\text{dist}}(d) + f_{\text{ang}}(\theta) + g(d, w) + u_r\right),$$

where d is the distance to the posts, θ is the kick angle, $f_{\text{dist}}(\cdot)$ and $f_{\text{ang}}(\cdot)$ are B-spline terms, $g(d, w)$ captures interactions between wind and kick geometry, and

$$u_r \sim \mathcal{N}(0, \sigma_u^2)$$

is a kicker-specific random effect.

The spline terms allow the relationship between success probability and kick geometry to vary flexibly, while the random effects capture persistent differences in kicker ability.

B.19.2 Kick Placement

Following a try, the kicking team may choose the placement of the ball along the line extending back from the grounding point. This creates a trade-off between distance and angle. For wide tries, stepping back reduces the angle but increases the distance of the kick.

I model this choice by selecting the feasible placement that maximises predicted conversion probability. This ensures that the value assigned to a try reflects both its grounding location and the kicker's ability to optimise the attempt.

B.19.3 Role in the Corner Sequence

The conversion model is used within the simulation to assign the value of a try conditional on where it is scored. This is important because tries grounded near the touchline have lower expected value than centrally located tries, even before accounting for the remainder of the match. By conditioning on try location, the model links the spatial dynamics of attacking play to the expected points generated by the sequence.

B.20 Validation

This appendix presents validation results for the conversion model. The sample contains 1,031 conversion attempts from 59 kickers, with an overall success rate of 71.0%.

Table 17: Conversion Model Performance

Metric	Model	Overall Mean	Kicker Mean
Brier score (IS)	0.155	0.206	0.188
Brier score (OOS)	0.155	0.207	0.217

B.20.1 Predictive Performance

Table 17 reports in-sample and out-of-sample predictive performance.

The model improves substantially on both baselines. Relative to the overall mean, the out-of-sample Brier score improves by roughly 25%. It also outperforms the kicker-specific baseline, indicating that distance, angle, and wind contribute predictive information beyond persistent kicker ability alone.

B.20.2 Calibration

Figure 33 shows calibration across probability bins. Predicted probabilities track observed success rates closely, with an expected calibration error of approximately 0.026.

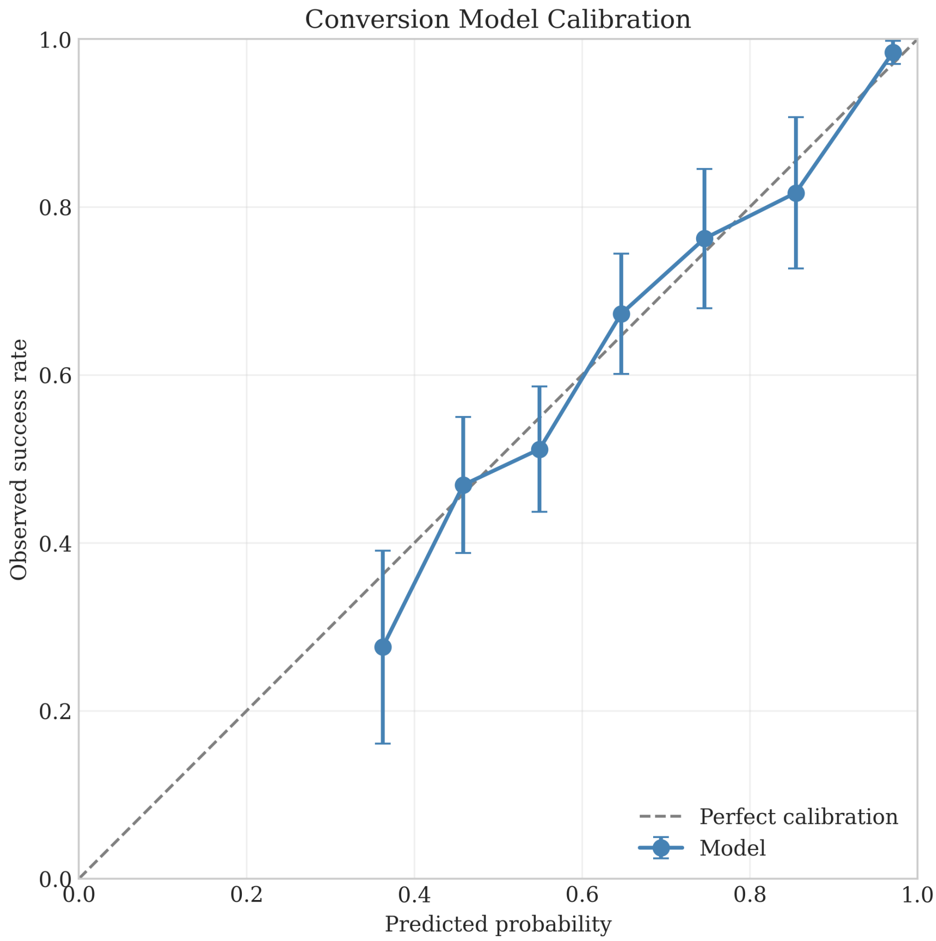

Figure 33: Calibration of Conversion Model

Notes: Observed success rates against predicted probabilities by bin. Error bars show binomial uncertainty.

Calibration matters more than exact classification here, since the model is used to assign expected value to tries scored at different locations.

B.20.3 Distance and Angle Effects

Figure 34 shows predicted success probability as a function of distance for central and wide kicks. Success declines with distance, and the decline is steeper for wider angles.

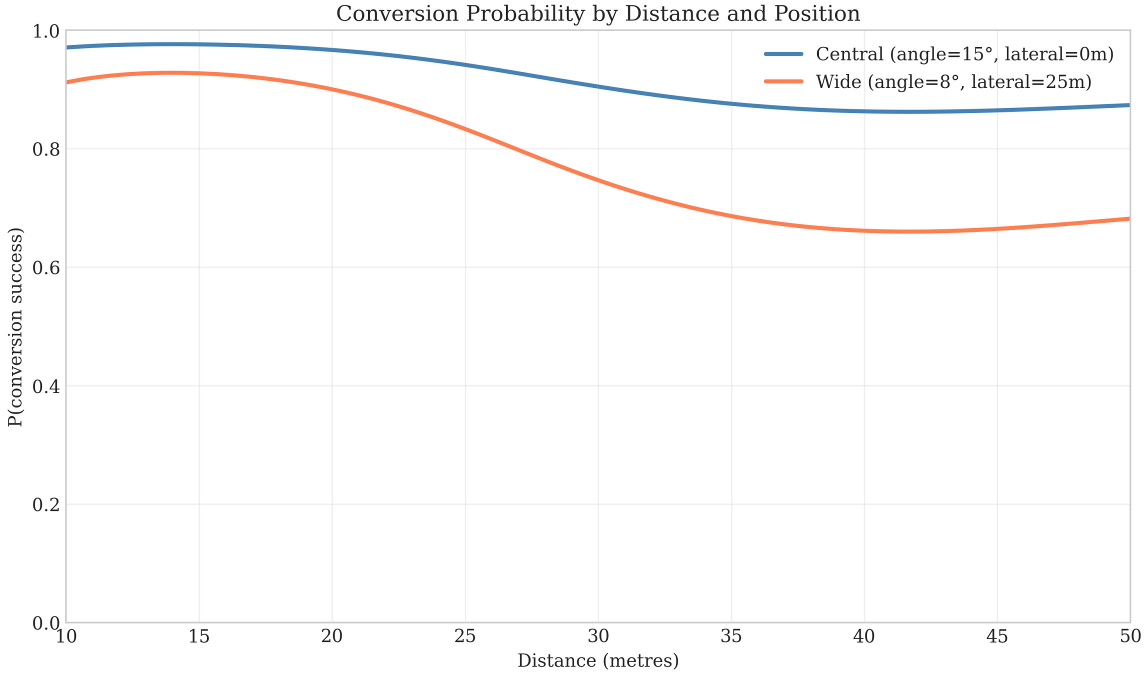

Figure 34: Conversion Probability by Distance and Angle

Figure 35 reports success probability as a function of try location after optimising kick placement. Success is highest for centrally located tries and declines toward the touchlines.

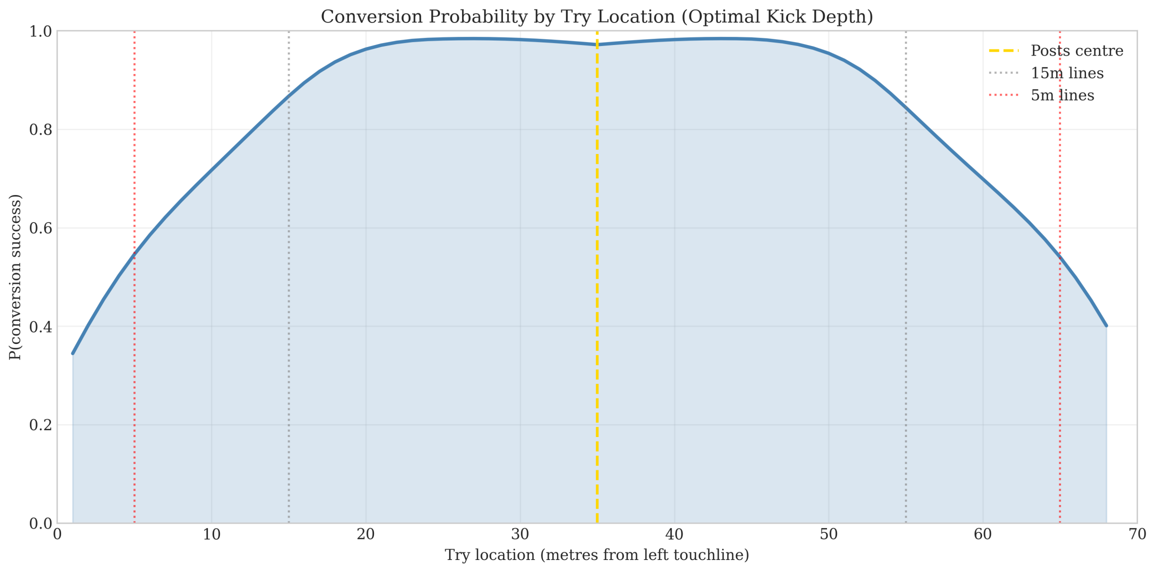

Figure 35: Conversion Probability by Try Location

These gradients are consistent with the geometry of goal kicking and confirm that the model captures the relationship between try location and conversion difficulty.

B.20.4 Kicker Effects

Figure 36 shows the estimated kicker random effects. There is substantial variation across players, with stronger kickers outperforming the baseline by a meaningful margin and weaker kickers falling below it.

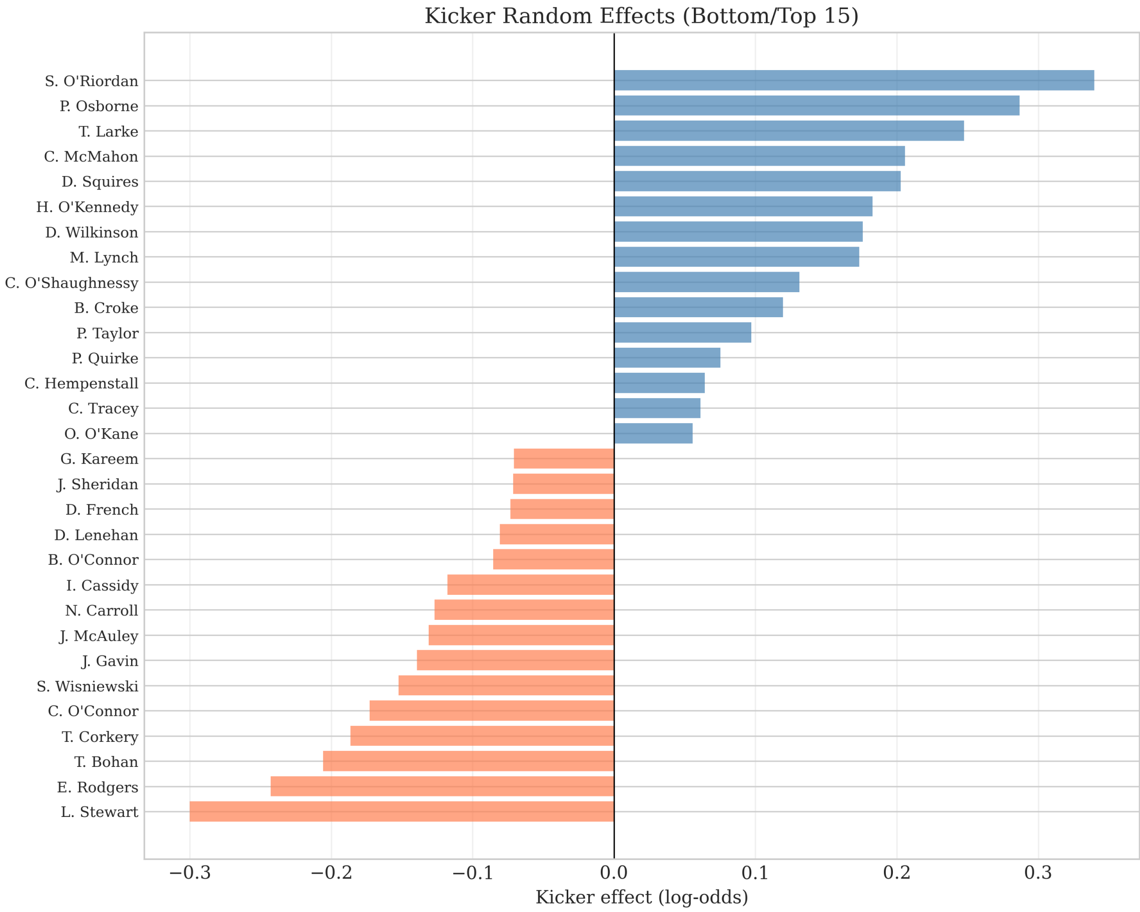

Figure 36: Kicker Random Effects

This heterogeneity matters for expected value calculations, since the value of a wide try depends not only on geometry but also on who is taking the kick.

B.20.5 Summary

The conversion model captures the main determinants of success: distance, angle, and kicker ability. Predicted probabilities are well calibrated and exhibit sensible gradients with respect to field position. This ensures that the value of a try is adjusted appropriately for its location within the simulation.

C Kick at Goal Model

This appendix provides the full technical details for the kick-at-goal model introduced in Section 5.3. The purpose of the model is to estimate the unconditional probability that a penalty kick would be successful if taken. This is not straightforward, because many kickable penalties are never attempted, and the observed sample of kicks is therefore selected. The model corrects for this informative missingness by jointly modelling kick success and the decision to attempt the kick.

The appendix proceeds in two parts. Appendix C.1 presents the full model specification, including the geometric covariates, the hierarchical probit success equation, the selection equation, and the estimation strategy. Appendix C.2 then reports validation evidence, comparing the informative missingness model with a naive alternative estimated on attempted kicks only.

C.1 Model Specification

This appendix sets out the full specification of the kick-at-goal model. The aim is to estimate the unconditional probability that a penalty kick would be successful if taken, while correcting for the fact that many kickable penalties are never attempted. Since teams are more likely to shoot when a kick is favourable, the observed sample of attempts is selected. A model estimated on attempted kicks alone would therefore overstate success probabilities, especially for kicks near the tactical margin.

I address this using a hierarchical probit model with informative missingness, adapted from Lock2014 to the rugby penalty context. The model contains two linked components: a success equation for whether the kick would go over, and a selection equation for whether the team chooses to attempt it.

C.1.1 Geometry and Covariates

Let $\ell = (x, y)$ denote the penalty location, where x is the distance upfield from the kicking team's try line and y is the lateral position measured from the left touchline. The opposition posts are located at $x = 100$, with uprights at $y_L = 31.2$ and $y_R = 36.8$ metres.

The distance to the try line is

$$d(\ell) = 100 - x.$$

The angle subtended by the posts is

$$\theta(\ell) = \left| \arctan\left(\frac{y_R - y}{d(\ell)}\right) - \arctan\left(\frac{y_L - y}{d(\ell)}\right) \right|.$$

The fixed-effect covariates are constructed from transformed versions of distance and angle:

$$\log d, \qquad \theta, \qquad \theta^2, \qquad (\log d)\theta.$$

These are standardised to zero mean and unit variance before estimation. The logarithmic distance term reflects diminishing marginal difficulty with range, while the interaction term allows the effect of distance to vary with angle.

C.1.2 Success Equation

Let $Y_i \in \{0, 1\}$ indicate whether kick i is successful. The latent success index is

$$U_i = X_i^\top \beta + Z_i^\top v_{r(i)} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, 1), \tag{1}$$

with

$$Y_i = \mathbf{1}\{U_i > 0\}. \tag{2}$$

Here X_i contains the fixed-effect covariates, Z_i contains the subset entering the random effects, and $r(i)$ indexes the kicker for observation i. Kicker-specific effects are modelled as

$$v_r \overset{iid}{\sim} \mathcal{N}(0, \Sigma_v), \tag{3}$$

which induces partial pooling across kickers. This is important because some players have many observed kicks while others have relatively few.

C.1.3 Selection Equation

Let $A_i \in \{0, 1\}$ indicate whether kick i is attempted. For penalties where the tactical breakeven threshold is defined, the attempt decision is modelled through a latent selection index:

$$M_i^* = \alpha_0 + \alpha_1 U_i + \alpha_2 q_i^* + \nu_i, \qquad \nu_i \sim \mathcal{N}(0, 1), \tag{4}$$

with

$$A_i = \mathbf{1}\{M_i^* > 0\}. \tag{5}$$

The term q_i^* is a transformed version of the tactical breakeven success probability:

$$q_i^* = \Phi^{-1}\left(\text{clip}(p_i^*, 0.01, 0.99)\right), \tag{6}$$

where

$$p_i^* = \frac{V(\text{corner}_i) - V(\text{miss}_i)}{V(\text{make}_i) - V(\text{miss}_i)}. \tag{7}$$

This is the success probability at which kicking at goal and kicking to the corner have equal expected value. The dependence of M_i^* on U_i is what generates informative missingness. The attempt decision reveals information about latent kick quality that is not fully captured by the observed geometry.

C.1.4 Data Structure

The data are organised into three tiers:

1. **Tier 1:** Kicks at goal with observed outcomes but no defined tactical threshold. These inform the success equation only.
2. **Tier 2:** Penalties with defined p_i^*, including both attempted kicks ($A_i = 1$, Y_i observed) and foregone kicks ($A_i = 0$, Y_i unobserved). These inform the selection equation, and attempted kicks also inform the success equation.
3. **Tier 3:** Additional kicks at goal with observed outcomes but no defined threshold. These inform the success equation only.

The selection correction operates through Tier 2. Foregone kicks reveal that teams preferred the corner despite the kick's observed geometry, which provides information about latent difficulty beyond the observed covariates alone.

C.1.5 Likelihood and Priors

Conditional on the latent variables, the complete-data likelihood is

$$L = \prod_{i=1}^{n} \phi\left(U_i - X_i^\top \beta - Z_i^\top v_{r(i)}\right) \times \prod_{i \in T_2} \phi\left(M_i^* - \alpha_0 - \alpha_1 U_i - \alpha_2 q_i^*\right) \times \prod_{r=1}^{R} \phi_3(v_r; 0, \Sigma_v), \tag{8}$$

where $\phi(\cdot)$ is the standard normal density and T_2 indexes Tier 2 observations.

I use weakly informative priors:

$$\beta \sim \mathcal{N}(0, 16 I_5), \qquad \alpha \sim \mathcal{N}(0, 4 I_3), \qquad \Sigma_v^{-1} \sim W(5, 0.5 I_3). \tag{9}$$

These priors are diffuse enough to let the data dominate, while still providing regularisation for the hierarchical structure.

C.1.6 Estimation

The posterior is sampled via Gibbs sampling with data augmentation. Each iteration updates the latent variables, fixed effects, selection parameters, kicker effects, and covariance matrix from their respective full conditional distributions. Convergence is assessed using trace plots and effective sample sizes.

The informative missingness model is estimated jointly on all relevant observations. For comparison, I also estimate a naive hierarchical probit model on attempted kicks only. That benchmark ignores the attempt decision entirely and therefore treats the observed sample as representative.

C.1.7 Prediction and Use in the Decision Framework

For a kick at location ℓ by kicker r, posterior draws $\{(\beta^{(s)}, v_r^{(s)})\}_{s=1}^{S}$ imply

$$p^{(s)}(\ell, r) = \Phi\Big(X(\ell)^\top \beta^{(s)} + Z(\ell)^\top v_r^{(s)}\Big). \tag{10}$$

The posterior mean provides a point estimate of success probability, while posterior quantiles provide credible intervals.

Within the decision framework, the key quantity is not just the expected success probability itself, but whether that probability exceeds the tactical breakeven threshold. This is summarised by

$$\Pr(p(\ell, r) > p^* \mid \text{data}) \approx \frac{1}{S} \sum_{s=1}^{S} \mathbf{1}\{p^{(s)}(\ell, r) > p^*\}. \tag{11}$$

This probability is used directly in the decision framework to classify states as favouring a kick at goal, favouring a kick to the corner, or remaining too uncertain to call decisively.

C.2 Validation

The kick-at-goal model translates penalty locations into success probabilities while accounting for the selection bias induced by teams' strategic choice between kicking at goal and kicking to the corner. Before using it in the decision framework, I compare the informative missingness model against a naive hierarchical probit estimated on attempted kicks only.

C.2.1 Comparison Framework

The sample comprises 2,013 kicks after removing 10 angle outliers. Of these, 1,382 were attempted and 631 were foregone. The attempted kicks include 1,291 from the 33 kickers with at least 10 attempts, yielding a realised success rate of 73.2% among observed kicks.

The central empirical issue is that attempted kicks are a selected sample. If teams disproportionately attempt kicks they expect to make, a model trained only on those observations will learn the relationship between geometry and success conditional on attempt, rather than the unconditional relationship required for counterfactual evaluation. The informative missingness model addresses this by jointly modelling the attempt decision and the success probability.

C.2.2 In-Sample and Out-of-Sample Performance

Both models are estimated using MCMC. For validation, I compare them on the 1,200 kicks from kickers common to both models. Table 18 reports in-sample performance.

Table 18: In-Sample Performance: Kick at Goal Models

Metric	Selection-Corrected	Naive
Brier Score	0.1527	0.1515
Log Loss	0.4566	0.4510
Accuracy	76.3%	76.2%
AUC-ROC	0.8038	0.8062
ECE	0.0239	0.0269

The naive model fits the observed attempted sample slightly better in-sample, which is unsurprising because it is estimated directly on that selected sample. The more relevant question is whether this apparent advantage survives out-of-sample evaluation.

Table 19: Out-of-Sample Performance: 5-Fold Cross-Validation

Metric	Selection-Corrected	Naive
Brier Score	0.1527	0.1620
Log Loss	0.4566	0.4799
Accuracy	76.3%	74.8%
AUC-ROC	0.8038	0.7757
ECE	0.0239	0.0247

I assess generalisation using 5-fold cross-validation, partitioning kicks so that all kicks from the same match remain in the same fold. Table 19 reports out-of-sample performance.

Out of sample, the pattern reverses. The selection-corrected model improves Brier score by 5.7%, improves log loss, and delivers higher discrimination. This indicates that the naive model's in-sample fit does not carry over once the model is evaluated away from the selected sample on which it was trained.

C.2.3 Generalisation Gap

A useful summary diagnostic is the change in Brier score from in-sample to out-of-sample evaluation. Table 20 reports this generalisation gap.

Table 20: Generalisation Gap: In-Sample to Out-of-Sample

Model	Brier (IS)	Brier (OOS)	Change
Selection-Corrected	0.1527	0.1527	+0.0000
Naive	0.1515	0.1620	+0.0105

The informative missingness model shows no deterioration in Brier score, while the naive model worsens materially out of sample. This is consistent with the naive specification overfitting to the selected sample of attempted kicks.

C.2.4 Performance by Difficulty

If the benefit of the informative missingness model comes from correcting selection bias, the gains should be largest where selection is strongest, namely among difficult kicks. Table 21 reports out-of-sample Brier scores by difficulty stratum.

Table 21: Out-of-Sample Brier Score by Kick Difficulty

Stratum	n	Selection-Corrected	Naive
Hard ($p \leq 0.50$)	203	0.2336	0.2534
Medium ($0.50 < p \leq 0.75$)	419	0.2260	0.2397
Easy ($p > 0.75$)	578	0.0712	0.0736

The improvement is largest in the hard-kick regime, smaller for medium kicks, and smallest for easy kicks. This is exactly the pattern one would expect if the value of the model comes from correcting the bias introduced by selective attempts.

C.2.5 Calibration

Figure 37 plots calibration curves for the two models. Both are reasonably well calibrated, with expected calibration errors close to 0.02. The informative missingness model performs slightly better in the middle-probability range, where selection effects are most relevant.

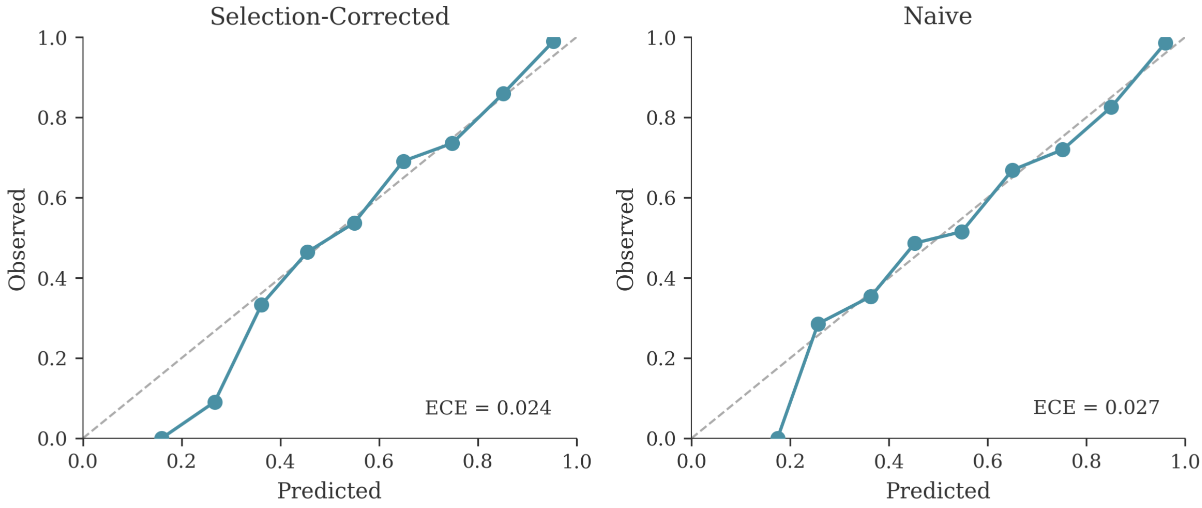

Figure 37: Calibration Curves: Kick at Goal Models

Notes: Predicted probabilities are grouped into bins and compared with realised success frequencies. The dashed line indicates perfect calibration. Based on out-of-sample predictions from 5-fold cross-validation.

C.2.6 Non-Attempted Kicks

A direct implication of informative missingness is that non-attempted kicks should be recognised as systematically harder than their observed geometry alone suggests. For the 416 non-attempted kicks from kickers common to both models, the informative missingness model predicts a lower success probability than the naive model in 59.4% of cases. Mean predicted success is 0.570 under the informative missingness model and 0.577 under the naive model.

The difference in means is modest, but the direction is informative. The selection-corrected model learns from the team's revealed preference to decline the kick, rather than treating all kicks with the same observed covariates as equally makeable.

C.2.7 Combined Validation Summary

Figure 38 summarises the validation evidence across calibration, generalisation, performance by difficulty, and predictions for non-attempted kicks.

C.2.8 Summary

The validation results support three conclusions. First, accounting for informative missingness improves out-of-sample predictive performance relative to a naive model estimated on attempted kicks only. Second, the gains are concentrated among difficult kicks, where selection is strongest and where the decision between kicking at goal and kicking to the corner is most consequential. Third, the model recognises that non-attempted kicks are systematically harder than their geometry alone would imply.

Taken together, these results indicate that the selection-corrected model provides a more credible estimate of the unconditional kick success probability required for counterfactual decision analysis.

D League Simulation and Season-Level Valuation

D.1 League Simulation Procedure

Season-level objectives are evaluated by embedding each post-decision state within a simulation of the remaining league season. All fixtures in the current and subsequent rounds, excluding the match under

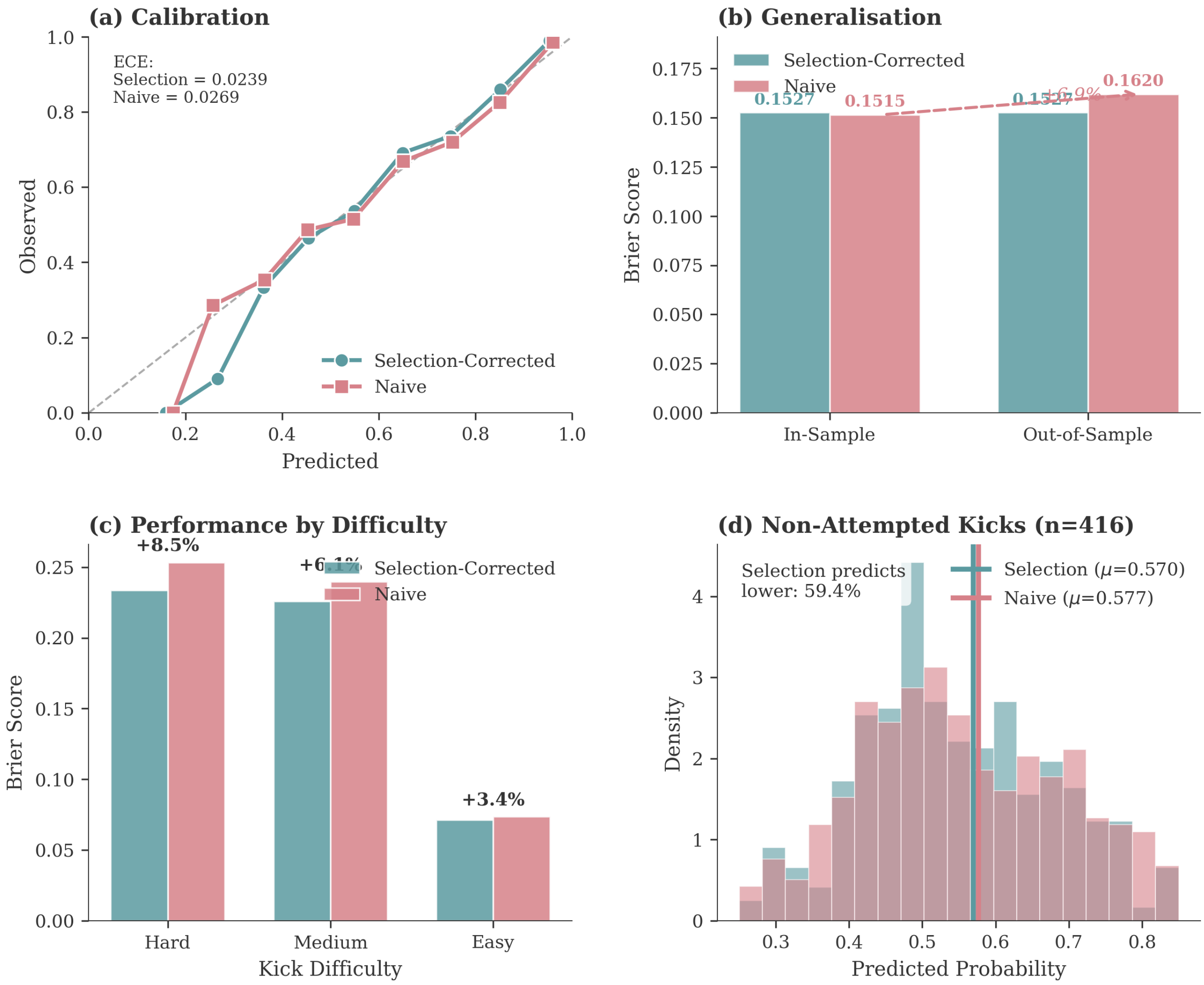

Figure 38: Kick at Goal Model Validation: Combined Results

Notes: Panels summarise calibration, in-sample versus out-of-sample Brier scores, performance by difficulty stratum, and predictions for non-attempted kicks.

analysis, are simulated in advance and cached. Each draw produces a complete set of results for all other teams.

For each post-decision state, the simulated outcome of the focal match is inserted into these pre-simulated seasons and final league tables are computed.

Each state is evaluated using 5,000 simulated seasons. This is sufficient to reduce Monte Carlo error to a negligible level relative to other sources of uncertainty.

D.2 Playoff Structure

Promotion and relegation are determined through a playoff system:

- Semi-final A: 2nd hosts 3rd,
- Semi-final B: 9th from the division above hosts 4th,
- Final: hosted by the winner of Semi-final B if that team is from the higher division, otherwise by the winner of Semi-final A.

The winner is promoted or survives relegation.

D.3 Strength Assignment for Ninth Place

The ninth-placed team from the higher division is assigned strength parameters equal to the average of second and third place in the lower division. This reflects its intermediate position between divisions and produces competitive playoff matches.

Alternative calibrations were tested, including assigning strengths equal to the best and fourth-best teams in the lower division. These adjustments do not materially affect decision recommendations.

D.4 Playoff Simulation

Playoff outcomes are simulated using the scoring model, incorporating team strengths and home advantage. For each relevant league position, the probability of winning the playoff bracket is estimated and used to compute continuation values.